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Draft Standard for Information technology— Telecommunications and information exchange between systems Local and metropolitan area networks— Specific requirements

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 5: Enhanced Service with Privacy Protection

Prepared by the 802.11 Working Group of the

LAN/MAN Standards Committee
of the IEEE Computer Society

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Abstract: This amendment defines modifications to the medium access control (MAC) sublayer that enhance user privacy protection.

Keywords: EPP, enhanced privacy protection

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Introduction

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This amendment defines modifications to both the IEEE 802.11 physical layer (PHY) and the IEEE 802.11 medium access control (MAC) sublayer to enhance user privacy protection.

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Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) specifications

Amendment 5: Enhanced Service with Privacy Protection

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D0.2	2024-02-07	Includes texts motioned on 1/31/2024
D0.3	2024-03-26	Includes texts motioned on 3/14/2024
D0.4	2024-05-28	Includes texts motioned on 5/16/2024
D0.5	2024-07-29	Includes texts motioned on 7/18/2024
D0.6	2024-09-24	Includes texts motioned on 9/12/2024
D0.7	2024-12-02	Includes texts motioned on 11/14/2024. Align the texts with IEEE P802.11REVme D7.0, IEEE P802.11bh D6.0, and IEEE P802.11be D7.0. Update values based on ANA assignments.
D1.0	2025-01-27	Includes texts motioned on 1/15/2025
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D2.0	2025-08-11	Includes texts motioned on 7/28/2025 and 7/31/2025.
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3. Definitions, acronyms, and abbreviations

3.2 Definitions specific to IEEE 802.11

Insert the following definitions (maintaining alphabetical order):

basic service set (BSS) privacy enhancements (BPE): [BPE] BSS privacy features for access point (AP) multi-link devices (MLDs) and non-AP MLDs, including client privacy enhancements **(**CPE) features.

basic service set (BSS) privacy enhancements (BPE) access point (AP) multi-link device (MLD): [BPE AP MLD] An AP MLD activating BPE FA features.

basic service set (BSS) privacy enhancements (BPE) non–access point (non-AP) multi-link device (MLD): [BPE non-AP MLD] A non-AP MLD activating BPE FA features.

client privacy enhancements (CPE): [CPE] Features for enhancing privacy of non–access point (AP) stations (STAs) and non-AP multi-link devices (MLDs).

client privacy enhancements (CPE) access point (AP) multi-link device (MLD): [CPE AP MLD] An AP MLD activating CPE FA features.

client privacy enhancements (CPE) non–access point (non-AP) multi-link device (MLD): [CPE non-AP MLD] A non-AP MLD activating CPE FA features.

distribution system (DS) medium access control (MAC) address: [DS MAC address] A MAC address used by an enhanced privacy protection (EPP) access point (AP) or an EPP AP multi-link device (MLD) as the address to notify the DS and establish the destination mapping for an EPP non-AP STA or an EPP non-AP MLD after (re)association.

enhanced privacy protection (EPP) access point (AP): [EPP AP] An AP with support for at least one of the EPP features.

enhanced privacy protection (EPP) non–access point (AP) station (STA): [EPP non-AP STA] A non-AP STA with support for at least one of the EPP features.

enhanced privacy protection (EPP) access point (AP) multi-link device (MLD): [EPP AP MLD] An AP MLD with support for at least one of the EPP features.

enhanced privacy protection (EPP) non–access point (AP) multi-link device (MLD): [EPP non-AP MLD] A non-AP MLD with support for at least one of the EPP features.

enhanced privacy protection (EPP) epoch: [EPP epoch] The operating period for an EPP group bounded by two consecutive EPP epoch start times and during which a set of FA parameters for that epoch and that group apply.

enhanced privacy protection (EPP) epoch interval: [EPP epoch interval] The nominal period (in TU) between two planned consecutive epoch start times within an EPP epoch sequence.

enhanced privacy protection (EPP) epoch sequence: [EPP epoch sequence] An ordered set of consecutive EPP epochs for a particular EPP group (identified by its EPP Group ID), beginning at the first epoch TSF start time and numbered from 0, that operate under a common EPP epoch settings configuration and Group Epoch Seed.

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enhanced privacy protection (EPP) epoch settings: [EPP epoch settings] A set of parameters characterizing an EPP epoch.

enhanced privacy protection (EPP) group: [EPP group] A construct for representing a collection of non–access point (non-AP) multi-link devices (MLDs) associated with a single AP MLD that apply identical EPP epoch settings. An EPP group includes zero or more members.

frame anonymization: [FA] An enhanced privacy protection (EPP) mechanism for multi-link devices (MLDs) that periodically anonymizes values transmitted unencrypted in MAC headers and control frames that identify an MLD either directly or indirectly.

frame anonymization parameters: [FA parameters] Per-EPP-epoch parameters used by both the AP MLD and non-AP MLD when CPE FA or BPE FA is activated. If a non-AP MLD has CPE FA activated, then these parameters comprise a per-non-AP-MLD AID and a CPE MHA parameter set. If an AP MLD has BPE FA activated, then these parameters comprise a per-non-AP-MLD AID and CPE MHA parameter set, and a per-AP-MLD BPE MHA parameter set.

identity key: A random value, assigned by the basic service set (BSS) privacy enhancements (BPE) access point (AP) multi-link device (MLD), that is used to detect the identity of the BPE AP MLD.

medium access control (MAC) header anonymization: [MHA] A frame anonymization mechanisms applied to selected fields of the MAC header.

over the air (OTA) medium access control (MAC) address: [OTA MAC address] A MAC address in the MAC header transmitted over the wireless medium.

over-the-air packet number: [OPN] The value transmitted in a Counter Mode (CTR) with cipher-block chaining message authentication code (CBC-MAC) protocol (CCMP) header or Galois/Counter Mode (GCM) protocol (GCMP) header in the place of the packet number as part of frame anonymization.

over-the-air sequence number: [OSN] The value transmitted in a medium access control (MAC) protocol data unit (MPDU) header in the place of the sequence number as part of frame anonymization.

privacy group temporal key (PGTK): [PGTK] A random value, assigned by an access point (AP) multi-link device (MLD), shared to all non–access point (non-AP) multi-link devices (MLDs) associated to the AP MLD for frame anonymization.

Change the following definition:

wireless network management (WNM) sleep mode: [WNM sleep mode] An extended power save mode for non–access point (non-AP) stations (STAs) and non-AP multi-link devices (non-AP MLDs) whereby a non-AP STA or non-AP STAs affiliated with a non-AP MLD need not listen for every delivery traffic indication map (DTIM) beacon and do not perform group temporal key/integrity group temporal key/beacon integrity group temporal key (GTK/IGTK/BIGTK) updates, and the non-AP MLDs do not perform privacy group temporal key (PGTK) updates.

3.4 Abbreviations and acronyms

Insert the following acronym definitions (maintaining alphabetical order):

BPE	BSS privacy enhancements
CPE	client privacy enhancements
EPP	enhanced privacy protection

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EPPKE	enhanced privacy protection key exchange
FA	frame anonymization
MHA	MAC header anonymization
OPN	over-the-air packet number
OSN	over-the-air sequence number
OTA	over-the-air

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4. General description

4.2 How wireless local area networks (WLANs) are different

4.2.5 Interaction with other IEEE 802® layers

Change the second paragraph as follows:

In a robust security network association (RSNA), IEEE Std 802.11 provides functions to protect Data frames, IEEE Std 802.1X-2020 provides authentication and a Controlled Port, and IEEE Std 802.11 and IEEE Std 802.1X-2020 collaborate to provide key management. All STAs in an RSNA have a corresponding IEEE 802.1X entity that handles these services. This standard defines how an RSNA utilizes IEEE Std 802.1X-2020 to access these services. Within IEEE Std 802.11, EAPOL PDUs are carried as MSDUs within one or more Data frames, as described in Clause 12 of IEEE Std 802.1X-2020. Within this standard, Data frames used for this purpose are generally referred to as *EAPOL-Key frames, EAPOL-Key request frames, and EAPOL-Start frames*. Within IEEE Std 802.11, EAPOL PDUs can also be carried within Authentication frames (see 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames)).

4.3 Components of the IEEE 802.11 architecture

4.3.8 Robust security network association (RSNA)

Change the third paragraph as follows:

The first component is an IEEE 802.1X port access entity (PAE). A PAE is present on each STA or each MLD in an RSNA to process the EAPOL PDUs and control the forwarding of data to and from the medium access control (MAC). An AP or an AP MLD always implements the Authenticator PAE and Extensible Authentication Protocol (EAP) Authenticator roles, and a non-AP STA or a non-AP MLD always implements the Supplicant PAE and EAP peer roles. In an IBSS or PBSS, each STA implements both the Authenticator PAE and Supplicant PAE roles and both EAP Authenticator and EAP peer roles.

4.5 Overview of the services

4.5.4 Access control and data confidentiality services

4.5.4.2 Authentication

Change the third, fourth, sixth paragraph as follows:

IEEE Std 802.11 defines the following ~~five~~ IEEE 802.11 authentication methods: ~~Open System authentication, FT authentication, simultaneous authentication of equals (SAE), FILS authentication, and preassociation security negotiation (PASN) authentication.~~

Change the description to bullet as follows:

— Open System authentication admits any STA to the DS.
— FT authentication relies on keys derived during the initial mobility domain association to authenticate the stations as defined in Clause 13 (Fast BSS transition).
— SAE authentication uses finite field cryptography to prove knowledge of a shared password.
— IEEE 802.1X authentication uses EAP to authenticate STAs and the AS with one another.

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- FILS authentication allows for faster connection to the network for FILS non-AP STAs by providing authentication, association, and key confirmation information in an efficient number of frame exchanges (see 4.10.3.6 (AKM operations using FILS authentication)).
- PASN authentication allows for the protection of Management frames without association by establishing a PTKSA using authentication frames.
- EPPKE authentication allows for the protection of (Re)Association Request/Response frame by establishing a PTKSA using authentication frames.

The IEEE 802.11 authentication mechanism also allows definition of new authentication methods, or any combination of these authentication methods.

An RSNA might support one or more of the following authentication methods: SAE authentication, IEEE 802.1X authentication, FILS authentication, ~~or~~ PASN authentication, or EPPKE authentication. An RSNA also supports authentication based on IEEE Std 802.1X-2020, or preshared keys (PSKs) after Open System authentication. ~~IEEE 802.1X authentication utilizes the EAP to authenticate STAs and the AS with one another.~~ This standard does not specify an EAP method that is mandatory to implement for authentication based on IEEE Std 802.1X-2020. See 12.6.4 (RSNA policy selection in an IBSS) for a description of the IEEE 802.1X authentication and PSK usage within an IEEE 802.11 IBSS.

SAE authentication, IEEE 802.1X authentication, and Open System IEEE 802.11 authentication are used by STAs in an RSN for an infrastructure BSS. FILS authentication can be used by FILS STAs in an RSN for an infrastructure BSS. SAE authentication, Open System IEEE 802.11 authentication, or no IEEE 802.11 authentication is used in an RSN for an IBSS. SAE authentication is used for an MBSS. In an RSN for DMG BSS, Open System IEEE 802.11 authentication is not used (12.2.4 (RSNA establishment)).

Change the last paragraph as follows:

PASN authentication or EPPKE authentication is used in an RSN for an infrastructure BSS when it is based on a PMKSA established by another RSN authentication protocol. Otherwise, it does not guarantee mutual authentication, and can be used as a non-RSN protocol in an infrastructure BSS.

4.5.4.10 MAC privacy enhancements

Change the first paragraph as follows:

When a non-AP STA searches for, and connects to, an infrastructure BSS, IBSS, or PBSS or attempts to discover services on a network before ~~pre~~association, it selects a MAC address for the particular connection. If ~~the~~a STA uses a fixed MAC address in these transactions, then it is trivial to track ~~the~~ that STA. An MSDU transmitted by a STA is assigned a sequence number that, if never reset, can also be used to track a device irrespective of the MAC address. If OFDM is used, the PHY DATA scrambler used can enable tracking of a device irrespective of the MAC address if it is not reseeded. The dynamic nature of BSS membership combined with this tracking information allows for construction of a network of connections, locations, and behavior. This network can be used to glean private and sensitive information regarding the individual behind the device.

Change the third paragraph as follows:

To mitigate traffic analysis and tracking, a non-AP STA or a non-AP MLD can support the ability to periodically and randomly change its MAC addresses and reset counters and seeds prior to association. Additional mechanisms to increase the privacy for associated STAs and non-AP MLDs can be provided by EPP features. For example, EPP provides MLDs with frame anonymization mechanisms hiding identifiable information in the frame headers by periodically changing the MAC addresses, AIDs, and other header parameters for groups of affiliated STAs while associated (see 4.5.4.10a (Enhanced Privacy Protection (EPP) enhancements)). A non-AP MLD can also change the MAC addresses of its affiliated STAs prior to

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an association. Such a non-AP STA or a non-AP MLD using the device ID mechanism, upon reconnecting to a network, can provide either a device ID or a PASN ID previously provided by the network. Such a non-AP STA using the identifiable random MAC address (IRM) mechanism, upon reconnecting to the network can provide the IRM that the STA or the non-AP MLD previously provided to the network. Such a non-AP STA can use both device ID and IRM mechanisms concurrently. Such a STA or the non-AP MLD can also use a measurement ID, previously provided by the network, to assist while performing beacon report measurement procedures. These mechanisms allow the network to recognize the STA or the non-AP MLD while providing protection against third party tracking or traffic analysis. When the network can recognize the STA, it can map already established shared identity state (see 12.2.13 (Identifying a non-AP STA or a non-AP MLD with changing MAC address)) to this STA, which can allow the network to provide network acquisition steering and selection, or allow the network to connect transactional information obtained preassociation or in a prior association to the device that is associating. While discovering networks, a non-AP STA can refrain from gratuitously transmitting Probe Request frames containing SSIDs of favored BSS networks.

Insert the following new subclause after 4.5.4.10 (MAC privacy enhancements):

4.5.4.10a Enhanced Privacy Protection (EPP) enhancements

Third parties observing the wireless medium might seek to track device locations and device activity. Using EPP features, a STA or MLD can modify the amount of information disclosed in several ways. Using EPP client privacy enhancements (CPE), a non-AP STA or non-AP MLD can modify the content of messages sent before and during association to reduce the opportunity to fingerprint the non-AP STA or non-AP MLD through its messages outside of a secured connection.

Additional threats exist while a secured connection is established. When an 802.11 device transmits static or predictable values associated with or assigned to that device in unencrypted fields and elements, those values can be used by a third party observer to determine the long-term presence of that non-AP STA or non-AP MLD at a location, even if the identity of the person related to that device cannot be determined.

Examples of unencrypted fields and elements that contain static or predictable values assigned to the transmitter or receiver include: transmitter address (TA), receiver address (RA); sequence number (SN); packet number (PN); timestamp; association identifier (AID) and fields and elements derived from the AID. A third-party observer can monitor these values and, as long as the values remain static or predictable, the third-party observer can determine that the corresponding 802.11 device continues to be present.

Frame anonymization (FA), available when MLO is enabled, improves user privacy by restricting the periods within which unencrypted fields and elements remain static or predictable. The periods, called EPP epochs, can be relatively short in duration when compared to the typical lifetime of an association.

The CPE FA mechanisms anonymize fields that identify a non-AP MLD (such as the MAC address(es) of its affiliated STAs) during an association.

The BPE FA mechanisms anonymize fields that identify an AP MLD and its set of associated non-AP MLDs (such as the MAC address(es) of its affiliated APs, the ESS that it belongs to , and its group addresses). A BPE AP MLD can protect the content of its Beacon frames and only be discoverable by BPE non-AP MLDs that are configured to recognize the BPE AP MLD. A BPE AP MLD and its associated non-AP MLDs can change their OTA MAC addresses and other trackable fields for both individually addressed and group addressed transmissions.

An AP MLD or non-AP MLD implementing EPP features can support CPE FA mechanisms or BPE FA mechanisms. While a CPE AP MLD implements one or more EPP groups, a BPE AP MLD implements a single EPP group. As such, an AP MLD or non-AP MLD implementing BPE FA mechanisms implements CPE FA mechanisms within a single EPP group without multi-group management.

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NOTE—EPP features might be generally described with MLO terminology, but separate descriptions are provided for individual features to explain when the behavior of MLO and non-MLO devices differs.

A third party can also determine the long-term presence of an MLD by exploiting other unencrypted fields and elements that contain static or predictable values that are not assigned to the transmitter or receiver. For example, the SA (or DA respectively), when present, provides the MAC address of the source (or destination respectively) of the frame. These MAC addresses can remain unchanged for a relatively long time or might never change. If SA and/or DA are not encrypted, these values might be used to profile the transmitter and/or receiver. An AP MLD or non-AP MLD implementing FA features can transmit MSDUs in A-MSDUs (with encrypt SA and DA) to mitigate such profiling.

4.10 IEEE Std 802.11 and IEEE Std 802.1X-2020

4.10.2 IEEE 802.11 usage of IEEE Std 802.1X-2020

Change the first paragraph as follows:

IEEE Std 802.11 depends upon IEEE Std 802.1X-2020 to control the flow of MAC service data units (MSDUs) between the DS and STAs by use of the IEEE 802.1X Controlled/Uncontrolled Port model. IEEE 802.1X EAPOL PDUs are transmitted either in one or more IEEE 802.11 Data frames and passed via the IEEE 802.1X Uncontrolled Port or in one or more Authentication frames. The IEEE 802.1X Controlled Port is blocked from passing general data traffic between two STAs until an IEEE 802.1X authentication procedure completes successfully over the IEEE 802.1X Uncontrolled Port or over the Authentication frame exchange carrying EAPOL PDUs. It is the responsibility of both the Supplicant and the Authenticator to implement port blocking. Each association between a pair of STAs creates a unique pair of IEEE 802.1X Ports, and authentication takes place relative to those ports alone or through the exchange of Authentication frames carrying EAPOL PDUs.

4.10.3 Infrastructure functional model overview

4.10.3.2 AKM operations with AS

Change item b) of the first paragraph as follows (not all lines are shown):

The following AKM operations are carried out when an IEEE 802.1X AS is used:

b) A STA discovers the AP's security policy through passively monitoring Beacon frames or through active probing. If IEEE 802.1X authentication is used, the EAP authentication process starts when the Authenticator sends the EAP-Request or the Supplicant sends the EAPOL-Start PDU (in one or more EAPOL-Start frames or in an Authentication frame). EAP messages pass between the Supplicant and AS via the Authenticator and Supplicant's Uncontrolled Ports as described in 12.7 (Keys and key distribution). EAP messages pass between the Supplicant and AS via the Authentication frame exchange carrying EAPOL PDUs as described in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames).

Change the second paragraph as follows:

The following is achieved by a~~A~~ 4-way handshake or FT 4-way handshake utilizing EAPOL-Key PDUs that is initiated by the Authenticator ~~to do the following~~or by an IEEE 802.1X Authentication frame exchange followed by an encrypted (Re)Association Request/Response frame exchange:

— Confirm that a live peer holds the PMK.

— Confirm that the PMK is current.

— In the case of fast BSS transition, derive PMK-R0s and PMK-R1s.

— Derive a fresh pairwise transient key (PTK) from the PMK or, in the case of fast BSS transition, from the PMK-R1~~,~~. ~~t~~The derived PTK includes~~ing~~ the key derivation key (KDK) under the conditions defined in 12.6.1.1.6 (PTKSA)~~if WUR frame protection is negotiated~~.

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- ~~If WUR frame protection is negotiated, d~~Derive a fresh WTK from the KDK under the conditions defined in 12.6.1.1.6 (PTKSA).
- Install the pairwise encryption and integrity keys, and ~~if WUR frame protection is negotiated,~~ the WTK if derived.
- Transport the group keys and sequence number from Authenticator to Supplicant and install the group keys and sequence number in the STA and, if not already installed, in the AP.
- Verify that the RSN capabilities negotiated are valid as defined in 9.4.2.23.4 (RSN capabilities).
- Confirm the cipher suite selection.

Change the fourth paragraph as follows:

Upon successful completion of the 4-way handshake or the encrypted (Re)Association Request/Response frame exchange, the Authenticator and Supplicant have authenticated each other; and the IEEE 802.1X Controlled Ports are unblocked to permit general data traffic as described in 12.7 (Keys and key distribution).

4.10.7 PMKSA caching

Change the second paragraph as follows (not all lines are shown):

A STA can supply a list of PMK identifiers in the (Re)Association Request frame ~~or~~ , in the first FILS Authentication frame of a FILS Authentication frame exchange, in the first IEEE 802.1X Authentication frame of an IEEE 802.1X Authentication frame exchange, or in the first EPPKE Authentication frame of an EPPKE Authentication frame exchange. Each PMK identifier names a PMKSA. The Authenticator can specify the selected PMK identifier in message 1 of the 4-way handshake ~~or the~~, in the second FILS Authentication frame of a FILS Authentication frame exchange, in the second IEEE 802.1X Authentication frame of an IEEE 802.1X Authentication frame exchange, or in the second EPPKE Authentication frame of an EPPKE Authentication frame exchange. The selection of the PMK identifiers to be included by the STA and Authenticator is out of the scope of this standard.

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6. Layer management

6.4 Table of MLME SAP interfaces

change Table 6-1 as follows:

Table 6-1—MLME SAP interface

Service Name	MLME-XXX	Type	References	Comments
...	...	...	...	...
EAPOL	EAPOL	5	12.7.2 (EAPOL-Key frames) 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames)	

6.5 MLME SAP primitives

6.5.3

6.5.3.3 MLME-SCAN.confirm

6.5.3.3.2 Semantics of the service primitive

change the first paragraph and table as follows:

The primitive parameters are as follows:

MLME-SCAN.confirm(
BSSDescriptionSet,
BSSDescriptionFromMeasurementPilotSet,
BSSDescriptionFromFDSet,
BSSDescriptionFromPrivacyBeacons,
ResultCode,
Multi-band local,
Multi-band peer,
ScannedChannelList,
VendorSpecificInfo
)

Name	Type	Valid range	Description
....	...	...	...
BSSDescriptionFromFDSet	Set of BSSDescriptionFromFDs	N/A	The BSSDescriptionFromFDSet is returned to indicate the results of the scan request derived from FILS Discovery frames. It is a set containing zero or more instances of a BSSDescriptionFromFD. Present if dot11FILSActivated is true; otherwise not present.

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Name	Type	Valid range	Description
BSSDescriptionFrom PrivacyBeaconSet	Set of BSSDescriptionFrom PrivacyBeacons	N/A	The BSSDescriptionFrom PrivacyBeaconSet is returned to indicate the results of the scan request derived from the Privacy Beacon frames. It contains zero or more instances of a BSSDescriptionFrom PrivacyBeacon. Present if dot11EPPEpochActivated is true, otherwise not present.
ResultCode	Enumeration	SUCCESS, INTERMEDIATE_SCAN_RESULT, NOT_SUPPORTED, PARTIAL_SCAN	Indicates the result of the MLME-SCAN.confirm primitive. The INTERMEDIATE_SCAN_RESULT is used to report the discovered BSSs when the value of the ReportingOption parameter in the MLME-SCAN.request primitive is CHANNEL_SPECIFIC or IMMEDIATE and is valid if dot11FILSActivated is true. The PARTIAL_SCAN is used to report that not all channels have been scanned as specified in the ChannelList parameter, if present, of the corresponding MLME-SCAN.request primitive. The ScannedChannelList parameter contains the list of scanned channels.
...	...	...	...

Insert the following at the end:

The BSSDescriptionFromPrivacyBeaconSet parameter is present if dot11EPPEpochActivated is true; otherwise, it is not present. Each BSSDescriptionFromPrivacyBeacon consists of the parameters shown in the following table:

Name	Type	Valid range	Description
Address 2	MAC address	N/A	The transmitter address of the Privacy Beacon frame.
Identity Hash	MAC address	N/A	The Identity Hash of the Privacy Beacon frame.

6.5.5 Authenticate

6.5.5.2 MLME-AUTHENTICATE.request

6.5.5.2.2 Semantics of the service primitive

Modify MLME-AUTHENTICATE.request and the table as follows (not all lines shown):

The primitive parameters are as follows:

MLME-AUTHENTICATE.request(
...,
Content of 802.1X Authentication frame,
Content of EPPKE Authentication frame,
VendorSpecificInfo
)

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Name	Type	Valid range	Description
.....	.....	.....	.....
AuthenticationType	Enumeration	OPEN_SYSTEM, SHARED_KEY, FAST_BSS_TRANSITION, SAE, FILS_SHARED_KEY WITHOUT_PFS, FILS_SHARED KEY_WITH_PFS, FILS_PUBLIC_KEY, 802_1X, EPPKE	Specifies the type of authentication algorithm to use during the authentication process.
.....	.....	.....	.....
Content of 802.1X Authentication frame	Sequence of elements and fields	As defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames), 12.16.8.3 (IEEE 802.1X), 9.4.1.82 (Encapsulation Length field), 9.4.1.83 (Encapsulation field), 9.4.2.295 (AKM Suite Selector element), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.188 (FILS Nonce element), 9.4.2.312 (Diffie-Hellman Parameter element).	The set of elements and fields to be included in the 802.1X Authentication frame. Present if AuthenticationType indicates 802_1X and dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated is true, otherwise not present.
Content of EPPKE Authentication frame	Sequence of elements and fields	As defined in 12.16.9.3.2 (EPPKE frame construction and processing), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.186 (Wrapped Data element), 9.4.2.305 (PASN Parameters element), 9.4.2.47 (Timeout Interval element)	The set of elements and fields to be included in the EPPKE Authentication frame. Present if AuthenticationType indicates EPPKE and dot11EPPKEActivated is true, otherwise not present.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.24 (Vendor Specific element)	Zero or more elements.

6.5.5.3 MLME-AUTHENTICATE.confirm

6.5.5.3.2 Semantics of the service primitive

Modify MLME-AUTHENTICATE.confirm and the table as follows (not all lines shown):

The primitive parameters are as follows:

MLME-AUTHENTICATE.confirm(
....
Content of 802.1X Authentication frame,
Content of EPPKE Authentication frame,
VendorSpecificInfo
)

Name	Type	Valid range	Description
.....	.....	.....	.....

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Name	Type	Valid range	Description
AuthenticationType	Enumeration	OPEN_SYSTEM, SHARED_KEY FAST_BSS_TRANSITION, SAE, FILS_SHARED KEY_WITHOUT_PFS, FILS_SHARED_KEY_WI TH_PFS, FILS_PUBLIC_KEY, PASN, 802_1X, EPPKE	Specifies the type of authentication algorithm that was used during the authentication process. This value matches the AuthenticationType parameter specified in the corresponding MLME-AUTHENTICATE.request primitive.
.....	.....	.....	.....
Content of 802.1X Authentication frame	Sequence of elements and fields	As defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames), 12.16.8.3 (IEEE 802.1X), 9.4.1.82 (Encapsulation Length field), 9.4.1.83 (Encapsulation field), 9.4.2.295 (AKM Suite Selector element), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.188 (FILS Nonce element), 9.4.2.312 (Diffie-Hellman Parameter element).	The set of elements and fields received in the 802.1X Authentication frame. Present if AuthenticationType indicates 802_1X and dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated is true, otherwise not present.
Content of EPPKE Authentication frame	Sequence of elements and fields	As defined in 12.16.9.3.2 (EPPKE frame construction and processing), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.186 (Wrapped Data element), 9.4.2.305 (PASN Parameters element), 9.4.2.47 (Timeout Interval element)	The set of elements and fields received in the EPPKE Authentication frame. Present if AuthenticationType indicates EPPKE and dot11EPPKEActivated is true, otherwise not present.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.24 (Vendor Specific element)	Zero or more elements.

6.5.5.4 MLME-AUTHENTICATE.indication

6.5.5.4.2 Semantics of the service primitive

Modify MLME-AUTHENTICATE.indication and the table as follows (not all lines shown):

The primitive parameters are as follows:

MLME-AUTHENTICATE.indication(

....

Content of 802.1X Authentication frame,

Content of EPPKE Authentication frame,

VendorSpecificInfo

)

Name	Type	Valid range	Description
.....	.....	.....	.....

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Name	Type	Valid range	Description
AuthenticationType	Enumeration	OPEN_SYSTEM, SHARED_KEY, FAST_BSS_ TRANSITION, SAE, FILS_SHARED_KEY_WI THOUT_PFS, FILS_SHARED_KEY_WI TH_PFS, FILS_PUBLIC_KEY, PASN, 802_1X, EPPKE	Specifies the type of authentication algorithm that was used during the authentication process.
.....	.....	.....	.....
Content of 802.1X Authentication frame	Sequence of elements and fields	As defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames), 12.16.8.3 (IEEE 802.1X), 9.4.1.82 (Encapsulation Length field), 9.4.1.83 (Encapsulation field), 9.4.2.295 (AKM Suite Selector element), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.188 (FILS Nonce element), 9.4.2.312 (Diffie-Hellman Parameter element).	The set of elements and fields received in the 802.1X Authentication frame. Present if AuthenticationType indicates 802_1X and dot11EPPIEEE8021XUtilizingAuthen ticationFrameActivated is true, otherwise not present.
Content of EPPKE Authentication frame	Sequence of elements and fields	As defined in 12.16.9.3.2 (EPPKE frame construction and processing), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.186 (Wrapped Data element), 9.4.2.305 (PASN Parameters element), 9.4.2.47 (Timeout Interval element)	The set of elements and fields received in the EPPKE Authentication frame. Present if AuthenticationType indicates EPPKE and dot11EPPKEActivated is true, otherwise not present.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.24 (Vendor Specific element)	Zero or more elements.

6.5.5.5 MLME-AUTHENTICATE.response

6.5.5.5.2 Semantics of the service primitive

Modify MLME-AUTHENTICATE.response and the table as follows (not all lines shown):

The primitive parameters are as follows:

MLME-AUTHENTICATE.response(

....

Content of 802.1X Authentication frame,

Content of EPPKE Authentication frame,

VendorSpecificInfo

)

Name	Type	Valid range	Description
.....	.....	.....	.....

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Name	Type	Valid range	Description
Content of 802.1X Authentication frame	Sequence of elements and fields	As defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames), 12.16.8.3 (IEEE 802.1X), 9.4.1.82 (Encapsulation Length field), 9.4.1.83 (Encapsulation field), 9.4.2.295 (AKM Suite Selector element), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.188 (FILS Nonce element), 9.4.2.312 (Diffie-Hellman Parameter element).	The set of elements and fields to be included in the 802.1X Authentication frame. Present if AuthenticationType indicates 802_1X and dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated is true, otherwise not present.
Content of EPPKE Authentication frame	Sequence of elements and fields	As defined in 12.16.9.3.2 (EPPKE frame construction and processing), 9.4.2.23 (RSNE), 9.4.2.240 (RSNXE), 9.4.2.186 (Wrapped Data element), 9.4.2.305 (PASN Parameters element), 9.4.2.47 (Timeout Interval element)	The set of elements and fields to be included in the EPPKE Authentication frame. Present if AuthenticationType indicates EPPKE and dot11EPPKEActivated is true, otherwise not present.
VendorSpecificInfo	A set of elements	As defined in 9.4.2.24 (Vendor Specific element)	Zero or more elements.

6.5.14 SetKeys

6.5.14.1 MLME-SETKEYS.request

6.5.14.1.2 Semantics of the service primitive

Modify the table at the end as follows:

Each SetKeyDescriptor consists of the following parameters:

Name	Type	Valid range	Description
Key	Bit string	N/A	The temporal key value
Length	Integer	N/A	The number of bits in the Key to be used.

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Name	Type	Valid range	Description
Key ID	Integer	0–3 shall be used with TKIP, CCMP, and GCMP; 4–5 with BIP for IGTK; 6-7 with BIP for BIGTK; 8–9 with BIP for WIGTK; and 10–4095 are reserved	Key identifier
Key Type	Enumeration	Group, Pairwise, PeerKey, IGTK, BIGTK, WIGTK, PGTK, Identity Key	Defines whether this key is a GTK, TK, TPK-TK, IGTK, BIGTK, ~~or~~ WIGTK, PGTK or Identity Key respectively.
Address	MAC address	Any valid individual address	This parameter is valid only when the Key Type value is one of: — Pairwise, — Group and the STA is in an IBSS or PBSS (but not an MBSS), — PeerKey.
Receive Sequence Counter	8 octets	N/A	Initialization value of the replay counter(s). This parameter is valid only when the Key Type is Group, IGTK, BIGTK, or WIGTK.

6.5.14.1.4 Effect of receipt

Change the first paragraph as follows:

When the Key Type is Group, IGTK, BIGTK, ~~or~~ WIGTK, or PGTK and the key matches the GTK, IGTK, BIGTK, or WIGTK, PGTK or Identity Key, if any, installed as a result of EAPOL-Key PDUs (see 12.7.7.4 (Group key handshake implementation considerations)) or exiting WNM sleep mode (see 11.2.3.15.1 (WNM sleep mode capability)) receipt of this primitive shall have no effect except updating the RSC(s) when they are greater than those currently stored. Otherwise, irrespective of the Key Type parameter, when the Key parameter is the same as a key installed as a result of EAPOL-Key PDUs or exiting WNM sleep mode, receipt of this primitive shall have no effect. Otherwise, receipt of this primitive causes the MAC to apply the keys as follows, subject to the MLME-SETPROTECTION.request primitive:

— The MAC uses the key and key ID for the transmission of subsequent frames to which the key and key ID apply (as defined by the Key Type and Address parameters).

— When the Key Type parameter is not PGTK, t~~T~~he MAC installs the key with the associated key ID such that received frames for that cipher, of the appropriate type, and containing the matching key ID are processed using that key and its associated state information. When the Key Type parameter is PGTK, the MAC installs the key, and the FA is processed using that key.

— When the Key Type parameter is Identity Key, the MAC installs the key if the key is not already installed such that the transmitted and received Privacy Beacon frames are processed by using that key. The identity key is maintained at least for the lifetime of the key that is configured together with the identity key. A non-AP MLD does not delete the installed identity key if the non-AP MLD is dis-associated or deauthenticated from the non-AP MLD.

— When the Key Type parameter is Pairwise or PeerKey, and the Key, Key ID, and Address (where valid) parameters identify a new key to be set, the MAC shall initialize the transmitter TSC/PN counter and the receiver replay counter(s) to 0. When the Key Type parameter is not Pairwise, Peer-Key, ~~or~~ BIGTK, PGTK, or Identity Key and the Key, Key ID, and Address (where valid) parameters identify a new key to be set, the MAC shall initialize, depending on the direction of the traffic, the transmitter TSC/PN/IPN/WIPN counter to 0 or 1 (see Clause 12 (Security) and Clause 29 (Wake-Up Radio (WUR) MAC specification(11ba))) or the receiver replay counter(s) to the value in the

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Receive Sequence Count parameter. When the Key Type parameter is BIGTK, and the Key and Key ID parameters identify a new key to be set, the MAC shall initialize, depending on the direction of the traffic, the transmitter BIPN counter as specified in 12.5.3.4 (BIP replay counters and packet numbers) or the receiver replay counter to the value in the Receive Sequence Count parameter. When the Key Type, Key, Key ID, and Address (where valid) parameters identify an existing key, the MAC shall not change the transmitter TSC/PN/IPN/BIPN/WIPN counter or the receiver replay counter(s) associated with that key.

6.5.15 DeleteKeys

6.5.15.1 MLME-DELETEKEYS.request

6.5.15.1.3 When generated

Modify the table at the end as follows:

Name	Type	Valid range	Description
Key ID	Integer	0–3 shall be used with TKIP, CCMP, and GCMP; 4–5 with BIP for IGTK; 6-7 with BIP for BIGTK; 8–9 with BIP for WIGTK; and 10–4095 are reserved	Key identifier.
Key Type	Enumeration	Group, Pairwise, PeerKey, IGTK, BIGTK, WIGTK, PGTK	Defines whether this key is a GTK, TK, TPK-TK, IGTK, BIGTK, ~~or~~ WIGTK or PGTK respectively.
Address	MAC address	Any valid individual address	This parameter is valid only when the Key Type value is one of: — Pairwise, — Group and the STA is in an IBSS or PBSS (but not an MBSS), — PeerKey.
Encapsulation Mode	Enumeration	Normal, BCE	This parameter is valid only when the Key Type value is BIGTK

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9. Frame formats

9.2 MAC frame formats

9.2.4 Frame fields

9.2.4.1 Frame Control field

9.2.4.1.3 Type and Subtype subfields

Insert new rows to Table 9-1 (Valid type and subtype combinations) (not all lines shown):

Table 9-1—Valid type and subtype combinations

Type value B3 B2	Type Description	Subtype value B7 B6 B5 B4	Subtype description
11	Extension	0010	Privacy Beacon

9.3 Format of individual frame types

9.3.3 (PV0) Management frames

9.3.3.5 Association Request frame format

Insert new rows to Table 9-64 (Association Request frame body) in numeric order (not all lines shown):

-- Editor Note: 802.11-2024 up to 60, 802.11bh-2024 61-62, 802.11be-2024 63-65, 802.11bk-2025 no addition, 802.11bf-2025 66-69

Table 9-64—Association Request frame body

Order	Information	Notes
…		
70	DS MAC Address	The DS MAC Address element is present if the Association Request frame is encrypted, dot11DSMACAddressActivated is true, and the peer indicates support for DS MAC Address in the RSNXE; otherwise, it is not present.
71	EPP	The EPP element is optionally present if the Association Request frame is encrypted and dot11EPPEpochActivated is true; otherwise, it is not present.

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Table 9-64—Association Request frame body *(continued)*

Order	Information	Notes
72	Fast BSS Transition	An FTE is present in an Association Request frame if dot11FastBSSTransitionActivated is true, encryption of the Association Request frame is used, and dot11RSNAAuthenticationSuiteSelected is equal to an AKM suite selector value with the Authentication type column indicating FT authentication. See Table 9-190 (AKM suite selectors) (i.e., part of a fast BSS transition in an RSN).
73	Nonce	The Nonce element is optionally present as defined in 12.16.7.2 (PMKID privacy); otherwise, it is not present.

9.3.3.6 Association Response frame format

Change rows in Table 9-65 (Association Response frame body) as follows (not all lines shown):

-- Editor Note: 802.11-2024 up to 77, 802.11bh-2024 78-80, 802.11be-2024 81-86, 802.11bk-2025 no addition, 802.11bf-2025 87-90,revmf D1.1 uses 92

Table 9-65—Association Response frame body

Order	Information	Notes
…		
10	RSN	The RSNE is present if dot11FILSActivated is true or if performing OWE or if the Association Response frame is encrypted; otherwise, it is not present.
...		
38	Key Delivery	The Key Delivery element is present if dot11FILSActivated is true or if the Association Response frame is encrypted; otherwise, it is not present.
...		
91	EPP	The EPP element is present if the Association Response frame is encrypted and dot11EPPEpochActivated is true; otherwise, it is not present.
93	Nonce	The Nonce element is optionally present as defined in 12.16.7.2 (PMKID privacy); otherwise, it is not present.
94	AID List	This element is present for CPE non-AP MLD if dot11EPPEpochActivated is true.

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9.3.3.7 Reassociation Request frame format

Insert new rows to Table 9-66 (Reassociation Request frame body) in numeric order (not all lines shown):

-- Editor Note: 802.11-2024 up to 64, 802.11bh-2024 no addition, 802.11be-2024 65-67, 802.11bk-2025 no addition, 802.11bf-2025 68-71, revmf D1.1 uses 74

Table 9-66—Reassociation Request frame body

Order	Information	Notes
…		
72	DS MAC Address	The DS MAC Address element is present if the Reassociation Request frame is encrypted, dot11DSMACAddressActivated is true, and the peer indicates support for DS MAC Address in the RSNXE; otherwise, it is not present.
73	EPP	The EPP element is optionally present if the Reassociation Request frame is encrypted and dot11EPPEpochActivated is true; otherwise, it is not present.
75	Nonce	The Nonce element is optionally present as defined in 12.16.7.2 (PMKID privacy); otherwise, it is not present.

9.3.3.8 Reassociation Response frame format

Change rows in Table 9-67 (Reassociation Response frame body) as follows (not all lines shown):

-- Editor Note: 802.11-2024 up to 80, 802.11bh-2024 no addition, 802.11be-2024 81-86, 802.11bk-2025 no addition, 802.11bf-2025 87-90, revmf D1.1 uses 92

Table 9-67—Reassociation Response frame body

Order	Information	Notes
…		
10	RSN	An RSNE is present in a Reassociation Response frame if dot11FastBSSTransitionActivated is true, dot11RSNAActivated is true, and this frame is a response to a Reassociation Request frame that contained an FTE (i.e., part of a fast BSS transition in an RSN)~~;~~ or if dot11FILSActivated is true~~;~~ or if performing OWE, or if the Reassociation Response frame is encrypted. Otherwise, it is not present.
…		

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Table 9-67—Reassociation Response frame body *(continued)*

Order	Information	Notes
41	Key Delivery	The Key Delivery element is present if dot11FILSActivated is true and FILS authentication is used or if the Reassociation Response frame is encrypted; otherwise, it is not present.
...		
91	EPP	This element is present if the Ressociation Response frame is encrypted and dot11EPPEpochActivated is true; otherwise, it is not present.
93	Nonce	The Nonce element is optionally present as defined in 12.16.7.2 (PMKID privacy); otherwise, it is not present.
94	AID List	This element is present for CPE non-AP MLD if dot11EPPEpochActivated is true.

9.3.3.11 Authentication frame format

-- Editor Note: 802.11-2024 up to order 27, 802.11bh-2024 no addition, 802.11be-2024 up to order 28, 802.11bk-2025 no addition, 802.11bf-2025 no addition

Modify Table 9-70 (Authentication frame body) as follows (not all lines shown):

Table 9-70—Authentication frame body

Order	Information	Notes
…		
9	Confirm	An unsigned integer encoded as described in 12.4.7.4 (Encoding and decoding of SAE Confirm messages). This is present only in certain Authentication frames as defined in Presence of fields and elements in Authentication frames.
9a	Encapsulation Length	This field indicates the number of octets in the Encapsulation field as described in 9.4.1.82 (Encapsulation Length field). This is present only in certain Authentication frames as defined in Table 9-71 (Presence of fields and elements in Authentication frames).
9b	Encapsulation	This field is used to carry an EAPOL PDU as described in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames). This is present only when the Encapsulation Length field is nonzero.
...		

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Table 9-70—Authentication frame body *(continued)*

Order	Information	Notes
17	~~FILS~~ Nonce	The ~~FILS~~ Nonce element is present only in certain ~~FILS~~ Authentication frames as defined in Table 9-71 (Presence of fields and elements in Authentication frames).
...		
25	PASN Parameters	A PASN Parameters element is present only in certain Authentication frames as defined in Table 9-71 (Presence of fields and elements in Authentication frames).
...		
29	Diffie-Hellman Parameter	A Diffie-Hellman Parameter element is present only in certain Authentication frames as defined in Table 9-71 (Presence of fields and elements in Authentication frames).

Change Table 9-71 (Presence of fields and elements in Authentication frames) and insert new rows at the end of Table 9-71 (Presence of fields and elements in Authentication frames) as follows (not all lines shown):

Table 9-71—Presence of fields and elements in Authentication frames

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
FT	1	Reserved	The MDE is present. The FTE and RSNE(s) are present if dot11RSNAActivated is true. The RSNXE is present if any subfield of the Extended RSN Capabilities field in this element is nonzero, except the Field Length subfield. The Diffie-Hellman Parameter element is optionally present as defined in 12.16.8.2 (FT protocol).
FT	2	Not REJECTED_WITH_SUGGESTED_BSS_TRANSITION	The MDE is present if the Status Code field is 0. The FTE and RSNE(s) are present if the Status Code field is 0 and dot11RSNAActivated is true. The Diffie-Hellman Parameter element is optionally present as defined in 12.16.8.2 (FT protocol).

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Table 9-71—Presence of fields and elements in Authentication frames *(continued)*

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
....			
FILS Shared Key authentication without PFS	1	Reserved	The RSNE is present. The MDE is present if the FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present. The FILS Session element is present. The Wrapped Data element is present.
FILS Shared Key authentication without PFS	2	Status	The RSNE is present. The MDE and the FTE are present if the Status Code field is 0 and FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present if the Status Code field is 0. The FILS Session element is present if the Status Code field is 0. The Wrapped Data element is present if the Status Code field is 0. The Association Delay Info element is present if the Status Code field is 0 and the AP expects that the (Re)Association Response frame will be transmitted more than 1 TU after the (Re)Association Request frame.
FILS Shared Key authentication with PFS	1	Reserved	The Finite Cyclic Group field is present. The FFE field is present. The RSNE is present. The MDE is present if the FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present. The FILS Session element is present. The Wrapped Data element is present.

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Table 9-71—Presence of fields and elements in Authentication frames *(continued)*

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
FILS Shared Key authentication with PFS	2	Status	The Finite Cyclic Group field is present if the Status Code field is 0. The FFE field is present if the Status Code field is 0. The RSNE is present. The MDE and the FTE are present if the Status Code field is 0 and FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present if the Status Code field is 0. The FILS Session element is present if the Status Code field is 0. The Wrapped Data element is present if the Status Code field is 0. The Association Delay Info element is present if the Status Code field is 0 and the AP expects that the (Re)Association Response frame will be transmitted more than 1 TU after the (Re)Association Request frame.
FILS Public Key authentication	1	Reserved	The Finite Cyclic Group field is present. The FFE field is present. The RSNE is present. The MDE is present if the FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present. The FILS Session element is present.

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Table 9-71—Presence of fields and elements in Authentication frames *(continued)*

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
FILS Public Key authentication	2	Status	The Finite Cyclic Group field is present if the Status Code field is 0. The FFE field is present if the Status Code field is 0. The RSNE is present. The MDE and the FTE are present if the Status Code field is 0 and FILS authentication is used for FT initial mobility domain association. The ~~FILS~~ Nonce element is present if the Status Code field is 0. The FILS Session element is present if the Status Code field is 0. The Association Delay Info element is present if the Status Code field is 0 and the AP expects that the (Re)Association Response frame will be transmitted more than 1 TU after the (Re)Association Request frame.
....			
IEEE 802.1X authentication	1	Reserved	The Encapsulation Length field is present. The Encapsulation field is present only when the Encapsulation Length field is nonzero. The AKM Suite Selector element is optionally present as defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames). The RSNE is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The RSNXE is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The Nonce element is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The Diffie-Hellman Parameter element is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The MDE is optionally present as defined in 12.16.8.4 (FT initial mobility domain association).

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Table 9-71—Presence of fields and elements in Authentication frames *(continued)*

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
IEEE 802.1X authentication	2	SUCCESS	The Encapsulation Length field is present. The Encapsulation field is present only when the Encapsulation Length field is nonzero. The AKM Suite Selector element is optionally present as defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames). The RSNE is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The Nonce element is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The Diffie-Hellman Parameter element is optionally present as defined in 12.16.8.3 (IEEE 802.1X). The MDE is optionally present as defined in 12.16.8.4 (FT initial mobility domain association). The FTE is optionally present as defined in 12.16.8.4 (FT initial mobility domain association). The MIC element is optionally present as defined in 12.16.8.3 (IEEE 802.1X).
IEEE 802.1X authentication	2	Not SUCCESS	The Encapsulation Length field is present. The Encapsulation field is present only when the Encapsulation Length field is nonzero.
IEEE 802.1X authentication	> =3	Status	The Encapsulation Length field is present. The Encapsulation field is present only when the Encapsulation Length field is nonzero.

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Table 9-71—Presence of fields and elements in Authentication frames *(continued)*

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements indicated as conditional in Table 9-70 (Authentication frame body)
EPPKE authentication	1	Reserved	The RSNE is present. The RSNXE is present if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero. The MDE is present if the Base AKMP is 00-0F-AC:9 or 00-0F-AC:25. The PASN Parameters element is present. The Timeout Interval element is optionally present. The Wrapped Data element is present if the Wrapped Data Format field in the PASN Parameters element is nonzero and not reserved.
EPPKE authentication	2	Status	The RSNE is present if the Status Code field is 0. The PASN Parameters element is present if the Status Code field is 0. The RSNXE is present if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero. The MDE and the FTE are present if the Base AKMP is 00-0F-AC:9 or 00-0F-AC:25. The Timeout Interval element is optionally present. The Wrapped Data element is present if the Wrapped Data Format field in the PASN Parameters element is nonzero and not reserved and the Status Code field is 0. The MIC element is present.
EPPKE authentication	3	Status	The PASN Parameters element is present if Status Code field is 0. The Wrapped Data element is present if the Wrapped Data Format field in the PASN Parameters element is nonzero and not reserved; and the Status Code field is 0. The MIC element is present.

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9.3.4 Extension frames

Insert the following new subclauses at the end of 9.3.4:

-- Editor Note: 802.11-2024 up to 9.3.4.3, 802.11be-2024, 802.11bh-2024, 802.11bk-2025, 802.11bf-2025 no new subclauses

-- Editor Note: 802.11-2024 up to Figure 9-135, 802.11be-2024, 802.11bh-2024, 802.11bk-2025, 802.11bf-2025 no new figure

-- Editor Note: 802.11-2024 up to Table 9-76, 802.11be-2024, 802.11bh-2024, 802.11bk-2025, 802.11bf-2025 no new table

9.3.4.4 Privacy Beacon frame format

The format of the Privacy Beacon frame is shown in Figure 9-135a (Privacy Beacon frame format).

	Frame Control	Duration	Address 1	Address 2	Identity Hash	Timestamp	Frame Body	FCS
Octets:	2	2	6	6	6	8	variable	4

Figure 9-135a—Privacy Beacon frame format

The Protected Frame field of the Frame Control field is set to 1 if the Privacy Beacon frame is protected, otherwise set to 0.

The Address 1 field is set to the broadcast address.

The Address 2 field is set to the anonymized MAC address of the AP sending the frame as defined in 10.71.5.4 (Addressing).

The Identity Hash field is set as described in 10.71.8.3 (BPE AP MLD beaconing).

The Timestamp field format is described in 9.4.1.10 (Timestamp field). The Timestamp field is anonymized as described in 10.71.5.5 (Timestamp anonymization).

The Privacy Beacon frame body is shown in Table 9-76a (Privacy Beacon frame body).

Table 9-76a—Privacy Beacon frame body

Order	Information	Notes
1	BSS Parameter Change Count (BPCC)	The BSS Parameter Change Count (BPCC) field, see 9.4.2.169.2 (Neighbor AP Information field), is present if the Privacy Beacon frame is protected, otherwise not present.
2	TIM	The TIM element, see 9.4.2.5 (TIM element), is present if the Privacy Beacon frame is protected, otherwise not present.

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Table 9-76a—Privacy Beacon frame body

Order	Information	Notes
3	Reduced Neighbor Report	The RNR element, see 9.4.2.169 (Reduced Neighbor Report element), is present if the Privacy Beacon frame is protected, otherwise not present.
4	Extended Channel Switch Announcement	The Extended Channel Switch Announcement element, see 9.4.2.51 (Extended Channel Switch element), is optionally present if the Privacy Beacon frame is protected and dot11SpectrumManagementRequired is true or dot11ExtendedChannelSwitchActivated is true.

9.4 Management and Extension frame body components

9.4.1 Fields that are not elements

9.4.1.1 Authentication Algorithm Number field

Change the first paragraph as follows (not all lines are shown):

-- Editor Note: ANA assignment is done

The Authentication Algorithm Number field indicates a single authentication algorithm. The Authentication Algorithm Number field is shown in Figure 9-136 (Authentication Algorithm Number field format). The following values are defined for authentication algorithm number:

Authentication algorithm number = 0: Open System

Authentication algorithm number = 2: Fast BSS Transition

Authentication algorithm number = 3: Simultaneous Authentication of Equals (SAE)

Authentication algorithm number = 4: FILS Shared Key authentication without PFS

Authentication algorithm number = 5: FILS Shared Key authentication with PFS

Authentication algorithm number = 6: FILS Public Key authentication

Authentication algorithm number = 7: PASN authentication

Authentication algorithm number = 8: IEEE 802.1X authentication

Authentication algorithm number = 9: EPPKE authentication

Authentication algorithm number = 65 535: vendor specific use

9.4.1.9 Status Code field

Insert the following new rows to Table 9-80 (Status codes) while maintaining the numerical order and updating the reserved range (not all lines shown):

-- Editor Note: ANA assignment and update are done

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Table 9-80—Status codes

Status code	Name	Meaning
…		
145	SUCCESS_SIMILAR_EPOCH	The request to join or create a group EPP epoch is successful but the epoch settings are not exactly those requested.
146	SUCCESS_ALREADY_EXISTING_EPOCH	The STA has successfully joined the requested group EPP epoch. The EPP group already exists, no new EPP group is created.
147	FAILURE_MAX_NUM_EPOCH_REACHED	Failure to create a group EPP epoch because the maximum number of group EPP epochs at the AP has been reached.
148	SUCCESS_AID_LIST_PARTIALLY_STORED	The AID assignment operation is successful but the AID list is too large and the CPE non-AP MLD has stored it partially.
149	FAILURE_AID_LIST_NOT_STORED	The AID assignment operation has failed. No AID value has been stored.
150	FAILURE_AID_STORAGE_TOO_SMALL	The request to join or create a group EPP epoch has failed, because the AID storage of the non-AP MLD is too small.
151	NO_ASSIGNED_AID	The non-AP MLD has no AID value for the current group EPP epoch.
152	802_1_X_AUTH_FAILED	IEEE 802.1X authentication failed.
153	802_1_X_AUTH_SUCCESS	IEEE 802.1X authentication is successful.
156	GROUP_INFORMATION_NOT_PROVIDED	The AP MLD does not provide information about this EPP group.

9.4.1.11 Action field

Insert the following new rows to Table 9-81 (Category values) while maintaining the numerical order and updating the reserved range (not all lines shown):

-- Editor Note: ANA assignment is done

Insert the following new subclauses at the end of 9.4.1:

-- Editor Note: 802.11-2024 up to 9.4.1.71, 802.11bh-2024 no new subclause, 802.11be-2024 up to 9.4.1.80, 802.11bk-2025 no new subclause, 802.11bf-2025 up to 9.4.1.81

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Table 9-81—Category values

Code	Meaning	See subclause	Robust	Group addressed privacy
…				
40	Protected HT	9.6.40 (Protected HT Action frame details)	Yes	No
41	Protected VHT	9.6.41 (Protected VHT Action frame details)	Yes	No
42	EPP	9.6.42 (EPP Action frame details)	Yes	No

-- Editor Note: 802.11-2024 up to Figure 9-207, 802.11bh-2024 no new figure, 802.11be-2024 up to Figure 9-207h, 802.11bk-2025 no new figure, 802.11bf-2025 up to Figure 9-207k

-- Editor Note: 802.11-2024 up to Table 9-129, 802.11bh-2024 no new table, 802.11be-2024 up to Table 9-129f, 802.11bk-2025 no new table, 802.11bf-2025 up to Table 9-129r

9.4.1.82 Encapsulation Length field

The Encapsulation Length field indicates the number of octets of the Encapsulation field (see 9.4.1.83 (Encapsulation field)). The format of the Encapsulation Length field is shown in Figure 9-207l (Encapsulation Length field format).

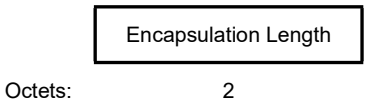


Figure 9-207l—Encapsulation Length field format

9.4.1.83 Encapsulation field

The Encapsulation field carries an EAPOL PDU. The format of the Encapsulation field is shown in Figure 9-207m (Encapsulation field format).

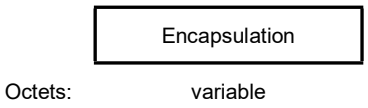


Figure 9-207m—Encapsulation field format

9.4.1.84 EPP Epoch Settings field

The EPP Epoch Settings field format is shown in Figure 9-207n (EPP Epoch Settings field format).

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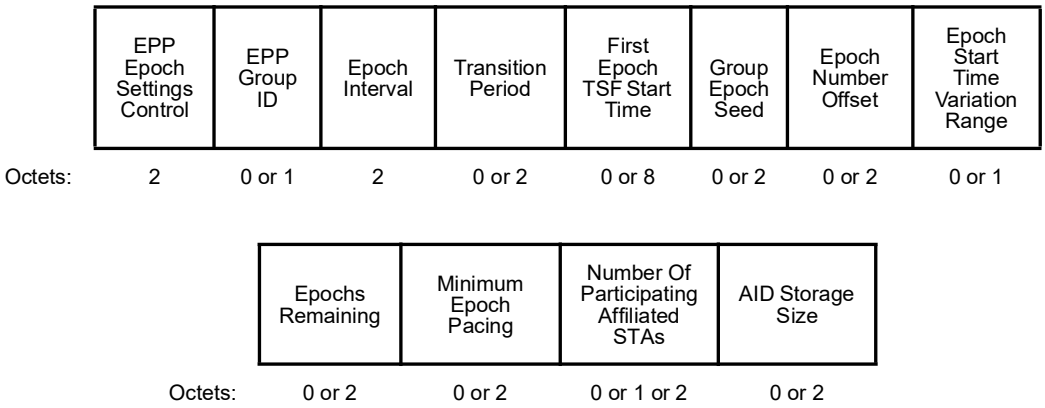


Figure 9-207n—EPP Epoch Settings field format

The EPP Epoch Settings field contains the EPP epoch settings of an EPP epoch sequence for a non-AP MLD.

The EPP Epoch Settings Control field format is shown in Figure 9-207o (EPP Epoch Settings Control field format).

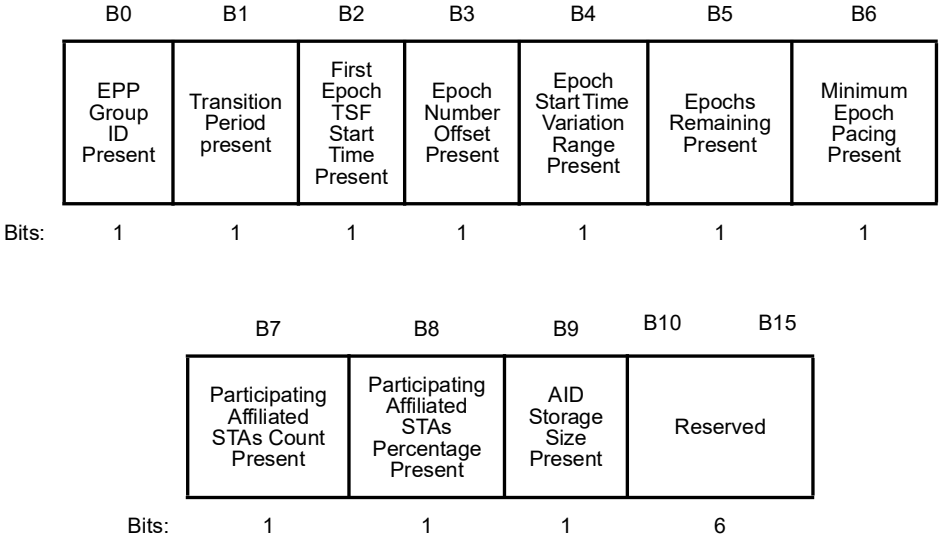


Figure 9-207o—EPP Epoch Settings Control field format

Each field in the EPP Epoch Settings Control field indicates the presence of the corresponding field in the EPP Epoch Settings field when equal to 1 and its absence when equal to 0.

The EPP Group ID field contains an identifier for the EPP group. The value 0 identifies the default EPP group. The value 255 is reserved.

The EPP Epoch Interval field format is shown in Figure 9-207p (Epoch Interval field format).

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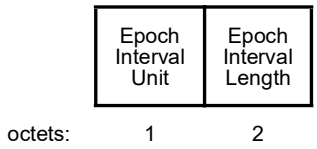


Figure 9-207p—Epoch Interval field format

The Epoch Interval Unit field indicates the units for the Epoch Interval Length field, and the Epoch Interval Length field contains the length of the EPP epoch as shown in Table 9-129s (Epoch Interval Units, Epoch Transition Period Units, and epoch durations).

Epoch Interval Length field value 0 is reserved.

Table 9-129s—Epoch Interval Units, Epoch Transition Period Units, and epoch durations

Epoch Interval Unit field value	Epoch Interval Unit	Epoch Transition Period Unit	Max Epoch Duration (approx.)
0	Reserved	Reserved	N/A
1	1000 TU (approx. 1 second)	10 TU (approx. 10ms)	Approx. 18 hours
2–255	Reserved	Reserved	N/A

The Transition Period field format is shown in Figure 9-207q (Transition Period field format).

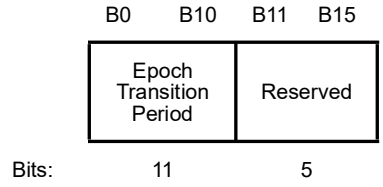


Figure 9-207q—Transition Period field format

The Epoch Transition Period field contains the duration of the transition period expressed in Epoch Transition Period Unit as defined in Table 9-129s (Epoch Interval Units, Epoch Transition Period Units, and epoch durations).

The First Epoch TSF Start Time field contains the first planned epoch TSF start time presented as the TSF timer value of the link that this field was transmitted on (see 10.71.2.4 (EPP Epoch Start Time Computation)).

The Group Epoch Seed field contains a seed selected by the AP for pseudo random computation (see 10.71.2.4 (EPP Epoch Start Time Computation)).

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The Group Epoch Seed field is present only if the First Epoch TSF Start Time field is present.

The Epoch Number Offset field contains the epoch number n of the epoch with a planned epoch start time set to the First Epoch TSF Start Time field value on the receiving link (see 10.71.2.4 (EPP Epoch Start Time Computation)).

The Epoch Number Offset field is present only if the First Epoch TSF Start Time field is present.

The Epoch Start Time Variation Range field contains the range of values, expressed in epoch interval units as defined in Table 9-129s (Epoch Interval Units, Epoch Transition Period Units, and epoch durations), used by the AP MLD and each non-AP MLDs that is a member of the EPP group to determine a random delay added to the EPP epoch planned start time as defined in 10.71.2.4 (EPP Epoch Start Time Computation). A value of 0 indicates that no random delay is used.

The Epochs Remaining field value indicates the number of EPP epochs left in the sequence after the current epoch finishes, except the value of 65535 indicates that the epoch sequence duration is unlimited.

The Minimum Epoch Pacing field indicates the minimum epoch duration the non-AP MLD can support. The format of the Minimum Epoch Pacing field is the same as the Epoch Interval field.

The Number of Participating Affiliated STAs field format is shown in Figure 9-207r (Number of Participating Affiliated STAs field format).

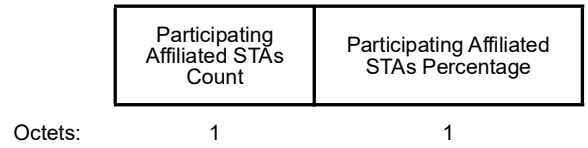


Figure 9-207r—Number of Participating Affiliated STAs field format

The Participating Affiliated STAs Count field represents an indication of the number of affiliated non-AP MLDs participating in the signaled EPP group on the MLD. This field provides the number of STAs in the EPP group rounded up to the nearest multiple of 10.

The Participating Affiliated STAs Percentage field value is the percentage of the associated affiliated non-AP MLDs participating to the signaled EPP group on the MLD. The Participating Affiliated STAs Percentage field can have a value from 0 to 100, and values 101–255 are reserved.

When transmitted by an AP MLD, the AID Storage Size field indicates the minimum number of AID values a non-AP MLD is capable of storing to be allowed to join in the EPP group.

When transmitted by a non-AP MLD, the AID Storage Size field indicates the number of AID values that the non-AP MLD can store.

The AID Storage Size field can have a value from 16 to 1024 and values 0–15 and 1025–65535 are reserved.

9.4.1.85 EPP Group Request field

The EPP Group field carries information on the action to be performed by the AP upon receiving an EPP Group Request frame.

The format of the EPP Group Request field is shown in Figure 9-207s (EPP Group Request field format).

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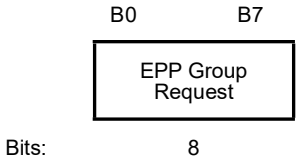


Figure 9-207s—EPP Group Request field format

The EPP Group Request field values are shown in Table 9-129t (Values for the EPP Group Request field).

Table 9-129t—Values for the EPP Group Request field

Value	Meaning
0	Reserved
1	Create/Join
2	All Groups Settings
3	Specific Group Settings
4–255	Reserved

9.4.1.86 Number Of Stored AIDs field

The Number Of Stored AIDs field indicates the number of AIDs that the non-AP MLD has stored.

The format of the Number Of Stored AIDs field is shown in Figure 9-207t (Number Of Stored AIDs field format).

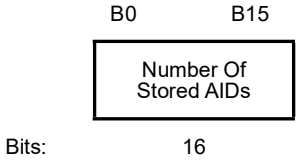


Figure 9-207t—Number Of Stored AIDs field format

9.4.2 Elements

9.4.2.1 General

Modify Table 9-130 (Element IDs) as follows:

-- Editor Note: ANA assignment and update are done

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Table 9-130—Element IDs

Element	Element ID	Element ID Extension	Extensible	Fragmentable
.....				
~~FILS~~ Nonce (see 9.4.2.188 (FILS Nonce element))	255	13	No	No
.....				
DS MAC Address (see 9.4.2.348 (DS MAC Address element))	255	146	No	No
EPP (see 9.4.2.349 (EPP element))	255	147	Yes	No
OTA MAC Collision Warning (see 9.4.2.350 (OTA MAC Collision Warning element))	255	148	Yes	No
AID List (see 9.4.2.351 (AID List element))	255	149	Yes	Yes
...				

NOTE 1—See 10.28.6 (Element parsing) on the parsing of elements.

NOTE 2—Yes for a Reassociation Response frame transmitted during ML resetup by an AP affiliated with an AP MLD, in response to a Reassociation Request frame received from a non-AP STA affiliated with a non-AP MLD. No otherwise.

9.4.2.5 TIM element

9.4.2.5.1 General

Insert the following note at the end:

NOTE 1—A CPE AP can set the bit number *N* corresponding to an unassigned AID to 1.

9.4.2.19 Measurement Request element

9.4.2.19.1 General

Change the fifth paragraph as follows (not all lines shown):

The Measurement Request Mode field (shown in Figure 9-242 (Measurement Request Mode field format)) is a bitmap with the following bits defined:

— The Parallel subfield is used to request that more than one measurement is to be started in parallel. The Parallel subfield is set to 1 to request that the measurement is to start at the same time as the measurement described by the next Measurement Request element in the same Radio Measurement Request frame. The Parallel subfield is set to 0 if the measurements are to be performed in sequence. The Parallel subfield is reserved when the Enable subfield is 1, in the last or only Measurement Request element in the frame, or when the Measurement Type field is 0, 1, ~~or~~ 2, 18 , or 19 (spectrum management measurements). See 11.10.16 (Requesting and reporting of measurements).

Change Table 9-136 (Measurement type definitions for measurement requests) as follows (not all lines shown):

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Table 9-136—Measurement type definitions for measurement requests

Name	Measurement type
...	...
Neighboring DMG APs	17
BPE AP MLD Discovery	18
CPE AP MLD Discovery	19
Reserved	2018–254
Measurement Pause	255

9.4.2.23 RSNE

9.4.2.23.3 AKM suites

Modify Table 9-190 (AKM suite selectors) as follows and update reserved suite type:

-- Editor Note: ANA assignment is done

Table 9-190—AKM suite selectors

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	0	Reserved	Reserved	Reserved	Reserved	Reserved
00-0F-AC	1	Authentication negotiated over IEEE Std 802.1X	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.2 (PRF)	0 (open) or 8 (IEEE 802.1X)	None
00-0F-AC	2	PSK	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.2 (PRF)	0 (open)	None

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	3	FT authentication negotiated over IEEE Std 802.1X	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) or 8 (IEEE 802.1X) for FT Initial Mobility Domain Association over IEEE Std 802.1X or PMKSA caching	None
00-0F-AC	4	FT authentication using PSK	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) for FT Initial Mobility Domain Association using PSK	None
00-0F-AC	5	Authentication negotiated over IEEE Std 802.1X	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	0 (open) or 8 (IEEE 802.1X)	None
00-0F-AC	6	PSK	RSNA Key Management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	0 (open)	None
00-0F-AC	7	TDLS	TPK handshake	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	N/A	None

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	8	SAE authentication	RSNA key management as defined in 12.7 (Keys and key distribution), or authenticated mesh peering exchange as defined in 14.6 (Authenticated mesh peering exchange (AMPE))	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm specified in 12.4.2 (Assumptions on SAE)	3 (SAE) for SAE Authentication 0 (open) for PMKSA caching	None
00-0F-AC	9	FT authentication over SAE	FT key management defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm specified in 12.4.2 (Assumptions on SAE)	3 (SAE) for FT Initial Mobility Domain Association 2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) for FT Initial Mobility Domain Association over PMKSA caching	None
00-0F-AC	10	APPeerKey Authentication with SHA-256	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	N/A	None
00-0F-AC	11	Authentication negotiated over IEEE Std 802.1X using a Suite B compliant EAP method supporting SHA-256	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	0 (open) or 8 (IEEE 802.1X)	Used only with cipher suite selector values 00-0F-AC:8 (GCMP-128) and 00-0F-AC:11 (BIP-GMAC-128)

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	12	Authentication negotiated over IEEE Std 802.1X using a CNSA Suite compliant EAP method	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	0 (open) or 8 (IEEE 802.1X)	Used only with cipher suite selector values 00-0F-AC:9 (GCMP-256), 00-0F-AC:10 (CCMP-256), 00-0F-AC:13 (BIP-CMAC-256), and 00-0F-AC:12 (BIP-GMAC-256)
00-0F-AC	13	FT authentication negotiated over IEEE Std 802.1X	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) or 8 (IEEE 802.1X) for FT Initial Mobility Domain Association over IEEE Std 802.1X or PMKSA caching	Used only with cipher suite selector values 00-0F-AC:9 (GCMP-256), 00-0F-AC:10 (CCMP-256), 00-0F-AC:13 (BIP-CMAC-256), and 00-0F-AC:12 (BIP-GMAC-256)
00-0F-AC	14	Key management over FILS using SHA-256 and AES-SIV-256, or authentication negotiated over IEEE Std 802.1X	FILS key management defined in 12.11.2.5 (Key establishment with FILS authentication) using SHA-256	Defined in 12.11.2.5 (Key establishment with FILS authentication) using SHA-256	4, 5 or 6 (FILS) for FILS Authentication 0 (open) or 8 (IEEE 802.1X) for IEEE Std 802.1X	None
00-0F-AC	15	Key management over FILS using SHA-384 and AES-SIV-512, or authentication negotiated over IEEE Std 802.1X	FILS key management defined in 12.11.2.5 (Key establishment with FILS authentication) using SHA-384	Defined in 12.11.2.5 (Key establishment with FILS authentication) using SHA-384	4, 5 or 6 (FILS) for FILS Authentication 0 (open) or 8 (IEEE 802.1X) for IEEE Std 802.1X	None

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	16	FT authentication over FILS with SHA-256 and AES-SIV-256 or authentication negotiated over IEEE Std 802.1X	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-256	4, 5 or 6 (FILS) for FT Initial Mobility Domain Association over FILS 2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) or 8 (IEEE 802.1X) for FT Initial Mobility Domain Association over IEEE Std 802.1X or PMKSA caching	None
00-0F-AC	17	FT authentication over FILS with SHA-384 and AES-SIV-512, or authentication negotiated over IEEE Std 802.1X	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	4, 5 or 6 (FILS) for FT Initial Mobility Domain Association over FILS 2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) or 8 (IEEE 802.1X) for FT Initial Mobility Domain Association over IEEE Std 802.1X or PMKSA caching	None
00-0F-AC	18	None	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm specified in Table 12-1	0 (open)	None

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	19	FT authentication using PSK	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) for FT Initial Mobility Domain Association using PSK	None
00-0F-AC	20	PSK	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	0 (open)	None
00-0F-AC	21	PASN	PASN	PASN key management defined in 12.13 (Preassociation security negotiation(11az))	Defined in 12.13.3 (Key establishment with PASN authentication)	None
00-0F-AC	22	FT authentication negotiated over IEEE Std 802.1X	FT key management as defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) or 8 (IEEE 802.1X) for FT Initial Mobility Domain Association over IEEE Std 802.1X or PMKSA caching	None
00-0F-AC	23	Authentication negotiated over IEEE Std 802.1X	RSNA key management as defined in 12.7 (Keys and key distribution)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using SHA-384	0 (open) or 8 (IEEE 802.1X)	None

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Table 9-190—AKM suite selectors *(continued)*

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1 (Authentication Algorithm Number field))	Cipher suite selector restriction
		Authentication type	Key management type	Key derivation type		
00-0F-AC	24	SAE authentication	RSNA key management as defined in 12.7 (Keys and key distribution), or authenticated mesh peering exchange as defined in 14.6 (Authenticated mesh peering exchange (AMPE))	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm specified in 12.4.2 (Assumptions on SAE)	3 (SAE) for SAE Authentication 0 (open) for PMKSA caching	None
00-0F-AC	25	FT authentication over SAE	FT key management defined in 12.7.1.6 (FT key hierarchy)	Defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm specified in 12.4.2 (Assumptions on SAE)	3 (SAE) for FT Initial Mobility Domain Association 2 (FT) for FT protocol reassociation as defined in 13.5 (FT protocol) 0 (open) for FT Initial Mobility Domain Association over PMKSA caching	None
.....						
00-0F-AC	29	EPPKE	EPPKE	EPPKE key management defined in 12.16.9 (Enhanced privacy protection key exchange)	Defined in 12.16.9.3 (Key establishment with EPPKE authentication)	

9.4.2.23.5 PMKID

Change third and fourth paragraph (not all shown) as follows:

A PMKID in the PMKID List field can refer to

a) The PMKID of a cached PMKSA that has been obtained through preauthentication with the target AP

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b) The PMKID of a cached PMKSA from an EAP, FILS, or SAE authentication

c) The PMKID of a PMKSA derived from a PSK for the target AP

ca) The changed PMKID if PMKSA caching privacy is used

d) The PMKR0Name of a PMK-R0 security association derived as part of an FT initial mobility domain association

da) The recomputed PMKR0Name as part of a fast BSS transition if PMKSA caching privacy is used

e) The PMKR1Name of a PMK-R1 security association derived as part of an FT initial mobility domain association or as part of a fast BSS transition.

See 12.7.1.3 (Pairwise key hierarchy), ~~and~~ 12.7.2.6.3 (PMK-R0), and 12.16.7.2 (PMKID privacy) for the construction of the PMKID, 13.8 (FT authentication sequence) for the population of PMKID List for fast BSS transitions, 12.6.8.3 (Cached PMKSAs and RSNA key management) for the population of PMKID List when using PMKSA caching, 13.4 (FT initial mobility domain association) for the population of PMKID List for FT initial mobility domain association, 12.11.2 (FILS authentication protocol) for the population of PMKID List with FILS authentication, ~~and~~ 12.7.1.6 (FT key hierarchy), and 12.16.7.3 (PMKR0Name privacy) for the construction of PMKR0Name and PMKR1Name.

9.4.2.25 Extended Capabilities element

Insert a new row to Table 9-192 (Extended Capabilities field) as follows (not all lines shown):

Table 9-192—Extended Capabilities field

Bit	Information	Notes
111	BPE Available	An AP STA sets the BPE Available to 1 when it belongs to an ESS that has one or more BPE APs available. A non-AP STA sets the BPE Available to 1 when it is BPE capable.

9.4.2.35 Neighbor Report element

Change Figure 9-423 (BSSID Information field format) as follows:

	B0	B1	B2	B3	B4	B9	B10	B11	B12	B13
	AP Reachability		Security	Key Scope	Capabilities		Mobility Domain	High Throughput	Very High Throughput	FTM
Bits:	2		1	1	6		1	1	1	1

	B14	B15	B16	B17	B18	B19	B20	B21
	High Efficiency	ER BSS	Colocated AP	Unsolicited Probe Responses Active	Member Of ESS With 2.4/5 GHz Colocated AP	OCT Supported With Reporting AP	Colocated With 6 GHz AP	Reserved
Bits:	1	1	1	1	1	1	1	1

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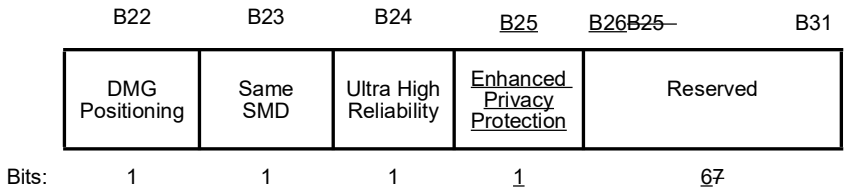


Figure 9-423—BSSID Information field format

Insert the following paragraphs after "The DMG Positioning field indicates that the reported AP is a DMG AP ..."

The Enhanced Privacy Protection field is set to 1 to indicate that the AP represented by this BSSID is an EPP AP that has set the Group EPP Epoch Supported field of the Extended RSN Capabilities field to 1. Otherwise, the Enhanced Privacy Protection field is set to 0.

Insert new rows to Table 9-212 (Optional subelement IDs for Neighbor Report) as follows (not all lines shown):

-- Editor Note: ANA assignment for subelement IDs is done

-- Editor Note: 802.11-2024 up to Figure 9-423

Table 9-212—Optional subelement IDs for Neighbor Report

Subelement ID	Name	Extensible
7	RSNE	No
8	RSNXE	Yes
9	BSSID Of The Next Epoch	Yes
10	Supported Rates and BSS Membership Selectors	No
11	Extended Supported Rates and BSS Membership Selectors	No
12	EPP	Yes

Insert the following paragraphs at the end

The RSNE subelement has the same format as the RSNE as defined in 9.4.2.23 (RSNE).

The RSNXE subelement has the same format as the RSNXE as defined in 9.4.2.240 (RSNXE).

The format of the BSSID Of The_Next Epoch subelement is shown in Figure 9-423a (BSSID Of The Next Epoch).

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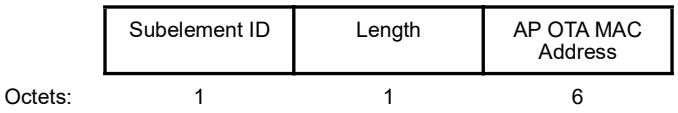


Figure 9-423a—BSSID Of The Next Epoch

The Subelement ID and Length fields are defined in 9.4.3 (Subelements).

The AP OTA MAC Address field specifies the OTA address that the reported AP affiliated with a BPE AP MLD uses in the next epoch.

The Supported Rates and BSS Membership Selectors subelement has the same format as the Supported Rates and BSS Membership Selectors element as defined in 9.4.2.3 (Supported Rates and BSS Membership Selectors).

The Extended Supported Rates and BSS Membership Selectors subelement has the same format as the Extended Supported Rates and BSS Membership Selectors element as defined in 9.4.2.11 (Extended Supported Rates and BSS Membership Selectors).

The Data field of the EPP subelement has the same format as the Information field of the EPP element defined in 9.4.2.349 (EPP element). The EPP subelement includes the EPP epoch settings of the default EPP epoch supported by the reported AP. The EPP Epoch Settings field included in the EPP subelement sent in a Neighbor Report element should only contain the EPP Epoch Settings Control, the EPP Group ID and the Epoch Interval fields.

9.4.2.46 FTE

Change the fifth paragraph as follows:

When using AKM 00-0F-AC:25 or if the FTE is included in a frame between the FTO with the (Re)Association Frame Encryption Support field in the RSNXE equal to 1 and the FTR with the (Re)Association Frame Encryption Support field in the RSNXE equal to 1, the MIC Length subfield defines the length of the MIC field, as defined in Table 9-220 (MIC Length subfield values). This subfield is reserved ~~for other AKMs~~otherwise.

Modify Table 9-220 (MIC Length subfield value) as follows:

Table 9-220—MIC Length subfield value

Value	MIC field length in octets
0	16
1	24
2	32
3	0
~~4~~3–7	Reserved

Modify Table 9-221 (Subelement IDs) (not all lines shown) as follows:

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-- Editor Note: ANA assignment is done

Table 9-221—Subelement IDs

Value	Subelement Name
…	
11	PGTK
13	Identity Key

-- Editor Note: 802.11-2024 up to Figure 9-444, 802.11be-2024 up to Figure 9-444c

Modify the paragraph "The GTK subelement contains the GTK, ..." as follows:

The GTK subelement contains the GTK, which is encrypted (see procedures in 13.8.5 (FT authentication sequence: contents of fourth message)) or is not encrypted (see procedures in 12.16.6 ((Re)Association Request/Response frame encryption)) and is defined in Figure 9-439 (GTK subelement format).

Modify Figure 9-439 (GTK subelement format) as follows:

	Subelement ID	Length	Key Info	Key Length	RSC	Wrapped Key/Key
Octets:	1	1	2	1	8	~~24~~ ~~40~~ variable

Figure 9-439—GTK subelement format

Modify the paragraph "The Wrapped Key field contains the wrapped GTK being distributed." as follows:

The Wrapped Key/Key field contains the wrapped GTK being distributed if the frame containing the FTE is not encrypted or the GTK being distributed if the frame containing the FTE is encrypted.

Modify Figure 9-441 (IGTK subelement format) as follows:

	Subelement ID	Length	Key ID	IPN	Key Length	Wrapped Key/Key
Octets:	1	1	2	6	1	~~24-40~~variable

Figure 9-441—IGTK subelement format

Modify the paragraph "The Wrapped Key field contains the wrapped IGTK being distributed." as follows:

The Wrapped Key/Key field contains the wrapped IGTK being distributed if the frame containing the FTE is not encrypted or the IGTK being distributed if the frame containing the FTE is encrypted.

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Modify Figure 9-443 (BIGTK subelement format) as follows:

	Subelement ID	Length	Key ID	BIPN	Key Length	Wrapped Key/Key
Octets:	1	1	2	6	1	~~24~~40variable

Figure 9-443—BIGTK subelement format

Modify the paragraph "The Wrapped Key field contains the wrapped BIGTK being distributed." as follows:

The Wrapped Key/Key field contains the wrapped BIGTK being distributed if the frame containing the FTE is not encrypted or the BIGTK being distributed if the frame containing the FTE is encrypted.

Modify Figure 9-444 (WIGTK subelement format) as follows:

	Subelement ID	Length	Key ID	WIPN	Key Length	Wrapped Key/Key
Octets:	1	1	2	6	1	~~24~~variable

Figure 9-444—WIGTK subelement format

Modify the paragraph "The Wrapped Key field contains the wrapped WIGTK being distributed." as follows:

The Wrapped Key/Key field contains the wrapped WIGTK being distributed if the frame containing the FTE is not encrypted or the WIGTK being distributed if the frame containing the FTE is encrypted.

Modify the paragraph "The MLO GTK subelement contains the GTK for a link, ..." as follows:

The MLO GTK subelement contains the GTK for a link, which is encrypted (see procedures in 13.8.5 (FT authentication sequence: contents of fourth message)) or is not encrypted (see procedures in 12.16.6 ((Re)Association Request/Response frame encryption))and is defined in Figure 9-444a (MLO GTK subelement format).

Modify Figure 9-444a (MLO GTK subelement format) as follows:

	Subelement ID	Length	Key Info	Link ID Info	Key Length	RSC	Wrapped Key/Key
Octets:	1	1	2	1	1	8	~~24–40~~variable

Figure 9-444a—MLO GTK subelement format

Modify the paragraph "The definitions of the Key Info, Key Length, RSC, and Wrapped Key fields are the same as in the GTK subelement." as follows:

The definitions of the Key Info, Key Length, RSC, and Wrapped Key/Key fields are the same as in the GTK subelement.

Modify Figure 9-444b (MLO IGTK subelement format) as follows:

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	Subelement ID	Length	Key ID	IPN	Link ID Info	Key Length	Wrapped Key/Key
Octets:	1	1	2	6	1	1	~~24–40~~variable

Figure 9-444b—MLO IGTK subelement format

Modify the paragraph "The definitions of the Key ID, IPN, Key Length, and Wrapped Key fields are the same as in the IGTK subelement." as follows:

The definitions of the Key ID, IPN, Key Length, and Wrapped Key/Key fields are the same as in the IGTK subelement.

Modify Figure 9-444c (MLO BIGTK subelement format) as follows:

	Subelement ID	Length	Key ID	BIPN	Link ID Info	Key Length	Wrapped Key/Key
Octets:	1	1	2	6	1	1	~~24–40~~variable

Figure 9-444c—MLO BIGTK subelement format

Modify the paragraph "The definitions of the Key ID, BIPN, Key Length, and Wrapped Key fields are the same as in the BIGTK subelement." as follows:

The definitions of the Key ID, BIPN, Key Length, and Wrapped Key/Key fields are the same as in the BIGTK subelement.

Insert the following at the end of 9.4.2.46 (FTE):

The PGTK subelement contains the PGTK, used to anonymize fields that are common for all STAs assigned to an EPP group. The PGTK subelement format is shown in Figure 9-444d (PGTK subelement format).

	Subelement ID	Length	PGTK Switch Time	Key Length	Wrapped Key/ Key
Octets:	1	1	8	1	variable

Figure 9-444d—PGTK subelement format

The Subelement ID and Length fields are defined in 9.4.3 (Subelements).

The PGTK Switch Time field is as defined in Figure 12-50i (PGTK KDE format).

The Key Length field is the length of the PGTK in octets, not including any padding (see 13.8.5 (FT authentication sequence: contents of fourth message)).

The Wrapped Key/Key field contains the wrapped PGTK being distributed if the frame containing the FTE is not encrypted or the PGTK being distributed if the frame containing the FTE is encrypted.

The Identity Key subelement contains the Identity Key, used to encrypt the identity of the AP MLD in the Privacy Beacons. The Identity Key subelement format is shown in Figure 9-444e (Identity Key subelement format).

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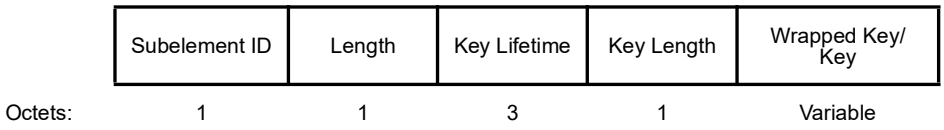


Figure 9-444e—Identity Key subelement format

The Subelement ID and Length fields are defined in 9.4.3 (Subelements).

The Wrapped Key/Key field contains the wrapped identity key being distributed if the frame containing the FTE is not encrypted or the identity key being distributed if the frame containing the FTE is encrypted.

The Key Length field is the length of the identity key in octets, not including any padding (see 13.8.5 (FT authentication sequence: contents of fourth message)).

The Key Lifetime contains the Lifetime of the Key in units of the seconds. Value 0 indicates that the key has no lifetime.

9.4.2.66 Event Report element

9.4.2.66.2 Transition event report

Insert new rows to Table 9-238 (Transition and Transition Query reasons) as follows (not all lines shown):

-- Editor Note: 802.11-2024 used up to 20

Table 9-238—Transition and Transition Query reasons

Transition Reason value	Description
21	BPE AP MLD Discovery

9.4.2.80 WNM Sleep Mode element

Modify Table 9-266 (WNM Sleep Mode Response Status definition) (not all lines shown) as follows:

Table 9-266—WNM Sleep Mode Response Status definition

Value	Description
0	Enter/Exit WNM sleep mode Accept.
1	Exit WNM sleep mode Accept, GTK/ IGTK/BIGTK/PGTK update required.

9.4.2.186 Wrapped Data element

Change the third paragraph as follows:

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The Wrapped Data field is the data used by the FILS authentication algorithm (see 12.11 (Authentication for FILS)), ~~and~~ PASN authentication algorithm (see 12.13 (Preassociation security negotiation)), and EPPKE algorithm (see 12.16.9 (Enhanced privacy protection key exchange)).

Modify 9.4.2.188 as follows:

9.4.2.188 ~~FILS~~ Nonce element

The ~~FILS~~ Nonce element is used for exchanging an additional source of randomness in an~~the FILS~~ authentication exchange or an encrypted (Re)Association Request/Response frame exchange. The format of the ~~FILS~~ Nonce element is shown in Figure 9-767 (FILS Nonce element format).

	Element ID	Length	Element ID Extension	~~FILS~~ Nonce
Octets:	1	1	1	16

Figure 9-767—~~FILS~~ Nonce element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1 (General).

The ~~FILS~~ Nonce field contains randomly generated data.

9.4.2.240 RSNXE

Insert the following new rows to Table 9-373 (Extended RSN Capabilities field) while maintaining the numerical order and updating the reserved range (not all lines shown):

-- Editor Note: ANA assignment and update are done

Table 9-373—Extended RSN Capabilities field

Bit	Information	Notes
…		
23	EPP Robust Individually Addressed Management Frame Support	An EPP STA sets the EPP Robust Individually Addressed Management Frame Support field to 1 if dot11EPPRobustIndividuallyAddressedManagementFrameActivated is true. Otherwise, this field is set to 0. See 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames).
24	EPP Robust Individually Addressed Beamforming/CSI/CQI Frame TB Tx Support	An EPP STA sets the EPP Robust Individually Addressed Beamforming/CSI/CQI Frame TB Tx Support field to 1 if dot11EPPRobustIndividuallyAddressedBFCSICQITBTxActivated is true. Otherwise, this field is set to 0. See 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames).

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Table 9-373—Extended RSN Capabilities field *(continued)*

Bit	Information	Notes
25	EPP Robust Individually Addressed Beamforming/CSI/CQI Frame Rx Support	An EPP STA sets the EPP Robust Individually Addressed Beamforming/CSI/CQI Frame Rx Support field to 1 if dot11EPPRobustIndividuallyAddressedBeamformingCSICQIFrameRxActivated is true. Otherwise, this field is set to 0. See 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames).
26	EPP Capabilities And Operation Parameters Request/Response Support	An EPP STA sets the EPP Capabilities And Operation Parameters Request/Response field to 1 if dot11EPPCapabilitiesAndOperationParametersExchangeActivated is true. Otherwise, this field is set to 0. See 12.16.4 (Capabilities and operation parameters request and response procedure).
27	(Re)Association Frame Encryption Support	An EPP STA sets the (Re)Association Frame Encryption Support field to 1 if dot11EPPReAssociation FrameEncryptionSupportActivated is true. Otherwise, this field is set to 0. See 12.16.6 ((Re)Association Request/Response frame encryption).
28	IEEE 802.1X Authentication Utilizing Authentication Frame Support	An EPP STA sets the IEEE 802.1X Authentication Utilizing Authentication Frame Support field to 1 if dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated is true. Otherwise, this field is set to 0.
29	PMKSA Caching Privacy Support	An EPP STA sets the PMKSA Caching Privacy Support field to 1 if dot11EPPPMKSACachingPrivacySupport Activated is true. Otherwise, this field is set to 0. See 12.16.7 (PMKSA caching privacy).
30	EPP Epoch Supported	An EPP STA sets the EPP Epoch Supported field to 1 when dot11EPPEpochActivated is true and sets it to 0 otherwise.
31	DS MAC Address Support	The DS MAC Address Support field is set to 1 when dot11DSMACAddressActivated is true and is set to 0 otherwise.
32	EPP Robust Individually Addressed Beamforming/CSI/CQI Frame Non-TB Tx Support	An EPP STA sets the EPP Robust Individually Addressed Beamforming/CSI/CQI Frame Non-TB Tx Support field to 1 if dot11EPPRobustIndividuallyAddressedBFCSICQINonTBTxActivated is true. Otherwise, this field is set to 0. See 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames).

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Table 9-373—Extended RSN Capabilities field *(continued)*

Bit	Information	Notes
34	SAE Password Identifier Change Support	A non-AP STA that supports changing the SAE Password Identifier sets this to 1. Otherwise, this is set to 0.
36	CPE Frame Anonymization Activated	A CPE AP MLD sets the CPE Frame Anonymization Activated field to 1. An EPP non-AP MLD sets the CPE Frame Anonymization Activated field to 1 to initiate activation of CPE FA as described in 10.71.1.3 (General CPE FA requirements). Otherwise, this is set to 0.

Insert the following new subclauses at the end of 9.4.2:

-- Editor Note: 802.11-2024 up to 9.4.2.315, 802.11bh-2024 up to 9.4.2.320, 802.11be-2024 up to 9.4.2.330, 802.11bk-2025 up to 9.4.2.331, 802.11bf-2025 up to 9.4.2.346 (should be 9.4.2.347)

-- Editor Note: 802.11-2024 up to Figure 9-1077###, 802.11bh-2024 up to Figure 9-1074h, 802.11be-2024 up to Figure 9-1074bm, 802.11bk-2025 up to Figure 9-1074bma (should be 9-1074bn), 802.11bf-2025 up to Figure 9-1074do (should be 9-1074dp)

-- Editor Note: 802.11-2024 up to Table 9-417, 802.11bh-2024 up to Table 9-417d, 802.11be-2024 up to Table 9-417x, 802.11bk-2025 no new table, 802.11bf-2025 up to Table 9-417ai

9.4.2.348 DS MAC Address element

The DS MAC Address element is used by either an EPP non-AP MLD or an EPP non-AP STA that is not affiliated with a non-AP MLD to provide the DS MAC address to an EPP AP MLD or an EPP AP, respectively, for the DS mapping.

The format of the DS MAC Address element is shown in Figure 9-1074dq (DS MAC Address element format).

	Element ID	Length	Element ID Extension	DS MAC Address
Octets:	1	1	1	6

Figure 9-1074dq—DS MAC Address element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1 (General).

The DS MAC Address field contains a MAC address that is used for the DS mapping.

9.4.2.349 EPP element

The EPP element provides the EPP epoch settings. The EPP element format is shown in Figure 9-1074dr (EPP element).

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1 (General).

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	Element ID	Length	Element ID Extension	EPP Epoch Settings
Octets:	1	1	1	variable

Figure 9-1074dr—EPP element

The EPP Epoch Settings field is defined in 9.4.1.84 (EPP Epoch Settings field).

9.4.2.350 OTA MAC Collision Warning element

The OTA MAC Collision Warning element is used when a frame anonymization OTA MAC address expected to be used by an EPP non-AP MLD in an upcoming epoch is calculated to collide with the OTA MAC address of another STA on a link in use by the non-AP MLD. The OTA MAC Collision Warning element format is shown in Figure 9-1074ds (OTA MAC Collision Warning element format).

	Element ID	Length	Element ID Extension	Collision Action	Colliding Epoch	Link ID Info	Non-AP MLD Specific Collision Epoch Offset
Octets:	1	1	1	1	1	1	1

Figure 9-1074ds—OTA MAC Collision Warning element format

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1 (General).

The possible values for the Collision Action field are listed in Table 9-417aj (Collision Action values).

Table 9-417aj—Collision Action values

Collision Action field value	Meaning
0	AP MLD informs the non-AP MLD that there is a future collision risk expected
1	Non-AP MLD indicates that it will take suggested action
2	Non-AP MLD indicates that it will not take suggested action
3–255	Reserved

The Colliding Epoch field indicates the epoch that is likely to have a MAC address collision. The value is indicated in units of epochs. The epochs are sequentially numbered starting with the value 1 for the epoch following the current epoch. The value 0 is reserved.

The Link ID Info field is defined in 9.4.1.77 (Link ID Info field) and indicates the link that is likely to have the collision.

The Non-AP MLD Specific Collision Epoch Offset field indicates the number of epochs that need to be skipped to avoid the OTA MAC address collision. The value 0 is reserved. The field is reserved when the Collision Action field value is 1 or 2.

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9.4.2.351 AID List element

The AID List element contains a sequence of AID values for the receiving EPP non-AP MLD to use in a sequence of contiguous EPP epochs for frame anonymization.

The format of the AID List element is shown in Figure 9-1074dt (AID List element).

	Element ID	Length	Element ID Extension	EPP Group ID	Start Epoch	Number Of Epochs	AID List Value
Octets:	1	1	1	1	2	2	variable

Figure 9-1074dt—AID List element

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1 (General).

The EPP Group ID contains an identifier of the EPP group. The value 0 indicates the default EPP group. The value 255 is reserved.

The Start Epoch field is the 2 least significant octets of the future closest EPP epoch number where the first AID in the AID List Value field is assigned.

The Number Of Epochs field indicates the number, n, of consecutive epochs with the provided AID fields.

The AID List Value field format is shown in Figure 9-1074du (AID List Value field format).

	AID12s	Padding
Bits	$n \times 12$	0 or 4

Figure 9-1074du—AID List Value field format

The number of AID12 fields is set to the value in the Number Of Epochs field.

Each AID12 field is 12 bits and indicates the 12 LSBs of an AID assigned to an epoch. The ith AID12 field is the AID for the ith epoch of the consecutive epochs.

A Padding field is optionally present to align the field to octet limits.

9.4.2.352 EPP Group Parameter Update element

The Group Parameter Update element is used by an EPP AP-MLD to inform non-AP MLDs member of an EPP group about upcoming EPP group parameters change. The EPP Group Parameter Update element format is shown in Figure 9-1074dv (EPP Group Parameters Update element).

	Element ID	Length	Element ID Extension	EPP Group Parameters Change Count	EPP Epoch Settings
Octets:	1	1	1	1	variable

Figure 9-1074dv—EPP Group Parameters Update element

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1 (General).

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The EPP Group Parameter Change Count field is set to the number of TBTTs until the AP MLD sending the EPP Group Parameter Update element changes the parameters of the group which ID is indicated in the Group ID field of the EPP Epoch Settings field. The value 1 indicates that the change occurs at the next TBTT, and the value 0 indicates that the change occurs at any time after the frame containing the element is transmitted.

The EPP Epoch Settings field is defined in 9.4.1.84 (EPP Epoch Settings field). When sent inside an EPP Group Parameters Update element, the EPP Group Id indicates the target group for which the update is intended, and the Minimum Epoch Pacing and Number of Participating Affiliated STAs fields are not present.

9.6 Action frame format details

9.6.6 Radio Measurement Action frame details

9.6.6.6 Neighbor Report Request frame format

Change Figure 9-1186 (Neighbor Report Request frame Action field format) as follows:

	Category	Radio Measurement Action	Dialog Token	SSID (optional)	LCI Measurement Request (optional)	Location Civic Measurement Request (optional)	Neighbor DMG Request (optional)
Octets:	1	1	1	variable	variable	variable	variable

	BPE AP MLD Discovery Request (optional)	CPE AP MLD Discovery Request (optional)
Octets:	variable	variable

Figure 9-1186—Neighbor Report Request frame Action field format

Insert the following paragraph at the end:

The BPE AP MLD Discovery Request field is optionally present. If present, it contains a Measurement Request element with Measurement Type field set to BPE AP MLD Discovery (see Table 9-136 (Measurement type definitions for measurement requests)). The element indicates a request for a Measurement Report subelement with Measurement Type field set to BPE AP MLD Discovery for each Neighbor Report element (see 11.10.10.2 (Requesting a neighbor report)). The Parallel, Enable, Request, Report and Duration Mandatory subfields of the Measurement Request Mode field of the Measurement Request element are reserved (see 9.4.2.19.1 (General)). The Measurement Request field is not present in the Measurement Request element.

The CPE AP MLD Discovery Request field is optionally present. If present, it contains a Measurement Request element with Measurement Type field set to CPE AP MLD Discovery (see Table 9-136 (Measurement type definitions for measurement requests)). The element indicates a request for the EPP element for each Neighbor report element ((see 11.10.10.2 (Requesting a neighbor report)). The Parallel, Enable, Request, Report and Duration Mandatory subfield of the Measurement Request Mode field of the Measurement Request element are reserved (see 9.4.2.19.1 (General)). The Measurement Request field is not present in the Measurement Request element.

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9.6.7 Public Action frame details

9.6.7.1 Public Action field

Modify Table 9-471 (Public Action field values) (not all lines shown) as follows:

-- Editor Note: ANA assignment is done

Table 9-471—Public Action field values

Public Action field Value	Description
65	Privacy Beacon Solicit Request (see 9.6.7.64 (Privacy Beacon Solicit Request frame format))

-- Editor Note: 802.11-2024 up to 9.6.7.55, 802.11be-2024 no new section, 802.11bk-2025 no new section, 802.11bf-2025 up to 9.6.7.63

-- Editor Note: 802.11-2024 up to Table 9-509, 802.11be-2024 no new table, 802.11bk-2025 no new table, 802.11bf-2025 no new table

9.6.7.64 Privacy Beacon Solicit Request frame format

The Privacy Beacon Solicit Request frame is transmitted as a unprotected Management frame to the broadcast address.

Table 9-509a—Privacy Beacon Solicit Request frame Action field format

Order	Meaning
0	Category
1	Public Action

The Category field is defined in 9.4.1.11 (Action field).

The Public Action field is defined in 9.6.7.1 (Public Action field).

9.6.13 WNM Action frame details

9.6.13.20 WNM Sleep Mode Response frame format

Change the sixth paragraph as follows:

-- Editor Note: 802.11-2024 up to Figure 9-1293###, 802.11be-2024 up to Figure 9-1290c, 802.11bk-2025 no new figure, 802.11bf-2025 no new figure

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The Key Data field contains zero or more subelements that provide the current GTK, IGTK, BIGTK to the STA and the current PGTK to the non-AP MLD. The format of these subelements is shown in Figure 9-1288 (WNM Sleep Mode GTK subelement format), Figure 9-1289 (WNM Sleep Mode IGTK subelement format), Figure 9-1290 (WNM Sleep Mode BIGTK subelement format), Figure 9-1290a (WNM Sleep Mode MLO GTK subelement format), Figure 9-1290b (WNM Sleep Mode MLO IGTK subelement format), and Figure 9-1290c (WNM Sleep Mode MLO BIGTK subelement format), and Figure 9-1290d (WNM Sleep Mode PGTK subelement format). The subelement IDs for these subelements are defined in Table 9- 540 (Optional subelement IDs for WNM Sleep Mode parameters). When management frame protection is not used, the Key Data field is not present.

change Table 9-540 (Optional subelement IDs for WNM Sleep Mode parameters) as follows:

-- Editor Note: 802.11-2024 0-2, 802.11be-2024 3-5, 802.11bk-2025 and 802.11bf-2025 no new row

Table 9-540—Optional subelement IDs for WNM Sleep Mode parameters

Value	Meaning
...	
5	MLO BIGTK
6	PGTK
~~6~~7–255	Reserved

Insert the following at the end of this subclause:

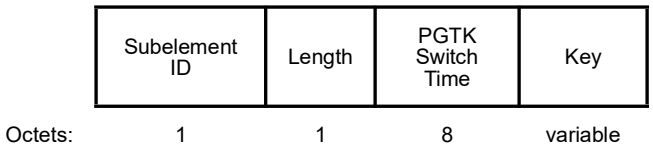


Figure 9-1290d—WNM Sleep Mode PGTK subelement format

The WNM Sleep Mode PGTK subelement contains the PGTK of the EPP AP MLD as shown in Figure 9-1290d (WNM Sleep Mode PGTK subelement format).

The Subelement ID field is defined in 9.6.13.20 (WNM Sleep Mode Response frame format).

The Length field is defined in 9.4.3 (Subelements).

The PGTK Switch Time field is defined in 12.7.2 (EAPOL-Key frames).

The Key field is the PGTK being distributed. The length of PGTK is equal to the value of PMK_bits (see 12.7.1.3 (Pairwise key hierarchy)).

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9.6.32 Protected HE Action frame details

9.6.32.1 Protected HE Action field

Modify Table 9-647 (Protected HE Action field values) while maintaining the numerical order and updating the reserved range (not all lines shown):

Table 9-647—Protected HE Action field values

Value	Meaning	Time priority
0	HE BSS Color Change Announcement	No
1	MU EDCA Reset	No
2	Protected HE Compressed Beamforming/ CQI	Yes
3	Protected Quiet Time Period	No
~~24~~–255	Reserved	

Insert the following subclause at the end of 9.6.32 as shown below:

-- Editor Note: 802.11-2024 up to 9.6.32.3, 802.11bh-2024, 802.11be-2024, 802.11bk-2025, 802.11bf-2025 no new subclause

9.6.32.4 Protected HE Compressed Beamforming/CQI frame format

The Protected HE Compressed Beamforming/CQI frame allows robust STA-STA communication of the same information that is conveyed in the HE Compressed Beamforming/CQI frame that is not robust (see 9.6.31.1 (HE Action field)).

The Action field of the Protected HE Compressed Beamforming/CQI frame has the same format as the Action field of the HE Compressed Beamforming/CQI frame (see 9.6.31.2 (HE Compressed Beamforming/ CQI frame format)), except that the Order 2 item is the Protected HE Action field, which is defined in 9.6.32.1 (Protected HE Action field), instead of the HE Action field.

9.6.32.5 Protected Quiet Time Period frame format

The Protected Quiet Time Period frame allows robust STA-STA communication of the same information that is conveyed in the Quiet Time Period frame that is not robust (see 9.6.31.1 (HE Action field)).

The Action field of the Protected Quiet Time Period frame has the same format as the Action field of the Quiet Time Period frame (see 9.6.31.3 (Quiet Time Period Action frame format)), except that the Order 2 item is the Protected HE Action field, which is defined in 9.6.32.1 (Protected HE Action field), instead of the HE Action field.

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9.6.38 Protected EHT Action frame details

9.6.38.1 Protected EHT Action field

Insert the following new rows to Table 9-658d (Protected EHT Action field values) while maintaining the numerical order and updating the reserved range (not all lines shown):

Table 9-658d—Protected EHT Action field values

Value	Meaning	Time Priority
...		
12	Link Reconfiguration Response	No
13	Protected EHT Compressed Beamforming/ CQI frame	Yes
~~13~~14–255	Reserved	

Insert the following subclauses at the end of 9.6.38 as shown below:

-- Editor Note: 802.11be-2024 up to 9.6.38.14, 802.11bk-2025, 802.11bf-2025 no new subclause

9.6.38.15 Protected EHT Compressed Beamforming/CQI frame format

The Protected EHT Compressed Beamforming/CQI frame allows robust STA-STA communication of the same information that is conveyed in the EHT Compressed Beamforming/CQI frame that is not robust (see 9.6.38.1 (EHT Action field)).

The Action field of the Protected EHT Compressed Beamforming/CQI frame has the same format as the Action field of the EHT Compressed Beamforming/CQI frame (see 9.6.37.2 (EHT Compressed Beamforming/CQI frame format)), except that the Order 2 item is the Protected EHT Action field, which is defined in 9.6.38.1 (Protected EHT Action field), instead of the EHT Action field.

Insert the following new subclauses at the end of 9.6:

-- Editor Note: 802.11-2024 up to 9.6.35, 802.11bh-2024 up to 9.6.36, 802.11be-2024 up to 9.6.38, 802.11bk-2025 no new subclause, 802.11bf-2025 up to 9.6.39

-- Editor Note: 802.11-2024 up to Table 9-658, 802.11bh-2024 up to Table 9-658a, 802.11be-2024 up to Table 9-658q, 802.11bk-2025 no new table, 802.11bf-2025 up to Table 9-658r

9.6.40 Protected HT Action frame details

9.6.40.1 Protected HT Action field

A Protected HT Action field, in the octet immediately after the Category field, differentiates the Protected HT Action frame formats. The Protected HT Action field values associated with each frame format within the Protected HT category are defined in Table 9-658s (Protected HT Action field values).

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Table 9-658s—Protected HT Action field values

Value	Meaning	Time Priority
0	Protected Notify Channel Width	No
1	Protected SM Power Save	No
2	Reserved	
3	Reserved	
4	Protected CSI	Yes
5	Protected Noncompressed Beamforming	Yes
6	Protected Compressed Beamforming	Yes
7–255	Reserved	

9.6.40.2 Protected Notify Channel Width frame format

The Protected Notify Channel Width frame allows robust STA-STA communication of the same information that is conveyed in the Notify Channel Width frame that is not robust (see 9.6.11.1 (HT Action field)).

The Action field of the Protected Notify Channel Width frame has the same format as the Action field of the Notify Channel Width frame (see 9.6.11.2 (Notify Channel Width frame format)), except that the Order 2 item is the Protected HT Action field, which is defined in 9.6.40.1 (Protected HT Action field), instead of the HT Action field.

9.6.40.3 Protected SM Power Save frame format

The Protected SM Power Save frame allows robust STA-STA communication of the same information that is conveyed in the SM Power Save frame that is not robust (see 9.6.11.1 (HT Action field)).

The Action field of the Protected SM Power Save frame has the same format as the Action field of the SM Power Save frame (see 9.6.11.3 (SM Power Save frame format)), except that the Order 2 item is the Protected HT Action field, which is defined in 9.6.40.1 (Protected HT Action field), instead of the HT Action field.

9.6.40.4 Protected CSI frame format

The Protected CSI frame allows robust STA-STA communication of the same information that is conveyed in the CSI frame that is not robust (see 9.6.11.1 (HT Action field)).

The Action field of the Protected CSI frame has the same format as the Action field of the CSI frame (see 9.6.11.4 (CSI frame format)), except that the Order 2 item is the Protected HT Action field, which is defined in 9.6.40.1 (Protected HT Action field), instead of the HT Action field.

9.6.40.5 Protected Noncompressed Beamforming frame format

The Protected Noncompressed Beamforming frame allows robust STA-STA communication of the same information that is conveyed in the Noncompressed Beamforming frame that is not robust (see 9.6.11.1 (HT Action field)).

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The Action field of the Protected Noncompressed Beamforming frame has the same format as the Action field of the Noncompressed Beamforming frame (see 9.6.11.5 (Noncompressed Beamforming frame format)), except that the Order 2 item is the Protected HT Action field, which is defined in 9.6.40.1 (Protected HT Action field), instead of the HT Action field.

9.6.40.6 Protected Compressed Beamforming frame format

The Protected Compressed Beamforming frame allows robust STA-STA communication of the same information that is conveyed in the Compressed Beamforming frame that is not robust (see 9.6.11.1 (HT Action field)).

The Action field of the Protected Compressed Beamforming frame has the same format as the Action field of the Compressed Beamforming frame (see 9.6.11.6 (Compressed Beamforming frame format)), except that the Order 2 item is the Protected HT Action field, which is defined in 9.6.40.1 (Protected HT Action field), instead of the HT Action field.

9.6.41 Protected VHT Action frame details

9.6.41.1 Protected VHT Action field

A Protected VHT Action field, in the octet immediately after the Category field, differentiates the Protected VHT Action frame formats. The Protected VHT Action field values associated with each frame format within the Protected VHT category are defined in Table 9-658t (Protected VHT Action field values).

Table 9-658t—Protected VHT Action field values

Value	Meaning	Time Priority
0	Protected VHT Compressed Beamforming	Yes
1	Protected Group ID Management	No
2	Protected Operating Mode Notification	No
3–255	Reserved	

9.6.41.2 Protected VHT Compressed Beamforming frame format

The Protected VHT Compressed Beamforming frame allows robust STA-STA communication of the same information that is conveyed in the VHT Compressed Beamforming frame that is not robust (see 9.6.22.1 (VHT Action field)).

The Action field of the Protected VHT Compressed Beamforming frame has the same format as the Action field of the VHT Compressed Beamforming frame (see 9.6.22.2 (VHT Compressed Beamforming frame format)), except that the Order 2 item is the Protected VHT Action field, which is defined in 9.6.41.1 (Protected VHT Action field), instead of the VHT Action field.

9.6.41.3 Protected Group ID Management frame format

The Protected Group ID Management frame allows robust STA-STA communication of the same information that is conveyed in the Group ID Management frame that is not robust (see 9.6.22.1 (VHT Action field)).

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The Action field of the Protected Group ID Management frame has the same format as the Action field of the Group ID Management frame (see 9.6.22.3 (Group ID Management frame format)), except that the Order 2 item is the Protected VHT Action field, which is defined in 9.6.41.1 (Protected VHT Action field), instead of the VHT Action field.

9.6.41.4 Protected Operating Mode Notification frame format

The Protected Operating Mode Notification frame allows robust STA-STA communication of the same information that is conveyed in the Operating Mode Notification frame that is not robust (see 9.6.22.1 (VHT Action field)).

The Action field of the Protected Operating Mode Notification frame has the same format as the Action field of the Operating Mode Notification frame (see 9.6.22.4 (Operating Mode Notification frame format)), except that the Order 2 item is the Protected VHT Action field, which is defined in 9.6.41.1 (Protected VHT Action field), instead of the VHT Action field.

9.6.42 EPP Action frame details

9.6.42.1 EPP Action field

An EPP Action field, in the octet immediately after the Category field, differentiates the EPP Action frame formats. The EPP Action field values associated with each frame format within the EPP category are defined in Table 9-658u (EPP Action field values).

Table 9-658u—EPP Action field values

Value	Meaning
0	EPP Capabilities And Operation Parameters Request
1	EPP Capabilities And Operation Parameters Response
2	EPP Groups Parameter
3	EPP Group Request
4	EPP Group Response
5	EPP Group Assignment
6	OTA MAC Collision Notification
7	OTA MAC Collision Response
8	AID Assignment Request
9	AID Assignment Response
10	EPP Group Parameter Update
11–255	Reserved

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9.6.42.2 EPP Capabilities And Operation Parameters Request frame format

The EPP Capabilities And Operation Parameters Request frame allows capabilities and operation parameters to be requested in a protected Action frame.

The Action field of the EPP Capabilities And Operation Parameters Request frame contains the information shown in Table 9-658v (EPP Capabilities And Operation Parameters Request frame Action field format for non-MLO) or Table 9-658w (EPP Capabilities And Operation Parameters Request frame Action field format for MLO).

Table 9-658v—EPP Capabilities And Operation Parameters Request frame Action field format for non-MLO

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token

Table 9-658w—EPP Capabilities And Operation Parameters Request frame Action field format for MLO

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Basic Multi-Link element

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The Basic Multi-Link element is defined in 9.4.2.322 (Multi-Link element) and is optionally present (see 12.16.4 (Capabilities and operation parameters request and response procedure)).

9.6.42.3 EPP Capabilities And Operation Parameters Response frame format

The EPP Capabilities And Operation Parameters Response frame allows capabilities and operation parameters to be provided in a protected Action frame.

The Action field of the EPP Capabilities And Operation Parameters Response frame contains the information shown in Table 9-658x (EPP Capabilities And Operation Parameters Response frame Action field format for non-MLO) or Table 9-658y (EPP Capabilities And Operation Parameters Response frame Action field format for MLO).

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Table 9-658x—EPP Capabilities And Operation Parameters Response frame Action field format for non-MLO

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Beacon Interval
4	Capability Information
5	Elements in order as defined in Table 9-67 (Probe Response frame body) excluding Multi-Link element and Multiple BSSID element

Table 9-658y—EPP Capabilities And Operation Parameters Response frame Action field format for MLO

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Basic Multi-Link element

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The Beacon Interval field is defined in 9.4.1.3 (Beacon Interval field).

The Capability Information field is defined in 9.4.1.4 (Capability Information field).

The Basic Multi-Link element is defined in 9.4.2.322 (Multi-Link element) and is optionally present (see 12.16.4 (Capabilities and operation parameters request and response procedure)).

9.6.42.4 EPP Groups Parameter frame format

The EPP Groups Parameter frame is used to carry the EPP epoch settings for one or more EPP groups using the procedures defined in 10.71.2.2 (EPP group operations) and 10.71.2.3 (EPP epoch transition operations). The EPP Groups Parameter frame contains the information shown in Table 9-658z (EPP Groups Parameter frame Action field format).

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

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Table 9-658z—EPP Groups Parameter frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Number of EPP Epoch Settings
4	EPP Epoch Settings List

The Number of EPP Epoch Settings field specifies the number of EPP Epoch Settings fields that are in the EPP Epoch Settings List field (i.e., the number of EPP groups with the conveyed parameters). The length of this field is one octet. The value 0 is reserved.

The EPP Epoch Settings List field contains one or more EPP Epoch Settings fields (as defined in 9.4.1.84 (EPP Epoch Settings field)) indicating the parameters of the EPP groups that the AP MLD wants to convey to the non-AP MLD.

9.6.42.5 EPP Group Request frame format

The EPP Group Request frame allows a CPE non-AP MLD to request a new EPP group creation, assignment to a particular EPP group, request information about one or more groups, or leave from the current EPP group, as defined in 10.71.2.1 (General) and 10.71.2.2 (EPP group operations).

Table 9-658aa—EPP Group Request frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	EPP Group Request
4	EPP Group Settings

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The EPP Group Request field is defined in 9.4.1.85 (EPP Group Request field).

The EPP Epoch Settings field is defined in 9.4.1.84 (EPP Epoch Settings field).

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9.6.42.6 EPP Group Response frame format

The EPP Group Response frame contains the response to an EPP Group Request or EPP Group Assignment frame.

Table 9-658ab—EPP Group Response frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Status Code
4	EPP Epoch Settings (Optional)

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The Status Code field is defined in 9.4.1.9 (Status Code field).

The EPP Epoch Settings field is optional and is defined in 9.4.1.84 (EPP Epoch Settings field).

9.6.42.7 EPP Group Assignment frame format

The EPP Group Assignment frame allows an EPP AP MLD to request a CPE non-AP MLD to transition to a different EPP group.

Table 9-658ac—EPP Group Assignment frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	EPP Epoch Settings

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

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The EPP Epoch Settings field is defined in 9.4.1.84 (EPP Epoch Settings field).

9.6.42.8 OTA MAC Collision Notification frame format

The OTA MAC Collision Notification frame warns of an upcoming MAC address collision with another STA in a given epoch.

Table 9-658ad—OTA MAC Collision Notification frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	OTA MAC Collision Warning element

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field).

The OTA MAC Collision Warning element is defined in 9.4.2.350 (OTA MAC Collision Warning element).

9.6.42.9 OTA MAC Collision Response frame format

The OTA MAC Collision Response frame indicates the action the STA will take based on the information on the OTA MAC Collision Notification frame.

Table 9-658ae—OTA MAC Collision Response frame Action field format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	OTA MAC Collision Warning element

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field).

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The OTA MAC Collision Warning element is defined in 9.4.2.350 (OTA MAC Collision Warning element). In the MAC Collision Response frame, the Colliding Epoch, Link ID Info and non-AP MLD Specific Collision Epoch Offset fields are reserved.

9.6.42.10 AID Assignment Request frame format

The AID Assignment Request frame is transmitted by a CPE AP MLD as part of setting up frame anonymization. The frame assigns AID values to the receiving CPE STA for the coming epochs.

Table 9-658af—AID Assignment Request frame format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	AID List

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The AID List element is defined in 9.4.2.351 (AID List element).

9.6.42.11 AID Assignment Response frame format

The format of the AID Assignment Response frame is shown in Table 9-658ag (AID Assignment Response frame format).

Table 9-658ag—AID Assignment Response frame format

Order	Meaning
0	Category
1	EPP Action
2	Dialog Token
3	Status Code
4	Number Of Stored AIDs

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

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The Dialog Token field is defined in 9.4.1.12 (Dialog Token field) and is set to a nonzero value to identify the request/response transaction.

The Status Code field indicates the result of the AID assignment request and is defined in 9.4.1.9 (Status Code field).

The Number Of Stored AIDs field is defined in 9.4.1.86 (Number Of Stored AIDs field) and is present if the Status Code field is equal to SUCCESS_AID_LIST_PARTIALLY_STORED, to indicate the number of AIDs that the CPE non-AP MLD has stored; otherwise not present.

9.6.42.12 EPP Group Parameter Update frame format

The EPP Group Parameter Update frame allows the AP MLD to inform the members of an EPP group about upcoming group parameters changes.

The EPP Group Parameter frame contains the information shown in Table 9-658ah (EPP Group Parameter Update frame format).

Table 9-658ah—EPP Group Parameter Update frame format

Order	Meaning
0	Category
1	EPP Action
2	EPP Group Parameter Update

The Category field is defined in 9.4.1.11 (Action field).

The EPP Action field is defined in 9.6.42.1 (EPP Action field).

The EPP Group Parameter Update element is defined in 9.4.2.352 (EPP Group Parameter Update element).

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10. MAC sublayer functional description

Insert the following new subclauses at the end of clause 10 (MAC sublayer functional description):

-- Editor Note: 802.11-2024 up to 10.70, 802.11bh-2024, 802.11be-2024, 802.11bk-2025 and 802.11bf-2025does not add new subclause

-- Editor Note: 802.11-2024 up to Figure 10-166

-- Editor Note: 802.11-2024 up to Table 10-40

-- Editor Note: 802.11-2024 up to Equation 10-27

10.71 Frame anonymization

10.71.1 General

10.71.1.1 Introduction

Frame anonymization (FA) is an optional EPP feature available during an association between MLDs. The objective of FA is to mitigate the privacy threat posed by static or predictable values associated or assigned to an MLD transmitted in unencrypted fields and elements. SA and DA can be protected by transmitting an MSDU in an A-MSDU that is encrypted together with its payload (see 10.11 (A-MSDU operation)) . FA transforms other values into “over-the-air” values that remain static or predictable only within configurable periods called EPP epochs, e.g., temporary MAC addresses. The over-the-air values can be transmitted in the clear while maintaining anonymity. These mechanisms mitigate third party attempts to determine the presence of an MLD across multiple FA epochs.

FA can be activated in one of two modes:

— Client privacy enhancement frame anonymization (CPE FA) comprises a set of FA features that mitigate tracking a non-AP MLD. When activated, CPE FA is applied to individually addressed frames transmitted to and from non-AP MLDs associated with an AP MLD.

— BSS privacy enhancement frame anonymization (BPE FA) comprises a set of FA features that mitigate tracking an AP MLD and its associated non-AP MLDs. Most of the features in CPE FA are also included in BPE FA. When activated, BPE FA is applied to

  • all non-broadcast, group addressed frames and Privacy Beacon frames transmitted by an AP MLD, and
  • individually addressed frames exchanged between a non-AP MLD and the AP MLD it is associated with.

Table 10-40a (Frame anonymization features for CPE FA and BPE FA) identifies the CPE FA and BPE FA features.

10.71.1.2 General requirements

Table 10-40a (Frame anonymization features for CPE FA and BPE FA) provides the requirement status (mandatory or not applicable) for the FA features when CPE FA is activated or BPE FA is activated.

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Table 10-40a—Frame anonymization features for CPE FA and BPE FA

FA Feature		Clause	Activated mode of FA	
			CPE FA	BPE FA
EPP epoch operations		10.71.2.2 (EPP group operations), 10.71.2.3 (EPP epoch transition operations), 10.71.2.4 (EPP Epoch Start Time Computation), 10.71.2.5 (OTA MAC address collision avoidance)	Mandatory	
EPP group operations	CPE EPP group operations	10.71.2.2.2 (CPE EPP group operations)	Mandatory	Not applicable
	BPE EPP group operations	10.71.2.2.3 (BPE EPP group operations)	Not applicable	Mandatory
AID Assignment		10.71.7 (Frame anonymization and AID)	Mandatory	
MAC header Anonymization	CPE MHA	10.71.3 (Establishing CPE MAC header anonymization parameter sets)	Mandatory	
	BPE MHA	10.71.4 (Establishing BPE MAC header anonymization parameter sets)	Not applicable	Mandatory

Table 10-40b (MAC header anonymization for CPE FA and BPE FA) identifies whether CPE MHA processing, BPE MHA processing or no MHA processing is applied to the value transmitted and received in the header fields impacted by MHA. The processing varies according to the activated mode of FA (CPE FA or BPE FA), the frame direction, and the frame addressing (i.e., whether the frame is individually addressed frame, non-broadcast, group addressed frame or a Privacy Beacon frame). For any frame that does not match a description in this table, MHA is not applied to the identified header fields.

Table 10-40b—MAC header anonymization for CPE FA and BPE FA

Activated mode of FA		**CPE FA**		**BPE FA**			
Frame Direction		From DS	To DS	From DS			To DS
Frame addressing		Individual	Individual	Privacy Beacon	Non-broadcast Group	Individual	Individual
Address field	Address 1	CPE MHA	No MHA	No MHA	BPE MHA	CPE MHA	BPE MHA
	Address 2	No MHA	CPE MHA	BPE MHA	BPE MHA	BPE MHA	CPE MHA

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Table 10-40b—MAC header anonymization for CPE FA and BPE FA

Activated mode of FA		CPE FA		BPE FA			
Sequence Number field	SNS1	No MHA	CPE MHA	N/A	N/A	CPE MHA	CPE MHA
	SNS3	CPE MHA	CPE MHA	N/A	N/A	CPE MHA	CPE MHA
	SNS9	CPE MHA	CPE MHA	N/A	N/A	CPE MHA	CPE MHA
	SNS10	CPE MHA	CPE MHA	N/A	N/A	CPE MHA	CPE MHA
	SNS11	N/A	N/A	N/A	BPE MHA	N/A	N/A
	SNS12	CPE MHA	CPE MHA	CPE MHA	N/A	CPE MHA	CPE MHA
Packet Number field		CPE MHA	CPE MHA	BPE MHA	BPE MHA	CPE MHA	CPE MHA
Timestamp field		N/A	N/A	BPE MHA	N/A	N/A	N/A

Only MLDs may implement FA mechanisms.

An MLD shall not simultaneously activate both CPE FA and BPE FA.

An AP MLD or non-AP MLD that has CPE FA or BPE FA activated shall set dot11DSMACAddressActivated to true.

TDLS shall not be used when CPE FA is activated or BPE FA is activated.

10.71.1.3 General CPE FA requirements

Activation of CPE FA is advertised by a CPE AP MLD in the CPE Frame Anonymization Activated field of the Extended RSN Capabilities field of the RSNXE field in Beacon and Probe Response frames.

A CPE AP MLD may permit non-AP STAs and non-AP MLDs to associate.

If the non-AP MLD determines that the CPE AP MLD has CPE FA activated, then the non-AP MLD may initiate activation of CPE FA by setting the CPE Frame Anonymization Activated field to 1 in the Extended RSN Capabilities field of the RSNXE field in the (Re)Association Request frame. CPE FA shall be activated for the session if the CPE AP MLD assigns the non-AP to an EPP group. (The non-AP MLD is now a CPE non-AP MLD).

If CPE FA is activated for an association, then:

— The CPE AP MLD and associated CPE non-AP MLD shall apply CPE MHA as indicated in Table 10-40b (MAC header anonymization for CPE FA and BPE FA) for CPE FA mode.

— The CPE AP MLD and associated CPE non-AP MLD shall use per-EPP-epoch AIDs as defined in 10.71.7 (Frame anonymization and AID).

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NOTE—Since an AP MLD that activates CPE FA also has dot11DSMACAddressActivated set to true, the AP MLD also advertises DS MAC address support in Beacon and Probe Response frames using the DS MAC Address Support field of the Extended RSN Capabilities field of the RSNXE field.

10.71.1.4 General BPE FA requirements

A BPE AP MLD shall apply BPE MHA operations as identified in Table 10-40b (MAC header anonymization for CPE FA and BPE FA) to all non-broadcast, group addressed frames and Privacy Beacons, even when there are no associated BPE non-AP MLDs.

A non-AP MLD shall activate BPE upon successful (re)association with a BPE AP MLD.

— The BPE AP MLD and associated BPE non-AP MLD shall apply BPE MHA operations and (for individually address frames) CPE MHA operations as indicated in Table 10-40b (MAC header anonymization for CPE FA and BPE FA) for BPE FA mode.

— The BPE AP MLD and associated BPE non-AP MLD shall use per-EPP-epoch AIDs as defined in 10.71.7 (Frame anonymization and AID).

An MLD that activates CPE FA or BPE FA shall transmit all MSDUs in A-MSDUs as defined in 10.11 (A-MSDU operation).

NOTE—A-MSDU operation allows the transmission of one MSDU in one A-MSDU or multiple MSDUs in one A-MSDU.

10.71.2 EPP epoch operation

10.71.2.1 General

If an MLD supports MAC Header Anonymization, then it shall support EPP epoch operation.

An AP MLD that enables BPE FA mechanisms shall support exactly one EPP group. An AP MLD that enables CPE FA mechanisms but not BPE FA mechanisms shall support at least one EPP group .
EPP epoch operation enables an AP MLD and non-AP MLDs to coordinate periodic changes to fields of individually addressed frames transmitted by those devices that might facilitate their identification (e.g., STA address, AID, PN, SN, etc.).

At any given time, an AP MLD shall not assign an associated non-AP MLD to more than one EPP group. A non-AP MLD shall not request to be assigned to more than one EPP group at a time.

A non-AP MLD that is a member of an EPP group and its associated AP MLD shall both anonymize the selected fields of the individually addressed frames according to the group EPP epoch settings as defined in 10.71.3 (Establishing CPE MAC header anonymization parameter sets), 10.71.4 (Establishing BPE MAC header anonymization parameter sets), 10.71.5 (MAC header anonymization and transmitting functions), 10.71.6 (MAC header anonymization and receiving functions), 10.71.7 (Frame anonymization and AID), and 10.71.8 (BSS privacy enhancements operations).

10.71.2.2 EPP group operations

10.71.2.2.1 General

The AP MLD may at any time decide to change the parameters of one or more CPE groups. An AP may use for example; the Minimum Pacing elements received in (re)association request frames, the number of non-AP MLDs in each existing group and other elements in the decision to change a group parameters. The algorithm to choose the new parameters of a group is beyond the scope of this standard.

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An AP MLD shall inform the non-AP MLDs that are member of a group which parameters will change, by transmitting the Group Parameter update element in one or more Group Parameter update frames prior to the intended parameters change time. The EPP group parameters change should be scheduled so that all members of the group, including non-AP MLDs in power save mode, have the opportunity to receive at least one Group Parameters update announcement frame before the change.

10.71.2.2.2 CPE EPP group operations

A CPE AP MLD advertises the support for group EPP epoch operations in Beacon and Probe Response frames by setting the Group EPP Epoch Supported field of the Extended RSN Capabilities field of the RSNXE to 1.

A non-AP MLD advertises the support for group EPP epoch operations in (Re)Association Request frames by setting the Group EPP Epoch Supported field of the Extended RSN Capabilities field of the RSNXE to 1.

A CPE non-AP MLD that supports group EPP epoch operations shall include in an encrypted (Re)Association Request frame an EPP element indicating the parameters for the EPP group it requests to join.

NOTE 1—The EPP parameters indicated in the (Re)Association Request frame correspond to the desired parameters derived by the CPE non-AP MLD. These parameters do not need to match the parameters of any EPP group existing in the AP MLD.

The first EPP epoch of an EPP epoch sequence is an EPP epoch numbered 0.

Within the EPP Epoch Settings field sent in encrypted (Re)Association Request frames, EPP Group Request frames and EPP Group Response frames, the CPE non-AP MLD indicates in the Epoch Interval field the desired duration of the EPP epoch the CPE non-AP MLD wants to join, and shall include an AID Storage Size field indicating the number of AIDs the CPE non-AP MLD is capable of storing.

The CPE non-AP MLD may also include within the EPP Epoch Settings field sent in the encrypted (Re)Association Request frames, EPP Epoch Request frames and EPP Epoch Response frames, the following fields:

— an EPP Group ID field indicating the identifier of the EPP Group the CPE non-AP MLD wants to join, and

— a Minimum Epoch Pacing field.

The rest of the fields in the EPP Epoch Settings field shall not be transmitted by a CPE non-AP MLD. The Minimum Epoch Pacing field includes the Minimum Epoch Pacing Parameters field, indicating the minimum epoch interval length supported by the CPE non-AP MLD. If the value resulting from the multiplication of the Epoch Interval Length field by the Epoch Interval Unit field included in the Minimum Epoch Pacing field is greater than the value resulting from the multiplication of the Epoch Interval Length field by the Epoch Interval Unit field for the default EPP group (group 0) or of any other EPP group already created, then the CPE non-AP MLD is not assigned to any EPP group at (re)association.

NOTE 2—The CPE non-AP MLD might remain associated without FA and might request the creation of a new EPP group (through the EPP Group Request frame (see 9.6.42.5 (EPP Group Request frame format))).

The CPE AP MLD, upon reception of the EPP element in an encrypted (Re)Association Request frame, shall assign the CPE non-AP MLD to the EPP group with parameters that best match the parameters requested, also considering the size of the available EPP groups, and shall include an EPP element in the encrypted (Re) Association Response frame, creating a new EPP group or assigning the CPE non-AP MLD to an already existing EPP group. The assigned EPP epoch duration shall not be shorter than indicated in the Minimum Epoch Pacing Parameters field.

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The CPE non-AP MLD, upon reception of the EPP element in a (Re)Association Response frame, EPP Epoch Response frames, EPP Epoch Assignment frames or EPP Groups Parameter frames, shall set the value of the dot11EPPEpochTransitionTime to the value in TUs of the Transition Period field if present.

The parameters of the assigned EPP group are provided to the CPE non-AP MLD through an EPP element in the encrypted (Re)Association Response frame. If no EPP element is included in the encrypted (Re)Association Response frame, the CPE non-AP MLD is not assigned to any EPP group.

A CPE AP MLD transmitting an EPP Epoch Settings field in an encrypted (Re)Association Response frame, EPP Group Response frame, EPP Group Assignment frame, or EPP Groups Parameter frame shall include the following fields:

- EPP Epoch Settings Control field indicating the optional fields included in the EPP Epoch Settings field.
- EPP Group ID field.
- Epoch Interval field.
- Transition Period field.
- First Epoch TSF Start Time.
- Group Epoch Seed field.
- Epoch Number Offset field.
- Epoch Start Time Variation Range field.
- Epochs Remaining field.

A CPE AP MLD transmitting EPP Epoch Settings fields in an encrypted (Re)Association Response frame, EPP Group Response frame, EPP Group Assignment frame, or EPP Groups Parameters frame may also include the Number Of Participating Affiliated STAs in the EPP Epoch Settings field.

The CPE AP MLD shall not include an AID Storage Size field or a Minimum Epoch Pacing field in transmitted EPP Epoch Settings fields.

Within an EPP element transmitted by an EPP AP MLD:

- If the Epoch Start Time Variation Range field is present, the Epoch Start Time Variation Range field value shall not exceed 20% of the Epoch Interval Length field value.
- If the Group Epoch Seed field is present, the Group Epoch Seed field value shall be the same for each transmission to any member of an EPP group for a given EPP Epoch sequence. An EPP AP shall select a new random Group Epoch Seed value for each EPP Epoch sequence.

An associated CPE non-AP MLD may request creation of a new EPP group by sending an EPP Group Request frame (see 9.6.42.5 (EPP Group Request frame format)) with an EPP Group Request field indicating “Create/Join” and indicating the parameters for the EPP group to be created in the EPP Epoch Settings field.

The CPE AP MLD may create a new EPP group with the received parameters indicated in EPP Group Request frame. Alternatively, the CPE AP MLD may assign the CPE non-AP MLD to an already existing EPP group with *similar* parameters. This may be signaled to the CPE non-AP MLD in an EPP Group Response frame setting the Status Code field to indicate SUCCESS_SIMILAR_EPOCH, and providing the EPP Epoch Settings field with the parameters of the EPP group. In this context, an EPP group with “similar parameters” refers to an existing EPP group whose epoch duration is equal or smaller than the one requested by the CPE non-AP MLD (i.e., most privacy-preserving). This choice is made while ensuring adherence to any pacing limits indicated in the Minimum Epoch Pacing Parameters field that the CPE non-AP MLD may specify in any transmitted EPP Epoch Settings field.

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Once the CPE non-AP MLD is associated and has been assigned an EPP group, it may request to join a different EPP group. Information on the available EPP groups may be distributed periodically by the CPE AP MLD transmitting EPP Groups Parameter frames. To join a different EPP group, the CPE non-AP MLD shall send an EPP Group Request frame (see 9.6.42.5 (EPP Group Request frame format)), indicating "Create/Join" in the EPP Group Request field and providing the EPP Epoch Settings field indicating the parameters of the EPP group it requests to join.

The CPE non-AP MLD may also request information about one or more EPP groups and their parameters by sending an EPP Epoch Request frame (see 9.6.42.5 (EPP Group Request frame format)) with a Epoch Request field indicating "All Groups Settings", when querying about all groups configured on the AP MLD, or "Specific Group Settings", when querying about a specific group configured on the AP MLD. When querying about a specific group, the non-AP MLD indicates the Group ID in the EPP Settings fields. The AP may refuse to provide the requested elements and respond with an EPP Epoch Response frame with status code set to GROUP_INFORMATION_NOT_PROVIDED. When the query is about all group settings, the AP can respond with an EPP Group parameter frame that contains the parameters of all groups.

Upon reception of the request, the CPE AP MLD may assign the CPE non-AP MLD to another EPP group and remove it from the previous EPP group. The result of the operation is indicated to the CPE non-AP MLD through an EPP Group Response frame. This frame indicates in the Status Code field "SUCCESS", and may include an optional EPP Epoch Settings field to indicate the parameters of the newly assigned EPP group.

The CPE AP MLD may request the associated CPE non-AP MLD to transition to a different EPP group, by sending an EPP Group Assignment frame to the associated CPE non-AP MLD including the EPP Epoch Settings field with the parameters of the suggested EPP group. The CPE non-AP MLD shall report the status of the operation by responding with an EPP Group Response frame. This operation allows the CPE AP MLD to reorganize the EPP groups in use. Following this reorganization, the resulting EPP groups should maintain the same level of restrictiveness or be even more restrictive, i.e., shorter epoch durations, while also respecting any pacing limits indicated in the Minimum Epoch Pacing Parameters field that the CPE non-AP MLD has specified in the encrypted (Re)Association Request frame.

Once the CPE non-AP MLD receives an EPP Group Response frame or transmits an EPP Group Response frame in response to an EPP Group Assignment frame signaling successful (Status code field set to SUCCESS, SUCCESS_SIMILAR_EPOCH, SUCCESS_ALREADY_EXISTING_EPOCH or SUCCESS_AID_LIST_PARTIALLY_STORED) reception of new EPP group parameters, the CPE non-AP MLD shall compute the new MHA parameters for the current EPP epoch, as per 10.71.3 (Establishing CPE MAC header anonymization parameter sets), and apply them in the next epoch starting time as indicated by the First Epoch TSF Start Time field of the EPP Epoch Settings field (considering the operations in 10.71.2.4 (EPP Epoch Start Time Computation) in the received EPP Group Response frame or EPP Group Assignment frame.

EPP epoch sequences may have a finite length indicated by the Epochs Remaining field included in the EPP Epoch Settings field.

Before the end of an EPP group epoch sequence, the EPP AP MLD may advertise the other EPP groups available through an EPP Groups Parameter frame. The AP MLD should schedule a transmission of an EPP Groups Parameter frame so that all members of the group, including non-AP MLDs in power save mode, have the opportunity to join or /create a new group before the EPP epoch sequence ends. A non-AP MLD whose EPP group epoch is nearing the end of its sequence may request to join or create a new EPP group through an EPP Group Request frame. In case the EPP non-AP MLD does not request the change to a different group before its current epoch sequence ends, the AP MLD shall assign the non-AP MLD that is a member of an EPP group whose EPP group epoch is nearing the end of its sequence to another EPP group with similar parameters, through an EPP Group Assignment frame.

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10.71.2.2.3 BPE EPP group operations

A BPE non-AP MLD advertises the support for EPP epoch group operations in (Re)Association Request frames by setting the Group EPP Epoch Supported field of the Extended RSN Capabilities field of the RSNXE to 1.

A BPE non-AP MLD shall include in an encrypted (Re)Association Request frame an EPP element indicating the number of AIDs the BPE non-AP MLD can store in the AID Storage Size field. Other fields of the EPP element are reserved.

A BPE AP MLD always assigns the BPE non-AP MLD to the unique EPP group used by the BPE AP MLD, regardless of the parameters indicated in a received encrypted (Re)Association Request frame.

A BPE non-AP MLD shall not transmit EPP Group Request frames or EPP Group Response frames. A BPE AP MLD shall not transmit EPP Group Response frames or EPP Group Assignment frames. A BPE AP MLD advertise EPP Group parameters to BPE non-AP MLD through EPP Groups Parameter frames.

10.71.2.3 EPP epoch transition operations

Each EPP epoch starts with a transition period.

During the transition period of an EPP epoch, the FA parameters assigned to a non-AP MLD during the preceding EPP epoch shall remain valid only for the following operations:

— Completion of a frame exchange sequence. whose first frame was transmitted in previous EPP epoch.

— Completion of a TXOP that started during preceding EPP epoch.

— Retransmissions of a frame that was transmitted in the preceding EPP epoch.

A transition period terminates at the end of a transition timeout interval or at the beginning of the next epoch margin, whichever comes first.

After the end of an EPP epoch transition period, the FA parameters assigned to a non-AP MLD during the preceding EPP epoch become invalid for both transmitted and received frames i.e., (re)transmissions with previous FA parameters are no longer allowed.

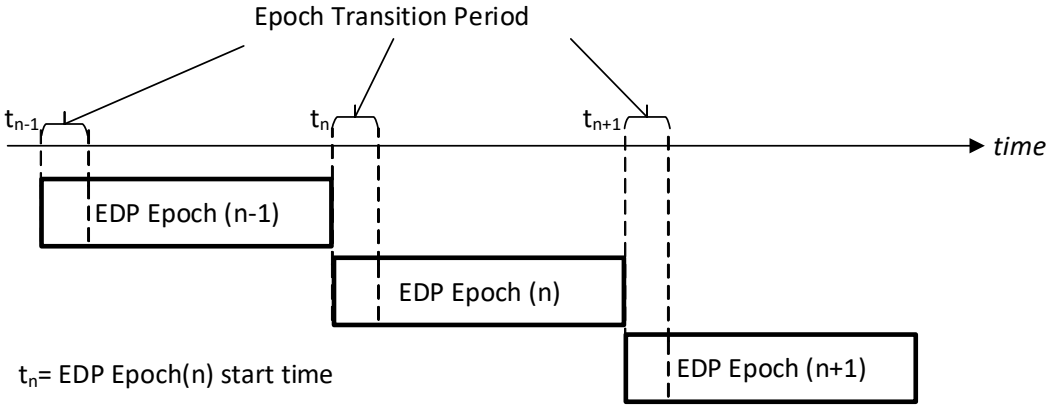


Figure 10-166a—Example of EPP epoch timeline

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Figure 10-166a (Example of EPP epoch timeline) shows an example EPP epoch sequence of consecutive EPP epochs with their associated EPP epoch start times t_n and transition period tp_n.

An overview of the EPP epoch is shown in Figure 10-166b (Overview of EPP epoch).

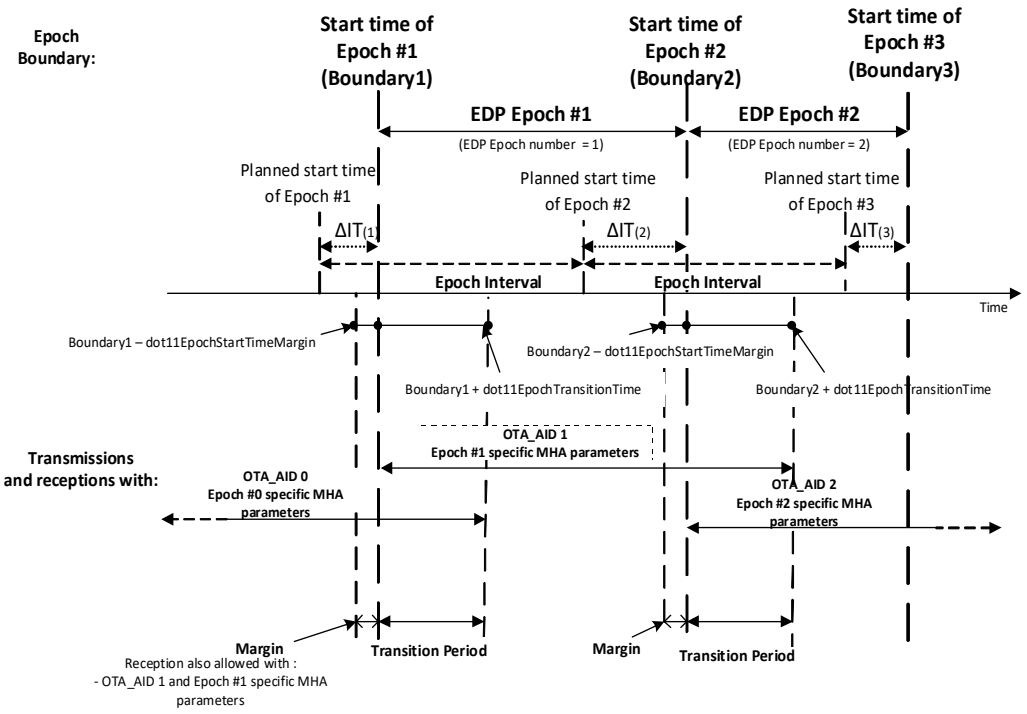


Figure 10-166b—Overview of EPP epoch

The next epoch boundary is derived from the value of the first epoch TSF start time (see 10.71.2.4 (EPP Epoch Start Time Computation)).

NOTE 1—The ΔIT(*n*) is a positive offset delaying the effective start of an EPP epoch boundary by a pseudo random value, so start times of EPP epochs (boundaries *n*) of a sequence are not occurring at a regular epoch interval. The time between two consecutive planned start times is constant and equal to Epoch Interval, while the duration of two consecutive epochs (time between Boundaries) may be different.

From the start of an EPP group's EPP epoch until the start of the next EPP epoch, the CPE AP MLDs and the associated CPE non-AP MLDs belonging to that group shall begin using the new FA parameters to anonymize selected OTA fields when transmitting new individually addressed frames. The CPE AP MLDs and the CPE non-AP MLDs shall continue using these anonymization parameters for the duration of the epoch.

If the start time of an EPP epoch occurs during an ongoing TXOP, the FA parameters corresponding to the EPP epoch apply at the end of that TXOP.

To account for clock drifts, the CPE non-AP MLD and CPE AP MLD shall begin to accept individually addressed frames that use the new FA parameters during a margin period for a duration equal to dot11EPPEpochStartTimeMargin before the start of the new epoch.

The CPE non-AP MLD and CPE AP MLD shall accept individually addressed frames with the old FA parameters for a duration equal to dot11EPPEpochTransitionTime after the start of the new epoch.

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NOTE 2—A CPE non-AP MLD exiting the doze state can use the TSF in a beacon sent by the affiliated AP of its associated AP MLD to realign its internal clock with that of the AP MLD, and confirm the EPP epoch number and delay until the next EPP epoch starts.

10.71.2.4 EPP Epoch Start Time Computation

To prevent an eavesdropper from easily determining the EPP epoch start time, a pseudo random offset is computed and used by both the AP and each non-AP STAs of the EPP group.

The start time of an epoch occurs at the same time on each link of an MLD.

Upon reception of an EPP Epoch Response frame, or of a encrypted (Re)Association Response frame containing an EPP element on a link, the non-AP STA of a non-AP MLD shall:

— Store the value of the first epoch TSF start time, the epoch interval, and set its epoch number *n* for this epoch to the value of the received epoch number offset.

— Construct the value of the corresponding first epoch TSF start time of its other links according to the formula:

The value of the first epoch TSF start time of another link shall be equal to the value of the first epoch TSF start time of the receiving link + TSF Offset value between the other link and the receiving link

NOTE 1—the TSF Offset value between the receiving link and the other links can be computed based on TSF Offset fields values of the latest Basic Multi-Link element exchange indicating TSF offsets between the reporting AP and reported APs received on any enabled link.

For a given link, for any EPP epoch number n ($n > 0$) in an EPP epoch sequence, the link TSF timer value corresponding to the start time of the EPP epoch number n is called EpochTSFStartTime(n) and is computed according to the formula:

EpochTSFStartTime(n) = PlannedEpochTSFStartTime(n) for the link + ΔIT(n)

where

PlannedEpochTSFStartTime(n) = FirstPlannedEpochTSFStartTime + (n – EpochNumberOffset) × EpochInterval mod 2^{64}

For BPE stations, ΔIT(n) is computed according to the following formula

ΔIT(n) =EpochStartTimeOffset(n) mod TimeRangeTU

With EpochStartTimeOffset(n) computed as defined in 10.71.4 (Establishing BPE MAC header anonymization parameter sets).

For CPE stations, ΔIT(n) is computed according to the following formula:

For all values of Epoch Start Time Variation Range field other than zero:

ΔIT(n) = int (KDF-*Hash-Length*(PGTK, "ERCM", Seed || AP MLD MAC || (n × EpochInterval))) mod TimeRangeTU

and where

n	is the current number of the EPP epoch in the EPP epoch sequence.
PlannedEpochTSFStartTime(n)	is the TSF timer value of the link corresponding to the nominal start time of the EPP epoch number n in the EPP epoch sequence. This planned start time occurs at a regular time interval equal to the

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	epoch interval.
EpochNumberOffset	is the value indicated in the Epoch Number Offset field of the EPP Epoch Settings field.
EpochInterval	is the value in TU corresponding to the Epoch Interval field of the EPP Epoch Settings field.
KDF-*Hash-Length*	is the key derivation function as defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm identified by the AKM suite selector (see Table 9-190 (AKM suite selectors)).
Length	is the number of bits to derive. 16 bits are derived for ΔIT.
FirstPlannedEpochTSFStartTime	is the value of the first epoch TSF start time, initialized, upon reception of an EPP element by the STA with the First Epoch TSF Start Time value of the EPP element of the received EPP Epoch Settings field.
TimeRangeTU	is the value in TU corresponding to the value of the Epoch Start Time Variation Range field multiplied by the number of TU in the Epoch Interval Unit field of the EPP Epoch Settings field.
PGTK	is the cryptographic key assigned by an EPP AP MLD that is used to manage the group EPP epoch, distributed to the EPP non-AP MLDs associated with the EPP AP MLD.
Seed	is the Group Epoch Seed field value of the received EPP Epoch Settings field.
AP MLD MAC	is the MLD MAC address of the CPE AP.

If the value of the Epoch Start Time Variation Range field is 0, then no computation of ΔIT(n) is needed, and ΔIT(n) = 0.

10.71.2.5 OTA MAC address collision avoidance

An EPP (CPE or BPE) AP MLD and an EPP (CPE or BPE) non-AP MLD anonymize selected OTA MAC header fields of individually addressed frames they transmit to each other.

An EPP AP MLD shall determine whether the OTA MAC address that an EPP non-AP MLD will use in a subsequent epoch on a link will cause a collision with the BSSID of the affiliated APs on that link of the associated AP MLD, the OTA MAC address of another non-AP MLD(s) or another STA on that link. The AP should then act as follows:

— When such a collision risk is anticipated with the OTA MAC of one or more non-CPE or non-BPE STA or non-AP MLD, the EPP AP MLD shall send to the EPP non-AP MLDs an OTA MAC Collision Notification frame before the epoch when the collision is anticipated to occur as indicated in the Colliding Epoch field, instructing the EPP non-AP MLDs to apply to all its links the Non-AP MLD Specific Collision Epoch Offset field value signaled in the AP MLD OTA MAC Collision Notification frame to avoid address collision.

— When such a collision risk is anticipated with the MAC address of one or more EPP STAs affiliated with EPP non-AP MLDs, the AP shall send the OTA MAC Collision Notification frame to the colliding EPP STAs affiliated with the EPP non-AP MLDs.

In general, the operation is as follows. If the collision is expected to occur m epochs (where m is less than or equal to 255) after the current epoch n (colliding epoch number $c = n + m$), then the EPP AP MLD sends an OTA Collision Warning element to the EPP non-AP MLD with the value of the Colliding Epoch field in the Collision Warning element equal to m, the Collision Action field set to 0, indicating the collision risk, and the Non-AP MLD Specific Collision Epoch Offset field set to q, where q is the epoch count that the non-AP MLD is requested to skip. The EPP AP MLD is therefore requesting that for epoch, the EPP AP MLD uses the EPP non-AP MLD CPE MHA parameters that the EPP non-AP MLD had planned to use for epoch $c + q$.

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Then, in the subsequent epoch *c*+1, the EPP non-AP MLD is expected to use the EPP non-AP MLD CPE MHA parameters that the EPP non-AP MLD had planned to use for epoch $c + q + 1$, unless the EPP AP MLD also signals a collision notification for that epoch. This computation occurs even if the EPP AP MLD has set a maximum number of epochs s for the group. if the EPP non-AP MLD has skipped *q* epochs at the current epoch *r*, the EPP non-AP MLD uses the CPE MHA parameters planned for epoch $r + q$, even if $r + q > s$. A non-AP MLD that received an OTA MAC Collision Notification frame shall respond with an OTA MAC Collision Response frame with the Collision Action field set to either 1, accepting the EPP AP MLD proposed remediation, thus applying the offset requested by the EPP AP MLD, or 2, rejecting the EPP AP MLD proposed remediation, thus using the EPP non-AP MLD CPE MHA parameters that the EPP non-AP MLD had planned to use for that epoch before receiving the EPP AP MLD OTA MAC Collision Notification frame. The AP may not accept traffic from, or forward traffic to, an EPP STA affiliated with the non-AP MLD that rejected the proposed remediation for the affected link, during the epoch when the collision occurs. Alternatively, the AP may disassociate an EPP non-AP MLD that rejected the proposed remediation.

NOTE 1—A non-AP MLD might decline to apply the requested offset for procedural reasons, e.g., the inability to skip epoch CPE MHA parameter sequences, or internal privacy configuration or policy reasons.

NOTE 2— Detection and remediation of a BPE AP MLD BSSID collision with MAC addresses other than the BPE AP MLD associated non-AP MLDs is outside the scope of this standard.

NOTE 3— The per-EPP-epoch AID values assigned by the EPP AP MLD to the EPP non-AP MLD as described in 10.71.7 (Frame anonymization and AID) are not impacted by OTA MAC address collision avoidance, that is, if an AID value is assigned for EPP epoch number *n*, then that value is used in EPP epoch number n even if the OTA MAC address collision avoidance instructs the EPP non-AP MLD to use the CPE MHA parameters that the EPP non-AP MLD planned to use for a subsequent EPP epoch.

10.71.3 Establishing CPE MAC header anonymization parameter sets

This subclause describes how an AP MLD and associated non-AP MLD establish the CPE MHA parameter set for each EPP epoch for the non-AP MLD. The creation of the BPE MHA parameter sets is described in 10.71.4 (Establishing BPE MAC header anonymization parameter sets).

The non-AP MLD and AP MLD establish the EPP epochs used for frame anonymization as described in 10.71.2 (EPP epoch operation).

For each EPP epoch of an EPP group that a non-AP MLD is assigned the non-AP MLD and the AP MLD shall generate a CPE_MHA_block as:

$$\text{CPE_MHA_block} = \text{KDF-}\textit{Hash-Length}(\text{KDK}, \text{“CPE_MHA_block”}, \text{Seed} \,||\, \textit{AP MLD MAC} \,||\, (((n + q) \times \text{EpochInterval}) \bmod 2^{64})$$

where

CPE_MHA_block	is the block of bits that is partitioned into the sets of all possible values for each CPE MHA parameter
KDF-*Hash-Length*	is the key derivation function as defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm identified by the AKM suite selector (see Table 9-190 (AKM suite selectors))
KDK	is the key derivation key
n	is the current number of the EPP epoch in the EPP epoch sequence as defined in 10.71.2.4 (EPP Epoch Start Time Computation)
Length	is the total number of bits to derive. A total of 1728 bits are derived for a CPE_MHA_block
q	is the value of the latest exchanged non-AP MLD Specific Collision Epoch Offset field if received and if *n* is greater or equal to colliding epoch number *c* (see 10.71.2.5 (OTA MAC address collision avoidance)); otherwise, *q* equals 0
Seed	is the value of the Group Epoch Seed field of the received EPP Epoch Settings

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	field
AP MLD MAC	is the MLD MAC address of the CPE AP.
EpochInterval	is the value in TU corresponding to the Epoch Interval field of the EPP Epoch Settings field.

The non-AP MLD and the AP MLD shall extract the CPE MHA parameters from the CPE_MHA_block as shown in the following tables:

— Table 10-40c (Extracting EPP_PN_offset values from the CPE_MHA_block).

— Table 10-40d (Extracting EPP_STA_address values from the CPE_MHA_block).

— Table 10-40e (Extracting EPP_SN_offset values for SNS1 and SNS 10 from the CPE_MHA_block).

— Table 10-40f (Extracting EPP_SN_offset values for SNS3 from the CPE_MHA_block).

— Table 10-40g (Extracting EPP_SN_offset values for SNS9 from the CPE_MHA_block).

— Table 10-40h (Extracting EPP_SN_offset values for SNS12 from the CPE_MHA_block).

Table 10-40c—Extracting EPP_PN_offset values from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Value
0:47	EPP_PN_offset for frames transmitted by non-AP MLD
48:95	EPP_PN_offset for frames transmitted by AP MLD

Table 10-40d—Extracting EPP_STA_address values from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:45]	Sub-block Bits [46:47]
96:143	EPP_STA_address [0:45] for link ID 0	Not used
144:191	EPP_STA_address [0:45] for link ID 1	Not used
192:239	EPP_STA_address [0:45] for link ID 2	Not used
240:287	EPP_STA_address [0:45] for link ID 3	Not used
288:335	EPP_STA_address [0:45] for link ID 4	Not used
336:383	EPP_STA_address [0:45] for link ID 5	Not used
384:431	EPP_STA_address [0:45] for link ID 6	Not used
432:479	EPP_STA_address [0:45] for link ID 7	Not used
480:527	EPP_STA_address [0:45] for link ID 8	Not used
528:575	EPP_STA_address [0:45] for link ID 9	Not used
576:623	EPP_STA_address [0:45] for link ID 10	Not used
624:671	EPP_STA_address [0:45] for link ID 11	Not used
672:719	EPP_STA_address [0:45] for link ID 12	Not used
720:767	EPP_STA_address [0:45] for link ID 13	Not used
768:815	EPP_STA_address [0:45] for link ID 14	Not used

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NOTE—Only 46 bits of each EPP_STA_address are extracted from the CPE_MHA_block. The generation of the full 48-bit EPP_STA_address at a transmitter is defined in 10.71.5.4 (Addressing). The generation of the full 48-bit EPP_STA_address at a receiver is defined in 10.71.6.1.1 (General).

Table 10-40e—Extracting EPP_SN_offset values for SNS1 and SNS 10 from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:11]	Sub-block Bits [12:23]	Sub-block Bits [24:35]	Sub-block Bits [36:47]
816:863	EPP_SN_offset value for SNS1 in frames transmitted by non-AP MLD	Not used	EPP_SN_offset value for SNS10 in frames transmitted by non-AP MLD	EPP_SN_offset value for SNS10 in frames transmitted by AP MLD

Table 10-40f—Extracting EPP_SN_offset values for SNS3 from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:11]	Sub-block Bits [12:23]	Sub-block Bits [24:35]	Sub-block Bits [36:47]
EPP_SN_offset values for SNS3 for frames transmitted by the non-AP MLD				
864:911	Value for TID 0	Value for TID 1	Value for TID 2	Value for TID3
912:959	Value for TID 4	Value for TID 5	Value for TID 6	Value for TID 7
960:1007	Value for TID 8	Value for TID 9	Value for TID 10	Value for TID 11
1008:1055	Value for TID 12	Value for TID 13	Value for TID 14	Value for TID 15
EPP_SN_offset values for SNS3 for frames transmitted by the AP MLD				
1056:1103	Value for TID 0	Value for TID 1	Value for TID 2	Value for TID3
1104:1151	Value for TID 4	Value for TID 5	Value for TID 6	Value for TID 7
1152:1199	Value for TID 8	Value for TID 9	Value for TID 10	Value for TID 11
1200:1247	Value for TID 12	Value for TID 13	Value for TID 14	Value for TID 15

Table 10-40g—Extracting EPP_SN_offset values for SNS9 from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:11]	Sub-block Bits [12:23]	Sub-block Bits [24:35]	Sub-block Bits [36:47]
EPP_SN_offset values for SNS9 for frames transmitted by the non-AP MLD				
1248:1295	Value for TID 0	Value for TID 1	Value for TID 2	Value for TID3
1296:1343	Value for TID 4	Value for TID 5	Value for TID 6	Value for TID 7
1344:1391	Value for TID 8	Value for TID 9	Value for TID 10	Value for TID 11
1392:1439	Value for TID 12	Value for TID 13	Value for TID 14	Value for TID 15

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Table 10-40g—Extracting EPP_SN_offset values for SNS9 from the CPE_MHA_block *(con-*

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:11]	Sub-block Bits [12:23]	Sub-block Bits [24:35]	Sub-block Bits [36:47]
EPP_SN_offset values for SNS9 for frames transmitted by the AP MLD				
1440:1487	Value for TID 0	Value for TID 1	Value for TID 2	Value for TID3
1488:1535	Value for TID 4	Value for TID 5	Value for TID 6	Value for TID 7
1536:1583	Value for TID 8	Value for TID 9	Value for TID 10	Value for TID 11
1584:1631	Value for TID 12	Value for TID 13	Value for TID 14	Value for TID 15

Table 10-40h—Extracting EPP_SN_offset values for SNS12 from the CPE_MHA_block

48-bit sub-block of the CPE_MHA_block	Sub-block Bits [0:11]		Sub-block Bits [12:23]		Sub-block Bits [24:35]		Sub-block Bits [36:47]	
	0:9	10:11	12:21	22:23	24:33	34:35	36:45	46:47
EPP_SN_offset values for SNS12 for frames transmitted by the non-AP MLD								
1632:1679	Value for ACI 0	Not used	Value for ACI 1	Not used	Value for ACI 2	Not used	Value for ACI 3	Not used
EPP_SN_offset values for SNS12 for frames transmitted by the AP MLD								
1680:1727	Value for ACI 0	Not used	Value for ACI 1	Not used	Value for ACI 2	Not used	Value for ACI 3	Not used

10.71.4 Establishing BPE MAC header anonymization parameter sets

This subclause describes how a BPE AP MLD and associated BPE non-AP MLD establish the BPE MHA parameter set for each EPP epoch for the BPE AP MLD and the BPE non-AP MLD. The BPE non-AP MLD and BPE AP MLD establish the EPP epochs used for frame anonymization as described in 10.71.8 (BSS privacy enhancements operations).

All associated BPE non-AP MLDs and the BPE AP MLD shall generate BPE MHA parameters for a given EPP epoch by computing a single pseudorandom BPE_MHA_block that is partitioned into a set of EPP BP frame anonymization parameters according to the following tables.

For a given EPP epoch, the BPE_MHA_block shall be generated as:

BPE_MHA_block = KDF-*Hash-Length* (PGTK, "BPE_MHA_block", Seed || AP MLD MAC || ((n × EpochInterval) mod 2^{64})),

where

KDF-*Hash-Length* is the key derivation function as defined in 12.7.1.6.2 (Key derivation function (KDF)) using the hash algorithm identified by the AKM suite selector (see Table 9-190 (AKM suite selectors))

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PGTK	is the privacy group transient key
n	is the current number of the EPP epoch in the EPP epoch sequence as defined in 10.71.2.4 (EPP Epoch Start Time Computation)
Length	is the total number of bits to derive. A total of 960 bits are derived for a BPE_MHA_block.
Seed	is the value of the Group Epoch Seed field received by the non-AP MLD or transmitted by the AP MLD.
AP MLD MAC	is the MLD MAC address of the BPE AP.

The BPE offsets for the Group PN, SNS1 DL, SNS11 DL and Timestamp fields together with the anonymized BPE AP link addresses are created from the BPE_MHA_block. The offsets and the BPE AP link addresses have static assignments within the BPE_MHA_block as shown in Table 10-40i (Extracting EPP_Group_PN_offset values from the BPE_MHA_block), Table 10-40j (Extracting EPP_AP_address values from the BPE_MHA_block), Table 10-40k (Extracting EPP_Group_Address_Anonymization_Offset from the BPE_MHA_Block), Table 10-40l (Extracting EPP_SN_offset values for SN1 and SNS 11 from the BPE_MHA_block), and Table 10-40m (Extracting Timestamp offset and EPP epoch start time offset from the BPE_MHA_block).

Table 10-40i—Extracting EPP_Group_PN_offset values from the BPE_MHA_block

48-bit sub-block of the BPE_MHA_block	Value
0:47	EPP_Group_PN_offset

Table 10-40j—Extracting EPP_AP_address values from the BPE_MHA_block

48-bit sub-block of the BPE_MHA_block	Sub-block Bits [0:45]	Sub-block Bits [46:47]
48:95	EPP_AP_address [0:45] for link ID 0	Not used
96:143	EPP_AP_address [0:45] for link ID 1	Not used
144:191	EPP_AP_address [0:45] for link ID 2	Not used
192:239	EPP_AP_address [0:45] for link ID 3	Not used
240:287	EPP_AP_address [0:45] for link ID 4	Not used
288:335	EPP_AP_address [0:45] for link ID 5	Not used
336:383	EPP_AP_address [0:45] for link ID 6	Not used
384:431	EPP_AP_address [0:45] for link ID 7	Not used
432:479	EPP_AP_address [0:45] for link ID 8	Not used
480:527	EPP_AP_address [0:45] for link ID 9	Not used
528:575	EPP_AP_address [0:45] for link ID 10	Not used
576:623	EPP_AP_address [0:45] for link ID 11	Not used
624:671	EPP_AP_address [0:45] for link ID 12	Not used
672:719	EPP_AP_address [0:45] for link ID 13	Not used
720:767	EPP_AP_address [0:45] for link ID 14	Not used

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NOTE—Only 46 bits of each EPP_AP_address are extracted from the BPE_MHA_block. The generation of the full 48-bit EPP_AP_address is defined in 10.71.5.4 (Addressing).The generation of the full 48-bit EPP_AP_address at a receiver is defined in 10.71.6.1.1 (General).

Table 10-40k—Extracting EPP_Group_Address_Anonymization_Offset from the BPE_MHA_Block

48-bit sub-block of the BPE_MHA_block	Sub-block Bits [0:45]	Sub-block Bits [46:47]
768:815	EPP_Group_Anonymization_Offset	Not used

Table 10-40l—Extracting EPP_SN_offset values for SN1 and SNS 11 from the BPE_MHA_block

48-bit sub-block of the BPE_MHA_block	Sub-block Bits [0:11]	Sub-block Bits [12:23]	Sub-block Bits [24:35]	Sub-block Bits [36:47]
	EPP_SN_offset values for SNS1		EPP_SN_offset values for SNS11	
816:863	Not used	Value for frames transmitted by BPE AP MLD	Not used	Value for frames transmitted by BPE AP MLD

Table 10-40m—Extracting Timestamp offset and EPP epoch start time offset from the BPE_MHA_block

48-bit sub-block of the BPE_MHA_block	Sub-block Bits [0:63]	Sub-block Bits [64:79]	Sub-block Bits [80:95]
864:959	Timestamp offset	EpochStartTimeOffset(n)	Not used

10.71.5 MAC header anonymization and transmitting functions

10.71.5.1 MAC header anonymization parameter set selection

When CPE FA is activated, then:

— when sending an individually addressed frame for the first time, the transmitting MLD shall use the CPE non-AP MLD's CPE MHA parameter set for the current EPP epoch.

NOTE 1—Completion of frame exchange sequences, TXOPs and retransmissions are addressed in 10.71.2.3 (EPP epoch transition operations).

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When BPE FA is activated, then:

— when sending an individually addressed frame for the first time, the transmitting MLD shall use the BPE non-AP MLD's CPE MHA parameter set and BPE AP MLD's BPE MHA parameter set for the current EPP epoch.

— when sending a group addressed frame, the BPE AP MLD shall use the BPE AP MLD's BPE MHA parameter set for the current EPP epoch.

10.71.5.2 Sequence number anonymization

NOTE 1—The sequence number spaces are defined in Table 10-5 (Transmitter sequence number spaces).

If the MAC header of the frame includes a Sequence Control field using:

— sequence number space SNS1 when the frame is transmitted by a non-AP MLD, or

— sequence number space SNS3 (Time Priority Management), or

— sequence number space SNS9 (MLD Individually addressed QoS Data frame), or

— sequence number space SNS10 (MLD Individually addressed Management frame),

then the transmitter shall compute an over-the-air SN (OSN) value from the sequence number SN assigned to the MPDU as follows:

$$OSN = (SN + EPP_SN_offset) \bmod 2^{12},$$

where EPP_SN_offset is selected from the applicable CPE MAC header anonymization parameter set for the frame, according to the following mechanisms specific to the sequence number space:

— In the case of SNS1 when the frame is transmitted by a non-AP MLD, the transmitter shall select the single defined EPP_SN_offset value for SNS1.

— In the case of SNS3, the transmitter shall select an EPP_SN_offset value for SNS3 according to the transmitting MLD (non-AP MLD or AP MLD) and the TID.

— In the case of SNS9, the transmitter shall select an EPP_SN_offset value for SNS9 according to the transmitting MLD (non-AP MLD or AP MLD) and the TID.

— In the case of SNS10, the transmitter shall select an EPP_SN_offset value for SNS10 according to the transmitting MLD (non-AP MLD or AP MLD).

If the MAC header of the frame includes a Sequence Control field using sequence number space SNS12 (IQMF), then the transmitter shall compute an OSN value from the sequence number SN assigned to the MPDU (defined in Figure-9-9 (Sequence Number field format in QMFs)) as follows:

$$OSN[10:11] = SN[10:11], \text{ and}$$

$$OSN[0:9] = (SN[0:9] + EPP_SN_offset) \bmod 2^{10},$$

where EPP_SN_offset is an EPP_SN_offset value for SNS12 selected from the applicable CPE MAC header anonymization parameter set for the frame according to the transmitting MLD (non-AP MLD or AP MLD) and the Access Class Index field (SN[10:11]).

If the AP MLD has BPE FA mechanisms enabled, and if the MAC header of a frame transmitted by the AP MLD includes a Sequence Control field using:

— sequence number space SNS1 or

— sequence number space SNS11 (Group addressed data),

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then the transmitter shall compute an over-the-air SN (OSN) value from the sequence number SN assigned to the MPDU as follows:

$$OSN = (SN + EPP_SN_offset) \bmod 2^{12},$$

where EPP_SN_offset is selected, from the applicable BPE MAC header anonymization parameter set for the frame, according to mechanisms specific to the sequence number space.

— In the case of SNS1, the transmitter shall select the single EPP_SN_offset value for SNS1.

— In the case of SNS11, the transmitter shall select an EPP_SN_offset value for SNS11 according to the transmitting MLD (non-AP MLD or AP MLD).

The transmitter shall transmit frames over the air using the OSN value in the Sequence Number field of the Sequence Control field (see 9.2.4.4 (Sequence Control field)).

10.71.5.3 Packet number anonymization

For encrypted individually addressed frames, the transmitter shall compute an over-the-air PN (OPN) value from the PN value in the CCMP header or GCMP header of the frame as follows:

$$OPN = (PN + EPP_PN_offset) \bmod 2^{48},$$

where EPP_PN_offset is selected from the applicable CPE MHA parameter set for the frame according to the transmitting MLD (non-AP MLD or AP MLD).

If the AP MLD has BPE FA mechanisms enabled, then for encrypted group addressed frames, the transmitter shall compute an over-the-air PN (OPN) value from the PN value in the CCMP header or GCMP header of the frame as follows:

$$OPN = (PN + EPP_Group_PN_offset) \bmod 2^{48},$$

using the EPP_Group_PN_offset value is the single EPP_Group_PN_offset value in the applicable BPE MHA parameter set for the frame.

The transmitter shall transmit frames over the air using the OPN value in fields PN0, PN1, PN2, PN3, PN4, PN5 of the CCMP header (see 12.5.2.2 (CCMP MPDU format)) or GCMP header (see 12.5.4.2 (GCMP MPDU format)).

10.71.5.4 Addressing

10.71.5.4.1 General

MLD addressing shall be applied per 35.3.2 (MLD addressing) with changes for CPE FA described in 10.71.5.4.2 (CPE FA addressing) and changes for BPE FA is defined in 10.71.5.4.3 (BPE FA addressing), using the following definitions for EPP_STA_address and EPP_AP_Address.

For both CPE FA and BPE FA, and EPP_STA_address of an affiliated STA on the link is the MAC address defined as follows:

— The Local/Global bit shall be set to value 1, local address.

— The Individual/Group bit shall be set to value 0, individual address.

— EPP_STA_address[0:45] shall be extracted from the CPE MAC header anonymization parameter set according to the link ID of the link on which the frame is transmitted.

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For BPE FA, an EPP_AP_address of an affiliated AP on the link on which a frame is transmitted is the MAC address defined as follows:

— The Local/Global bit shall be set to value 1, local address.

— The Individual/Group bit shall be set to value 0, individual address.

— EPP_AP_address[0:45] shall be extracted from the BPE MAC header anonymization parameter set according to the link ID of the link.

For BPE FA, the Address 3 field (if set to BSSID) is set to the link-specific EPP_AP_address value in all frames transmitted by the AP MLD or the non-AP MLD.

For BPE FA, an AP affiliated with an AP MLD shall set the Address 1 field value of a group addressed frame to:

— OTAGroupAddress = (group address + EPP_Group_Address_Anonymization_Offset) mod 2^{46}, if the group address is other than the broadcast address

— OTAGroupAddress = broadcast address, if the group address is the broadcast address

where group address is 46 bits of the group address excluding the local/global and individual/group bits and where EPP_Group_Address_Anonymization_Offset is the single EPP_Group_Address_Anonymization_Offset value extracted from the BPE MHA parameter set.

For BPE FA, an STA affiliated with a BPE non-AP MLD shall obtain the deanonymized group address from a received group address as:

— Group address = broadcast address, if the OTAGroupAddress is the broadcast address.

— Group address = (OTAGroupAddress – EPP_Group_Anonymization_Offset) mod 2^{46}, if the OTAGroupAddress is not the broadcast address.

where OTAGroupAddress is 46 bits of the received group address excluding the local/global and individual/group bits, and where EPP_Group Address_Anonymization_Offset is the single EPP_Group_Address_Anonymization_Offset value extracted from the BPE MHA parameter set.

10.71.5.4.2 CPE FA addressing

The Address 1 field of an individually addressed frame transmitted by an AP MLD shall be set to the EPP_STA_address value.

The Address 2 field of an individually addressed frame transmitted by a non-AP MLD shall be set to the EPP_STA_address value.

10.71.5.4.3 BPE FA addressing

The Address 1 field of an individually addressed frame transmitted by an AP MLD shall be set to the EPP_STA_address value.

The Address 2 field of an individually addressed frame transmitted by a non-AP MLD shall be set to the EPP_STA_address value.

The Address 2 field of a frame transmitted by an AP MLD shall be set to the EPP_AP_address value.

The Address 1 field of a frame transmitted by a non-AP AP MLD to an AP MLD shall be set to the EPP_AP_address value.

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10.71.5.5 Timestamp anonymization

For Privacy Beacon frames, the transmitter shall compute an over-the-air Timestamp (OTA_Timestamp) value from the Timestamp value of the frame as follows:

$$\text{OTA_Timestamp} = (\text{Timestamp} + \text{EPP_Timestamp_offset}) \bmod 2^{64},$$

where EPP_Timestamp_offset is the single EPP_Timestamp_offset value in the applicable BPE MHA parameter set.

The BPE AP shall transmit Privacy Beacon frames over the air with the Timestamp field set to the OTA_-Timestamp value (see 9.3.4.4 (Privacy Beacon frame format)).

NOTE—the sum Timestamp + EPP_Timestamp_offset may occasionally exceed 2^{64} and wrap. This event does not affect the BPE non-AP MLD, as it does not use OTSF, but the internal Timestamp value for its operations.

10.71.6 MAC header anonymization and receiving functions

10.71.6.1 Address filtering

10.71.6.1.1 General

Address filtering shall be applied per 10.2.8 (MAC data service) with changes for CPE FA described in 10.71.5.4.2 (CPE FA addressing), changes for BPE FA described in 10.71.5.4.3 (BPE FA addressing), and the addressing clarifications in 10.71.2.2.1 (General).

10.71.6.1.2 Address filtering for CPE FA

For a frame received on a given setup link of the non-AP MLD:

— During the margin period and the transition period of the EPP epoch of the non-AP MLD (see 10.71.2.1 (General) and 10.71.2.2 (EPP group operations)), the affiliated STA of the non-AP MLD and the affiliated AP of the AP MLD of the setup link shall perform address filtering using:

- the link-specific EPP_STA_address of the affiliated STA for the old EPP epoch (if any), and the nonanonymized address of the affiliated AP,
- the link-specific EPP_STA_address of the affiliated STA for the new EPP epoch, and the nonanonymized address of the affiliated AP, and
- the nonanonymized group address of the group addressed frame and the nonanonymized address of the affiliated AP.

— After this transition period and until the dot11EPPEpochStartTimeMargin period of the next EPP epoch of the non-AP MLD, the affiliated STA of the non-AP MLD and the affiliated AP of the AP MLD of the setup link shall perform address filtering using:

- the link-specific EPP_STA_address of the affiliated STA for the new EPP epoch (if any), and the nonanonymized address of the affiliated AP, and
- the nonanonymized group address of the group addressed frame and the nonanonymized address of the affiliated AP.

10.71.6.1.3 Address filtering for BPE FA

During the margin period and the transition period (see 10.71.2.1 (General) and 10.71.2.2 (EPP group operations)) from an old EPP epoch to a new EPP epoch of the BPE non-AP MLD, the affiliated STA of the BPE non-AP MLD and the affiliated AP of the BPE AP MLD (on a setup link of the BPE non-AP MLD) shall perform address filtering using:

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- the link-specific EPP_STA_address of the affiliated STA and link-specific EPP_AP_address of the affiliated AP for the old EPP epoch of the AP MLD,
- the broadcast address and the link-specific EPP_AP_address for the old EPP epoch of the AP MLD,
- the link-specific EPP_AP_address and deanonymized group address that is other than the broadcast address (obtained from the received group address) for the old EPP epoch of the AP MLD,
- the link-specific EPP_STA_MAC and link-specific EPP_AP_address for the new EPP epoch for the AP MLD,
- the broadcast address and the link-specific EPP_AP_address for the new EPP epoch of the AP MLD, and
- the link-specific EPP_AP_address and link-specific and deanonymized group address that is other than the broadcast address (obtained from the received group address) for the new EPP epoch of the AP MLD.

After this transition period, and until a dot11EPPEpochStartTimeMargin before the start of the transition period of the next EPP epoch, the affiliated STA of the BPE non-AP MLD and the affiliated AP of the BPE AP MLD (on a setup link of the BPE non-AP MLD) shall perform address filtering using:

- the link-specific EPP_STA_MAC and link-specific EPP_AP_address for the new EPP epoch of the AP MLD,
- the link-specific EPP_AP_address and the broadcast address, and
- and the link-specific EPP_AP_address and deanonymized group address that is other than the broadcast address for the new EPP epoch of the AP MLD.

10.71.6.1.4 MAC header anonymization parameter set selection

If an individually addressed frame is received by a non-AP MLD, then the non-AP MLD shall perform packet number deanonymization (10.71.6.3 (Packet number deanonymization)) and sequence number deanonymization (10.71.6.4 (Sequence number deanonymization)) using the CPE MHA parameter set containing the link-specific EPP_STA_address value matching the Address 1 field in the MAC header.

An AP MLD shall perform packet number deanonymization (10.71.6.3 (Packet number deanonymization)) and sequence number deanonymization (10.71.6.4 (Sequence number deanonymization)) using the applicable CPE MHA parameter set containing the link-specific EPP_STA_address value matching the Address 2 field in the MAC header.

The CPE MHA parameter set so identified is the applicable CPE MHA parameter set for the received frame.

If BPE FA is activated, then:

- If a group addressed frame is received by a BPE non-AP MLD, then the BPE non-AP MLD shall perform packet number deanonymization (10.71.6.3 (Packet number deanonymization)), sequence number deanonymization (10.71.6.4 (Sequence number deanonymization)), and group address deanonymization (10.71.6.1.1 (General)) using the applicable BPE MHA parameter set containing the link-specific EPP_AP_address value matching the Address 2 field in the MAC header of the group addressed frame.
- The BPE MHA parameter set so identified is the applicable BPE MHA parameter set for the received frame.
- If a Privacy Beacon is received by a BPE non-AP MLD, then the BPE non-AP MLD shall perform timestamp deanonymization (10.71.6.5 (Timestamp deanonymization)) using the BPE MHA parameter set containing the link-specific EPP_AP_address value matching the Address 2 field in the MAC header of the Privacy Beacon.

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10.71.6.2 Block ack scoreboarding

Block ack scoreboarding shall be applied per 35.3.8 (Block ack procedures in MLO), with the following clarifications:

— The values in the Address 1 field and Address 2 field of the (per-link) Block Ack frame shall be the values in the Address 2 field and Address 1 field (respectively) of MPDUs of the corresponding A-MPDU.

— The (per-link) Block Ack frame shall report the OSN values received in the Sequence Number field of the MPDU header within the MPDUs of the corresponding A-MPDU (rather than reporting the SN values recovered after SN deanonymization).

10.71.6.3 Packet number deanonymization

NOTE—The applicable CPE MHA parameter set and applicable BPE MHA parameter set (when the AP MLD enables BPE FA mechanisms) are determined in 10.71.6.1.4 (MAC header anonymization parameter set selection).

For encrypted individually addressed frames, the receiver shall obtain the original PN value (assigned by the transmitter) from the OPN value encoded in the fields PN0, PN1, PN2, PN3, PN4, PN5 of the CCMP header or GCMP header as follows:

$$PN = (OPN - EPP_PN_offset) \bmod 2^{48},$$

where EPP_PN_offset is selected from the applicable CPE MHA parameter set for the frame according to the transmitting MLD (non-AP MLD or AP MLD).

If the AP MLD has BPE FA mechanisms enabled, then for encrypted group addressed frames, the receiver shall obtain the PN value from the OPN value encoded in the fields PN0, PN1, PN2, PN3, PN4, PN5 of the CCMP header or GCMP header as follows:

$$PN = (OPN - EPP_Group_PN_offset) \bmod 2^{48},$$

where EPP_Group_PN_offset is the single EPP_Group_PN_offset in the applicable BPE MHA parameter set selected for the frame.

The PN value (nominally the PN value assigned by the transmitter) shall replace the OPN value in subsequent processing of the frame in the receiving MLD.

10.71.6.4 Sequence number deanonymization

NOTE 1—The sequence number spaces are defined in Table 10-5 (Transmitter sequence number spaces).

NOTE 2—The applicable CPE MHA parameter set is determined in 10.71.6.1.4 (MAC header anonymization parameter set selection). If the AP MLD has BPE FA mechanisms enabled, then the applicable BPE MHA parameter set is determined in 10.71.6.1.4 (MAC header anonymization parameter set selection).

If the MAC header of the frame includes a Sequence Control field using:

— sequence number space SNS1 when the frame is received by an AP MLD, or

— sequence number space SNS3 (Time Priority Management), or

— sequence number space SNS9 (MLD Individually addressed QoS Data frame), or

— sequence number space SNS10 (MLD Individually addressed Management frame),

then the receiver shall compute the SN value from the over-the-air value in the sequence number field, OSN, as follows:

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$$SN = (OSN - EPP_SN_offset) \bmod 2^{12},$$

where EPP_SN_offset is selected from the applicable CPE MAC header anonymization parameter set for the frame, according to mechanisms specific to the sequence number space.

- In the case of SNS1 when the frame is received by an AP MLD, the receiver shall select the single defined EPP_SN_offset value for SNS1.
- In the case of SNS3 and SNS9, the receiver shall select an EPP_SN_offset value according to the transmitting MLD (non-AP MLD or AP MLD) and the TID.
- In the case of SNS10, the receiver shall select an EPP_SN_offset value from Table 10-40e (Extracting EPP_SN_offset values for SNS1 and SNS 10 from the CPE_MHA_block) according to the transmitting MLD (non-AP MLD or AP MLD).

If the MAC header of the frames includes a Sequence Control field using sequence number space SNS12 (IQMF), then the receiver shall compute the original SN value from the over-the-air value in the Sequence Number field, OSN, (defined in Figure 9-9 (Sequence Number field format in QMFs)) as follows:

$$SN[10:11] = OSN[10:11], \text{ and}$$

$$SN[0:9] = (OSN[0:9] - EPP_SN_offset) \bmod 2^{10},$$

where EPP_SN_offset is an EPP_SN_offset value for SNS12 selected from the applicable CPE MAC header anonymization parameter set for the frame according to the transmitting MLD (non-AP MLD or AP MLD) and the Access Class Index field (SN[10:11]).

If the AP MLD has BPE FA mechanisms enabled, and if the MAC header of a frame received by a non-AP MLD includes a Sequence Control field using:

- sequence number space SNS1 or
- sequence number space SNS11 (Group addressed data),

then the receiver shall compute the SN value from the over-the-air value in the sequence number field, OSN, as follows:

$$SN = (OSN - EPP_SN_offset) \bmod 2^{12},$$

where EPP_SN_offset is selected from the applicable CPE MAC header anonymization parameter set for the frame, according to mechanisms specific to the sequence number space.

- In the case of SNS1, the receiver shall select the single EPP_SN_offset value for SNS1.
- In the case of SNS11, the receiver shall select an EPP_SN_offset value for SNS11 according to the transmitting MLD (non-AP MLD or AP MLD).

The SN value (nominally the SN value assigned by the transmitter) shall replace the OSN value in subsequent processing of the frame in the receiving MLD.

NOTE 3—If BPE FA mechanisms are not enabled, and a frame received by a non-AP MLD includes a Sequence Control field using sequence number space SNS1, then the sequence number is not deanonymized in that frame.

10.71.6.5 Timestamp deanonymization

NOTE—If the AP MLD has BPE FA mechanisms enabled, then the applicable BPE MHA parameter set is determined in 10.71.6.1.4 (MAC header anonymization parameter set selection).

For Privacy Beacon frames, the receiver shall recover the original Timestamp value (assigned by the transmitter) from the OTA_Timestamp value encoded in the Timestamp fields as follows:

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$$\text{Timestamp} = (\text{OTA_Timestamp} - \text{EPP_Timestamp_offset}) \bmod 2^{64},$$

where EPP_Timestamp_offset is the single EPP_Timestamp_offset value in the applicable BPE MHA parameter set selected for the frame.

The Timestamp value (nominally the Timestamp value assigned by the transmitter) shall replace the OTA_-Timestamp value in subsequent processing of the Privacy Beacon frame in the receiving MLD.

10.71.7 Frame anonymization and AID

Upon advertisement of EPP groups (as defined in 10.71.2.2 (EPP group operations)), the CPE AP MLD shall include an AID Storage Size field in the EPP Epoch Settings field it transmits.

A CPE non-AP MLD shall include an AID Storage Size field in the EPP Epoch Settings field of each (Re)Association Request frame and EPP Epoch Request frame it transmits.

If the AID Storage Size field indicated by the CPE non-AP MLD is lower than that indicated by the CPE AP MLD, then the CPE AP shall transmit an EPP Response frame with FAILURE_AID_STOR-AGE_TOO_SMALL in the Status field to indicate that the non-AP MLD is not allowed to join in the EPP group.

A CPE AP MLD generates a list of AIDs that an associated CPE non-AP MLD shall use in the subsequent epochs. While generating the list of AIDs, the AP MLD shall apply the rules in 35.3.5.1 (ML (re)setup pro-cedure) and ensure that, at any point of time, each non-AP MLD is assigned an unique AID value.

The AID list size (indicated by the Number of Epochs field in Figure 9-1074du (AID List Value field for-mat) shall be smaller than or equal to the value of AID Storage Size provided by the CPE non-AP MLD.

The CPE AP MLD shall include the AID List element in the (Re)Association Response frame.

The CPE non-AP MLD shall use the AID field in the (Re)Association Response for any communication in the EPP epoch where the (Re)Association Response is received.

For subsequent AID assignments, the CPE AP MLD shall transmit an AID Assignment Request frame to the non-AP MLD with the AID List element that contains the AID values. If the AID assignment operation has been successful, the CPE non-AP MLD and the CPE AP shall use the AIDs in the AID List element for any communications, starting from the epoch indicated in the Start Epoch field, for the number of epochs indi-cated in the Number Of Epochs field.

A receiving CPE non-AP MLD, that has not been able to store every AID of the AID list shall respond with an AID Assignment Response frame. The Status Code field in the AID Assignment Response frame shall be set according to the result of the AID assignment operation as follows:

— Status Code SUCCESS_AID_LIST_PARTIALLY_STORED indicates that the AID assignment operation has been successful but only the initial part of the AID list has been stored by the CPE non-AP MLD. In this case, the CPE non-AP MLD shall include the Number Of Stored AIDs field and set it to the number of the stored AID values.

— Status Code FAILURE_AID_LIST_NOT_STORED indicates that the AID assignment operation has failed. The non-AP MLD has not stored any AID in the AID list.

A CPE non-AP MLD that successfully stores the AID list may not respond with an AID Assignment Response frame.

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Before the end of all the epochs indicated in the Number of Epochs field, the CPE AP MLD shall generate a new list of AID values and send a new AID Assignment Request frame with the new AID List element to the associated CPE non-AP MLD.

The CPE AP MLD may generate a new AID list and send a new AID Assignment Request frame with the new AID List element to the associated CPE non-AP MLD at any time.

If the Start Epoch field of the AID List element indicates an epoch that has an assigned AID, the AIDs in the AID list shall override the previously assigned AIDs beginning from the epoch number value indicated by the received Start Epoch field of the AID List element. The AID value in use in the current EPP Epoch shall not be overridden.

Upon AID assignment failure, the CPE AP MLD may repeat the AID assignment operation. Upon subsequent failures, the CPE AP MLD may request the CPE non-AP MLD to join a different EPP group.

When the CPE non-AP MLD has no available AID for the current or the next EPP epoch, it should request a new AID list to stay associated with the current CPE AP MLD by transmitting an AID Assignment Response frame using the Status Code NO_ASSIGNED_AID.

After the CPE non-AP MLD has transmitted the AID Assignment Response frame, the non-AP MLD shall set at least one STA to active mode and stay awake until it receives the AID list.

The CPE AP MLD shall assign new AIDs for the coming EPP epochs by transmitting them in an AID Assignment frame. The CPE AP MLD shall not send any data frames to the CPE non-AP MLD not having any assigned AID in the current EPP Epoch.

A CPE non-AP MLD that does not have an assigned AID for the current EPP epoch shall not transmit any frames to its associated CPE AP MLD except the following:

— AID Assignment Response frames
— Disassociation frames
— Deauthentication frames
— Control frames

The CPE AP MLD should disassociate with the Status Code NO_ASSIGNED_AID, the CPE non-AP MLD that has no AID assigned for the current EPP epoch and that transmits a protected individually addressed frame of a type not specified in the previous list.

10.71.8 BSS privacy enhancements operations

10.71.8.1 General

BSS Privacy Enhancement (BPE) operations protect the privacy of BPE AP MLDs and associated BPE non-AP MLDs. The BPE AP MLD privacy is achieved by removing the need to send BPE AP MLD discovery information, e.g., SSID, capability or operation elements, over the air in the clear.

APs affiliated with a BPE AP MLD transmit Privacy Beacon frames 9.3.4.4 (Privacy Beacon frame format) instead of Beacon frames 9.3.3.2 (Beacon frame format). A BPE AP MLD is discoverable only by non-AP MLDs that have the preshared identity key of the BPE AP MLD as described in 10.71.8.2 (BPE AP MLD discovery).

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A BPE AP MLD does not use beacon protection with a BIGTK or BIP with IGTK. A BPE non-AP MLD obtains from the protected (Re)Association Response frame a GTK for each affiliated BPE AP with a setup link and the AP MLD specific PGTK, and Identity Key.

NOTE 1—An AP affiliated with a BPE AP MLD does not have BIGTK, because they do not transmit Beacon frames. Instead, they transmit Privacy Beacon frames (see 10.71.8.3 (BPE AP MLD beaconing)).

NOTE 2—An AP affiliated with a BPE AP MLD does not have IGTK, because all group addressed management frames are encrypted by using the group cipher suite indicated for the BSS. IGTK integrity protects group addressed management frames. Integrity protected group addressed management frames can be intercepted and read by eavesdroppers, which would reduce privacy of the BPE AP MLD.

A BPE AP MLD defines a single EPP group. All non-AP MLDs associated to the BPE AP MLD are part of that single EPP group as long as the BPE non-AP MLD is associated.

The BPE AP MLD shall place any successfully associating non-AP MLD into the single EPP group. A non-AP MLD that finds that the parameters of the single EPP group are incompatible with its own (for example the Minimum Epoch Pacing values) may disassociate.

The BPE AP MLD might at any time modify the parameters of the single EPP group. When doing so, the BPE AP MLD shall inform the associated non-AP MLDs by increasing the value of the BPCC at least one epoch before the time of the intended change.

At the beginning of each epoch, the BPE non-AP STA addresses and SN spaces and PNs of the individual frames are anonymized in all links by applying CPE MHA, using parameters described in 10.71.3 (Establishing CPE MAC header anonymization parameter sets). The BPE MLD affiliated AP addresses, the Timestamp field of the Privacy Beacons and the group frames are anonymized by applying BPE MHA, using parameters described in 10.71.4 (Establishing BPE MAC header anonymization parameter sets). The AIDs used by the associated BPE non-AP MLDs are assigned by the AP MLD, see 10.71.7 (Frame anonymization and AID).

10.71.8.2 BPE AP MLD discovery

Each BPE AP affiliated with the BPE AP MLD shall transmit Privacy Beacon frames. A BPE STA may discover a BPE AP through received Privacy Beacon frames as described in 10.71.8.3 (BPE AP MLD beaconing).

A BPE AP shall not respond to Probe Request frames and a BPE AP shall not transmit Probe Response frames. A BPE MLD shall not transmit unprotected GAS frames.

A BPE non-AP MLD may transmit unprotected Privacy Beacon Solicit Request frames (see 9.6.7.64 (Privacy Beacon Solicit Request frame format)) to solicit unprotected, broadcast addressed, Privacy Beacon frames from BPE APs. A BPE non-AP STA shall use a random transmitter address when it transmits Privacy Beacon Solicit Request frames. This random transmitter address shall be changed at least for each TXOP with Privacy Beacon Solicit Request frames transmission.

A BPE non-AP STA may detect from a received Privacy Beacon frames whether the transmitting AP MLD identity key is configured to the STA, as defined in 10.71.8.3 (BPE AP MLD beaconing). If the BPE STA has the configured identity key, it may authenticate and associate with the BPE AP MLD. On reception of a Privacy Beacon Solicit Request frame, a BPE AP should schedule an unprotected Privacy Beacon frame for transmission at least within a dot11PrivacyBeaconResponseTime.

NOTE 1—An unprotected Privacy Beacon frame is used only by unassociated STAs affiliated with a BPE non-AP MLD to check whether they have AP identity key and can associate with the AP, i.e., an unprotected Privacy Beacon frame has no frame body as described in Table 9-76a (Privacy Beacon frame body).

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The BPE AP should limit the unprotected Privacy Beacon frame transmissions to at most 1 response per 5 ms per AP by adding random delay to its responses or by responding to multiple requests with a single Privacy Beacon frame transmission. A BPE STA should not transmit a Privacy Beacon Solicit Request frame on a channel if the STA has already transmitted a request on the channel less than dot11PrivacyBeaconResponseTime ago.

NOTE 2—If the medium is congested, the transmission of a Privacy Beacon frame might take longer than the dot11PrivacyBeaconResponseTime.

A BPE STA may initiate authentication and association with a BPE AP by sending frames with receiver address set to the Address 2 of the Privacy Beacon frame.

An ESS may contain non-EPP APs, non-EPP AP MLDs, CPE AP MLDs and BPE AP MLDs. An AP MLD may recommend an associated non-AP MLD to operate with a BPE AP MLD as described in 10.71.8.5 (Steering to a BPE AP MLD).

10.71.8.3 BPE AP MLD beaconing

A BPE AP MLD shall indicate the status of buffered frames in the TIM element of a Privacy Beacon frame as specified in 35.3.12.4 (Traffic indications). The BPE non-AP MLD power management rules are specified in 35.3.12 (ML power management).

The frame body of a Privacy Beacon frame is encrypted by the GTK, and it can be decrypted only by the BPE non-AP MLDs associated with the BPE AP MLD of the transmitting BPE AP. If the BPE AP has no associated STAs, then the BPE AP should transmit only unprotected Privacy Beacon frames that have no frame body; otherwise the BPE AP may transmit unprotected and protected Privacy Beacon frames.

The MAC header of the Privacy Beacon frame contains a Timestamp field that is anonymized as described in 10.71.5.5 (Timestamp anonymization). A receiver deanonymizes the Timestamp field as described in 10.71.6.5 (Timestamp deanonymization).

NOTE 1—A BPE non-AP MLD exiting the doze state can use the MAC address of the AP affiliated with the AP MLD, legible in the privacy beacon header, to deduce the value of the current epoch and know which key to use to deanonymize the privacy beacon Timestamp field and realign its internal clock with that of the AP MLD.

A BPE non-AP MLD may discover a BPE AP MLD by using the configured identity key of that BPE AP MLD. The identity key may be configured on the STA from the received protected (Re)association Response frame or by using mechanisms that are out of the scope of the specification. When an identity key is configured to the BPE non-AP MLD, the parameters required to authenticate and associate with the identified AP MLD are configured at the same time. The configured parameters include at least RSNE, RSNXE, MLD address of the AP MLD, Supported Rates and BSS Membership Selectors and Extended Supported Rates and BSS Membership Selectors. A BPE Non-AP MLD may receive the identity key and the BPE AP MLD parameters in the protected (Re)Rssociation Response frame.

A BPE non-AP MLD has a set of configured identity keys and parameters required to authenticate and associate with BPE AP MLDs. When a STA affiliated with the BPE non-AP MLD receives a Privacy Beacon frame, the non-AP MLD may determine whether it has parameters required to authenticate and associate with BPE AP MLD of the AP that transmitted the Privacy Beacon. The STA uses Equation (10-28) with the Address 2 of the received Privacy Beacon and each identity key as input parameters to calculate Identity Hash alternatives for the BPE AP. If any of the calculated Identity Hash alternatives matches with the Identity Hash of the received Privacy Beacon frame, then the STA may use the Identity Key and the BPE AP MLD parameters to authenticate and (re)associate with the BPE AP MLD.

NOTE 2—This checking can be done for the protected and unprotected Privacy Beacon frames.

Identity Hash = Truncate-48(HMAC-SHA-256(Identity Key, “BPE AP MLD address resolution” || Address 2)) (10-28)

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where

Identity Hash	is the value of the Identity Hash field of the Privacy Beacon frame.
Identity Key	is a 128-bit Identity Key of the BPE AP MLD.
Address 2	is the Address 2 field of the Privacy Beacon frame.

A BPE AP may include an Extended Channel Switch Announcement element in the protected Privacy Beacon frames as described in 11.8.8.2(Selecting and advertising a new channel in a non-DMG infrastructure BSS).

A BPE AP MLD shall not be part of a Multiple BSSID set.

NOTE 3—The BPE AP identity key is AP MLD and single ESS specific.

An associated BPE non-AP MLD maintains a BSS Parameter Change Count (BPCC) value for each BPE AP that has a link with the BPE non-AP MLD. If an associated BPE non-AP MLD detects that a BPCC value of a BPE AP in a received Privacy Beacon frame is larger than the stored BPCC value of the AP, then the BPE non-AP MLD shall obtain the updated BSS parameter values of the AP before it sends data to the AP.

An associated BPE non-AP MLD may use the solicited and unsolicited procedure defined in 12.16.4.3 (MLO procedure) to obtain the capabilities and operation parameters of BPE AP MLD.

A BPE AP may broadcast unsolicited encrypted Capabilities And Operation Parameters Response frames to signal updated BSS parameter values to STAs affiliated with associated BPE non-AP MLDs.

10.71.8.4 Group addressed frames anonymization

A BPE AP MLD shall anonymize group addressed frames by using offsets as described in 10.71.4 (Establishing BPE MAC header anonymization parameter sets):

- The MAC header anonymization parameters are selected as described in 10.71.5.1 (MAC header anonymization parameter set selection).
- The receiver address and transmitter address are anonymized as described in 10.71.5.4 (Addressing).
- The SN is anonymized with the SNS1 DL offset as described in 10.71.5.2 (Sequence number anonymization).
- The PN is anonymized with the PN group offset as described in 10.71.5.3 (Packet number anonymization).

A BPE non-AP MLD affiliated STA shall deanonymize the received group frames by using the offsets as described in 10.71.4 (Establishing BPE MAC header anonymization parameter sets):

- The transmitter address is filtered as described in 10.71.6.1.3 (Address filtering for BPE FA).
- The receiver address is deanonymized as described in 10.71.6.1.3 (Address filtering for BPE FA).
- The PN is deanonymized with the PN group offset as described in 10.71.6.3 (Packet number deanonymization).
- The SN is deanonymized with the SNS1 DL offset as described in 10.71.6.4 (Sequence number deanonymization).

The BPE AP MLD shall use GTK to encrypt the payload of the group addressed Management frames.

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10.71.8.5 Steering to a BPE AP MLD

An non-BPE AP sets the BPE Available field of the Extended Capabilities element of the Beacon, Probe Response and (Re)Association Response frames to indicate that the ESS that the AP operates in contains one or more BPE AP MLDs. A BPE AP sets this field always to 1 in (Re)Association Response frames.

A BPE capable non-AP MLD sets the BPE Available field in the Extended Capabilities element of the (Re)Association Request frame to indicate that it is capable to associate to a BPE AP MLD and receive BPE AP MLDs information in the Neighbor Report elements in the encrypted individually addressed BTM Request, Neighbor Report Response or ANQP Response frames. An AP that sets the BPE Available field to 1 shall implement individually addressed encrypted GAS/ANQP.

NOTE—AP accepts and responds to encrypted ANQP requests from associated STAs.

A non-AP STA that has received a Beacon, Probe Response or (Re)Association Response frame with BPE Available field equal to 1 in the Extended Capabilities element from its associated AP may query information of available BPE AP MLDs by sending an encrypted BTM Query frame to its associated AP. The BTM Query frame with the BSS Transition Query Reason field equal to a BPE AP MLD indicates that the non-AP MLD is interested to only receive information of the available BPE AP MLDs.

An AP that has set the BPE Available field of the Extended Capabilities element on its (Re)Association Response frame to 1 may send an encrypted solicited or unsolicited BTM Request frame that contains information of BPE AP MLDs to an associated non-AP STA that has set the BPE Available field of the Extended Capabilities element on its (Re)Association Request frame to 1.

The non-AP STA may send an individually addressed encrypted Neighbor Report Request frame to request Neighbor Report elements of available BPE AP MLDs in the ESS when the Radio Measurement Action field of the frame is Neighbor Report Request and the frame includes BPE AP MLD Discovery Request field and no LCI Measurement Request field, Location Civic Measurement field, and Neighbor DMG Measurement field (see 9.6.6.6 (Neighbor Report Request frame format)).

The responding AP may include BPE AP MLD information in the Neighbor Report elements of the encrypted individually addressed Neighbor Report Response frame transmitted to the requesting STA.

The non-AP STA may send an individually addressed encrypted ANQP Request frame that requests Neighbor Reports to be included in the ANQP Response frame. The responding AP may send an individually addressed encrypted ANQP Response frames that may include BPE AP information in the Neighbor Report elements to the requesting STA.

Each AP affiliated with the BPE AP MLD should be reported in a separate Neighbor Report element. Such a Neighbor Report element shall include valid operating class and channel number values and at least the following subelements: BSSID of the current and next epoch, RSNE, RSNXE, Supported Rates and BSS Membership Selectors, Extended Supported Rates and BSS Membership Selectors, BSS Load, HT Capabilities, HT Operation, VHT Capabilities, VHT Operation, HE Capabilities, EHT Capabilities, EHT Operation and Basic Multi-link element.

A STA that has received a BPE AP information in a Neighbor Report element should detect the presence of a reported AP affiliated with BPE AP MLD before it authenticates and associates with the BPE AP MLD. The presence detection may be done by receiving a frame with transmitter address or receiver address in the MAC header by matching the transmitter address or the receiver address in the MAC header of a received frame to the BSSID reported in the Neighbor Report.

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11. MLME

11.2 Power management

11.2.3 Power management in a non-DMG infrastructure network

11.2.3.1 General

Change the eighth paragraph as follows:

WNM sleep mode enables an extended power save mode in which a non-AP STA needs not listen for every DTIM beacon, and does not need to perform GTK/IGTK/BIGTK updates. Further, a non-AP MLD need not perform PGTK updates. A STA in WNM sleep mode can transition to awake state as infrequently as once every WNM sleep interval to check whether its corresponding TIM bit is set or group addressed traffic is pending. The WNM sleep interval advertised by a non-AP STA affiliated with a non-AP MLD is applied at the MLD level and the WNM procedures described in this subclause and in 11.2.3.15 (WNM sleep mode) are performed at the MLD level and apply to all the STAs affiliated with the MLD.

11.2.3.3 AP TIM transmissions

Insert the following paragraph after the first paragraph:

A CPE AP may set additional bits to 1, in the TIM's partial virtual bitmap, corresponding to AIDs that are not assigned in the current EPP epoch, to prevent an eavesdropper from easily determining the number of CPE non-AP MLDs with buffered traffic.

11.2.3.15 WNM sleep mode

11.2.3.15.1 WNM sleep mode capability

Change the fourth paragraph as follows:

WNM sleep mode enables an extended power save mode for non-AP STAs in which a non-AP STA need not listen for every DTIM beacon, and need not perform GTK/IGTK/BIGTK updates. Further, a non-AP MLD need not perform PGTK updates. A non-AP STA can sleep for extended periods as indicated by the WNM Sleep Interval field of the WNM Sleep Mode element, which is present in WNM Sleep Mode Request frames transmitted by the non-AP STA.

11.2.3.15.2 WNM sleep mode non-AP STA operation

Change the fifth paragraph as follows:

The receipt of an MLME-SLEEPMODE.confirm primitive with a valid SleepMode parameter indicates to the STA’s SME that the AP has processed the corresponding WNM Sleep Mode Request frame. The content of the WNM sleep mode parameter in the WNM Sleep Mode Response frame provides the status of WNM Sleep Mode elements processed by the AP. The non-AP STA shall delete the GTKSA if the response indicates success. If RSN is used with management frame protection, the non-AP STA shall delete the IGTKSA if the response indicates success, If RSN is used with beacon frame protection, the non-AP STA shall delete the BIGTKSA if the response indicates success. If the Group EPP Epoch Supported field in the RSNXE is 1 for both the APs affiliated with the AP MLD and the non-AP MLD, the non-AP MLD shall delete the PGTKSA if the response indicates success.

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11.2.3.15.3 WNM sleep mode AP operation

Change the last paragraph as follows:

For MLO, with RSN and a valid PTK is configured for the non-AP MLD:

— If management frame protection is negotiated for the MLDs, the current GTK, IGTK when management frame protection is negotiated, and BIGTK when beacon protection is negotiated for each setup link shall be included in the WNM Sleep Mode Response frame using the WNM Sleep Mode MLO GTK/IGTK/BIGTK subelement (see 9.6.13.20 (WNM Sleep Mode Response frame format)). If a GTK/IGTK/BIGTK update is in progress for one or more links, the pending GTK, IGTK when management frame protection is negotiated, and BIGTK when beacon protection is negotiated for each of the affected AP(s) shall be included in the WNM Sleep Mode Response frame using the WNM Sleep Mode MLO GTK/IGTK/BIGTK subelement (see 9.6.13.20 (WNM Sleep Mode Response frame format)). A non-AP MLD identifies the corresponding link to which the GTK/IGTK/BIGTK belongs based on the value of the Link ID subfield included in the subelement of the Key Data field.

— If management frame protection is not negotiated for the MLDs, the current GTK for each setup link shall be sent to the non-AP MLD using a group key handshake (see 12.7.7 (Group key handshake)) immediately following the WNM Sleep Mode Response frame. If a GTK update is in progress for a setup link, the pending GTK for the setup link shall be sent to the STA using another group key handshake immediately after the current GTK of the setup link has been sent.

— <u>If the Group EPP Epoch Supported field in the RSNXE is 1 for both the APs affiliated with the AP MLD and the non-AP MLD, the current PGTK shall be included in the WNM Sleep Mode Response frame. If a PGTK update is in progress, the pending PGTK shall be sent to the non-AP MLD using another group key handshake immediately after the current PGTK has been sent.</u>

11.3 STA authentication and association

11.3.2 State transition diagram for nonmesh STAs or nonmesh MLDs

Replace Figure 11-23 (Relationship between state and services between a given pair of nonmesh STAs or nonmesh MLDs) as follows:

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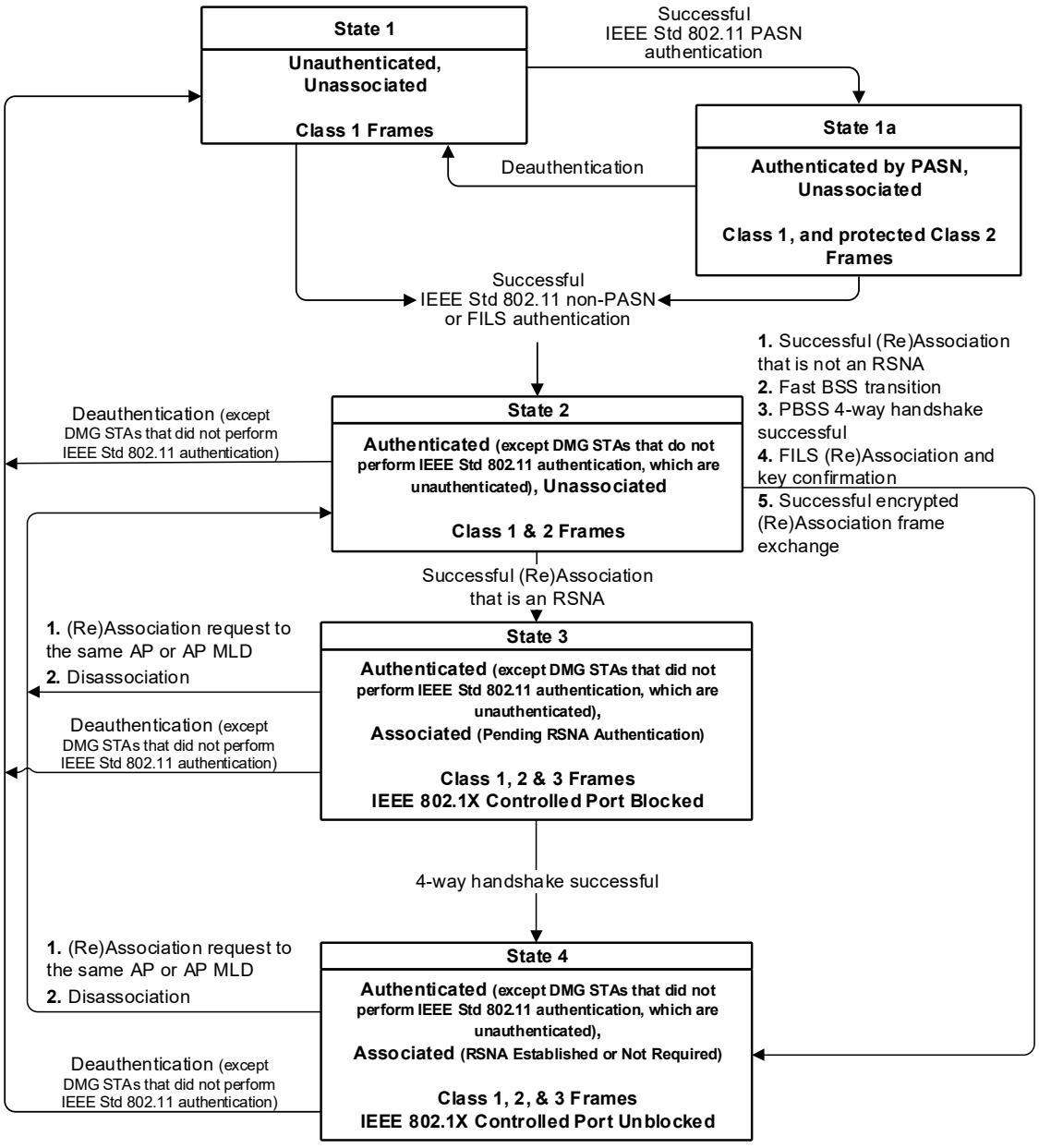


Figure 11-23—Relationship between state and services between a given pair of nonmesh STAs or nonmesh MLDs

11.3.4 Authentication and deauthentication

11.3.4.2 Authentication—originating STA or MLD

Change the second paragraph as follows (not all lines shown):

Upon receipt of an MLME-AUTHENTICATE.request primitive, the originating STA shall authenticate with the indicated STA using the following procedure:

b) The STA shall execute one of the following:

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5) For PASN authentication, the authentication mechanism described in 12.13 (Preassociation security negotiation(11az)).

6) For EPPKE authentication, the authentication mechanism described in 12.16.9 (Enhanced privacy protection key exchange).

11.3.4.3 Authentication—destination STA or MLD

Change the first paragraph as follows (not all lines shown):

Upon receipt of an Authentication frame with authentication transaction sequence number equal to 1, the destination STA shall authenticate with the originating STA using the following procedure:

g) Upon receipt of an MLME-AUTHENTICATE.response primitive, if the ResultCode is not SUCCESS, the MLME shall transmit an Authentication frame with the corresponding status code, as defined in 9.4.1.9 (Status Code field), and the state for the originating STA or MLD shall be left unchanged. The Authentication frame is constructed using the appropriate procedure in 12.3.3.2 (Open System authentication), 13.5 (FT protocol), 13.6 (FT resource request protocol), ~~or~~ 12.13 (Preassociation security negotiation(11az)) or 12.16.9 (Enhanced privacy protection key exchange).

11.3.5 Association, reassociation, and disassociation

11.3.5.2 Non-AP STA, non-AP MLD and non-PCP STA association initiation procedures

Change the first paragraph as follows:

The SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the AP MLD by using MLME-DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) before invoking MLME-ASSOCIATE.request primitive.

11.3.5.3 AP, AP MLD or PCP association receipt procedures

Change the first paragraph as follows (not all lines shown):

The following procedure shall be used by an AP or PCP upon receipt of an Association Request frame from a STA or by an AP MLD after an AP affiliated with the AP MLD receives an Association Request frame with Basic Multi-Link element from a non-AP STA affiliated with a non-AP MLD:

m) If the ResultCode in the MLME-ASSOCIATE.response primitive is SUCCESS, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, Identity Key SA and TPKSA (including temporal keys) held for communication with the STA or non-AP MLD by using the MLME-DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)).

11.3.5.4 Non-AP STA, non-AP MLD and non-PCP STA reassociation initiation procedures

Change the first paragraph as follows:

Except when the association is part of a fast BSS transition, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the AP, AP MLD, or PCP by using the MLME-DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) before invoking an MLME-REASSOCIATE.request primitive.

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11.3.5.5 AP, AP MLD, or PCP reassociation receipt procedures

Change the first paragraph as follows (not all lines shown):

The following procedure shall be used by an AP or PCP upon receipt of a Reassociation Request frame from a STA or by an AP affiliated with an AP MLD upon receipt of a Reassociation Request frame with Basic Multi-Link element from a non-AP STA affiliated with a non-AP MLD:

k) If management frame protection is not in use, or the ResultCode in the MLME-REASSOCIATE.response primitive is SUCCESS and the reassociation is not part of a fast BSS transition, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, Identity Key SA, and TPKSA (including temporal keys) held for communication with the STA or the non-AP MLD by using the MLME-DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)).

11.3.5.6 Non-AP STA, non-AP MLD, and non-PCP STA disassociation initiation procedures

Change the second paragraph as follows (not all lines shown):

Upon receipt of an MLME-DISASSOCIATE.request primitive, a non-AP STA, non-AP MLD, and non-PCP STA's MLME shall disassociate from an AP, AP MLD, or PCP, respectively, using the following procedure:

d) Upon receiving an MLME-DISASSOCIATE.confirm primitive, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the AP, AP MLD or PCP by using the MLME DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) and by invoking an MLME-SETPROTECTION.request(None) primitive. In the case of an MM-SME coordinated STA, the MLME shall perform this for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

11.3.5.7 Non-AP STA, non-AP MLD and non-PCP STA disassociation receipt procedure

Change the first paragraph as follows (not all lines shown):

Upon receipt of a Disassociation frame from an AP, AP MLD, or PCP for which the state is State 3 or State 4, if management frame protection was not negotiated when the PTKSA(s) were created, or if management frame protection was negotiated when the PTKSA(s) were created and the frame is not discarded per management frame protection processing, a non-AP STA, non-AP MLD, and non-PCP STA, respectively, shall disassociate from the AP, AP MLD, or PCP using the following procedure:

c) Upon receiving the MLME-DISASSOCIATE.indication primitive, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the AP, AP MLD or PCP by using the MLME DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) and by invoking an MLME-SETPROTECTION.request(None) primitive. The MM-SME shall perform this process for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

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11.3.5.8 AP, AP MLD, or PCP disassociation initiation procedure

11.3.5.8.1 General

Change the second paragraph as follows (not all lines shown):

Upon receipt of an MLME-DISASSOCIATE.request primitive, an AP, AP MLD, or PCP shall disassociate a STA (with respect to the AP or PCP) or a non-AP MLD (with respect to the AP MLD) using the following procedure:

d) Upon receiving an MLME-DISASSOCIATE.confirm primitive, the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the STA by using the MLME DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) and by invoking an MLME-SETPROTECTION.request(None) primitive. The MM-SME shall perform this process for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

11.3.5.9 AP, AP MLD, or PCP disassociation receipt procedure

Change the first paragraph as follows (not all lines shown):

Upon receipt of a Disassociation frame from a STA or a non-AP MLD for which the state is State 3 or State 4, if management frame protection was not negotiated when the PTKSA(s) were created, or if management frame protection was negotiated when the PTKSA(s) were created and the frame is not discarded per management frame protection processing, the AP or PCP (with respect to the STA) or AP MLD (with respect to the non-AP MLD) shall disassociate the STA or the non-AP MLD using the following procedure:

c) Upon receiving an MLME-DISASSOCIATE.indication primitive the SME shall delete any PTKSA, GTKSA, IGTKSA, BIGTKSA, WIGTKSA, WTKSA, PGTKSA, and TPKSA (including temporal keys) held for communication with the STA or the non-AP MLD by using the MLME DELETEKEYS.request primitive (see 12.6.16 (RSNA security association termination)) and by invoking an MLME-SETPROTECTION.request(None) primitive. The MM-SME shall perform this process for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

11.10 Radio measurement procedures

11.10.10 Usage of the neighbor report

11.10.10.2 Requesting a neighbor report

Insert the following paragraph after the paragraph "To request a list of neighboring APs that support DMG ..." as follows:

To request a list of neighboring BPE AP MLDs, the STA shall transmit an encrypted Neighbor Report Request frame that includes BPE AP MLD Discovery Request field as described in 10.71.8.5 (Steering to a BPE AP MLD). A STA shall not send this type of request to an AP unless it received an Extended Capabilities element from the AP with the BPE Available Information field equal to 1.

Insert the following paragraph at the end:

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To request the CPE parameters of neighboring AP MLDs, the STA shall transmit an encrypted Neighbor Report Request frame that includes a CPE AP MLD Request field with a Measurement Request element with the value of its Measurement Type field set to CPE AP MLD EPP.

11.10.10.3 Responding to a neighbor report request

Insert the following paragraph at the end:

A CPE AP that has set the Group EPP Epoch Supported to 1 in the RSNXE field and that receives an encrypted Neighbor Report Request frame that includes a Measurement Request element with the value of its Measurement Type field equal to CPE AP MLD EPP, shall respond with an encrypted Neighbor Report Response frame with a list of Neighbor Report elements of neighboring APs supporting CPE and their individual EPP elements.

11.21 Wireless network management procedures

11.21.14 Proxy ARP Service

Change the second paragraph as follows:

When the AP sets the Proxy ARP field to 1 in the Extended Capabilities element, the AP shall maintain a Hardware Address or a DS MAC address (if present) to Internet Address mapping for each associated STA and for each IPv4 and IPv6 address of the STA, and shall update the mapping when one of the addresses of the associated STA changes. When an AP MLD supports proxy ARP (see 35.3.22 (Proxy ARP service in AP MLDs)), the AP MLD shall maintain an MLD MAC address or a DS MAC address (if present) to Internet address mapping for each associated non-AP MLD and for each IPv4 and IPv6 address of the non-AP MLD, and shall update the mapping when one of the addresses of the associated non-AP MLD changes. A Proxy ARP service receives and processes three types of messages: IPv4 ARP requests, IPv6 ND address lookups, and IPv6 ND duplicate address detection (DAD) messages. These messages are all received as group addressed. If the target address is not known, the Proxy ARP service does not forward the request to the BSS. If the target address is known, the Proxy ARP service can either respond directly on behalf of a STA or forward the request as an individually addressed frame to the intended STA. For fixed devices in the doze state, a direct response is preferable. Otherwise, forwarding as an individually addressed frame is recommended, to avoid responding with misleading information.

Change the fourth paragraph as follows:

When an AP receives an IPv4 ARP request from one associated STA or from the DS with a target IPv4 address that corresponds to a second associated STA, the AP that decides to form a proxy ARP response frame shall insert the second STA's MAC address or the second STA's DS MAC address (if present) as the sender's MAC address in the ARP response. When an AP affiliated with an AP MLD receives an IPv4 ARP request from one associated STA, or from a non-AP STA affiliated with a non-AP MLD that is associated with the AP MLD, or from the DS, with a target IPv4 address that corresponds to a second associated STA, the AP shall insert the second STA's MAC address or the second STA's DS MAC address (if present) as the Sender's MAC Address in the ARP response packet. When an AP MLD receives an IPv4 ARP request from a STA associated with an affiliated AP, or from one associated non-AP MLD via any affiliated AP, or from the DS, with a target IPv4 address that corresponds to a second associated non-AP MLD, the AP MLD that decides to form a proxy ARP response shall insert the MLD MAC address or the DS MAC address (if present) of the second non-AP MLD as the Sender's MAC Address in the ARP response packet.

Change the tenth paragraph as follows:

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With the IPv6 ND proxy operation defined in IETF RFC 8929, the backbone router function at the AP typically operates as a bridging proxy though operation as a routing proxy is also possible. As a bridging proxy, the NS lookups are replied with the MAC address or the DS MAC address (if present) of the STA (or the non-AP MLD), and the packets to the STA (or the non-AP MLD) are bridged normally; as a routing proxy, the backbone router function replies to lookups from the wired backbone with its own MAC address and then routes to the STA (or the non-AP MLD) at the IP layer. The routing proxy isolates the layer-2 domains and hides the MAC address or the DS MAC address (if present) of the STA (or the non-AP MLD) in the wired backbone, for a better stability and scalability of the bridged domain. The Proxy ARP function shall support the bridging proxy and may support the routing proxy operation.

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12. Security

12.2 Framework

12.2.2 Security methods

Change the fifth bullet of the second paragraph as follows (not all lines are shown):

RSNA security comprises the following algorithms and procedures:

— RSNA establishment and termination procedures, including use of IEEE 802.1X authentication, described in 12.6 (RSNA security association management) and 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames), SAE authentication described in 12.4 (Authentication using a password), and OWE described in IETF RFC 8110.

12.2.4 RSNA establishment

Change the first bullet, the second bullet, the fifth bullet, and add a new bullet of the first paragraph as follows (not all lines are shown):

An SME establishes an RSNA in one of seven ways:

a) If an RSNA uses authentication negotiated over IEEE Std 802.1X or FILS authentication in an infrastructure BSS, an SME establishes an RSNA as follows:

1) It identifies the AP as an RSNA AP from the AP's Beacon, Privacy Beacon, DMG Beacon, Announce, Information Response, FILS Discovery, or Probe Response frames.

NOTE—As described in 10.71.8.5 (Steering to a BPE AP MLD), the STA might identify an AP affiliated with a BPE AP MLD by using the AP's anonymized address and the Neighbor Report element that is received from the associated AP.

2) It shall invoke Open System, IEEE 802.1X authentication, or FILS authentication if the STA is a non-DMG STA.

3) It negotiates cipher suites during the association process, as described in 12.6.2 (RSNA selection) and 12.6.3 (RSNA policy selection in an infrastructure BSS).

4) It uses IEEE Std 802.1X-2020 to authenticate if IEEE 802.1X authentication is not performed before association, as described in 12.6.8 (RSNA establishment in an infrastructure BSS) and 12.6.9 (RSNA authentication in an IBSS), FT protocol to authenticate as described in 13.5 (FT protocol) or uses FILS authentication to authenticate as described in 12.11 (Authentication for FILS).

10) If EPP epoch operation is supported by both the AP MLD and the non-AP MLD, the SME installs the PGTK into the MAC for frame anonymization.

11) If the AP is affiliated with a BPE AP MLD, the SME installs the identity key if the key is not already installed in the MAC to identify the AP MLDs with that configured identity key from its Privacy Beacon frames.

b) If an RSNA is based on a PSK or password in an infrastructure BSS, an SME establishes an RSNA as follows:

9) If EPP epoch operation is supported by both the AP MLD and the non-AP MLD, the SME installs the PGTK into the MAC for frame anonymization.

10) If the AP is affiliated with a BPE AP MLD, the SME installs the identity key if the key is not already installed in the MAC for identifying the AP MLDs with that configured identity key from its Privacy Beacon frames.

e) If an RSNA allows for confidentiality only (no authentication) in an infrastructure BSS, an SME establishes an RSNA as follows:

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9) If EPP epoch operation is supported by both the AP MLD and the non-AP MLD, the SME installs the PGTK into the MAC for frame anonymization.

10) If the AP is affiliated with a BPE AP MLD, the SME installs the identity key into the MAC for identifying the AP MLD from its Privacy Beacon frames.

h) If an RSNA uses PASN authentication, an RSNA capable the STA establishes an RSNA asdescribed in 12.13 (Preassociation security negotiation(11az)).

i) If an RSNA uses EPPKE authentication, an RSNA capable STA establishes an RSNA as described in 12.16.9 (Enhanced privacy protection key exchange).

12.2.7 Requirements for management frame protection

Change the first paragraph as follows:

The robust Management frames are~~: Disassociation, Deauthentication, robust Action, and robust Action No Ack frames.~~

- Disassociation frame
- Deauthentication frame
- robust Action frame
- robust Action No Ack frame
- if encryption of (Re)Association Request/Response frames is negotiated (see 12.16.6 ((Re)Association Request/Response frame encryption)):
  - Association Request frame
  - Association Response frame
  - Reassociation Request frame
  - Reassociation Response frame.

12.4 Authentication using a password

Modify 12.4.3 (Representation of a passwords and password identifiers) as follows:

12.4.3 Representation of ~~a~~ passwords and password identifiers

Modify the second paragraphs as follows:

Similarly, to address ambiguity when identifying passwords, a STA shall represent a character-based password identifier as a UTF-8 string that is processed according to the UsernameCasePreserved profile of IETF RFC 8265, the output of which is an octet string that is stored in the dot11RSNAConfigPasswordValueTable. When a password identifier is changed using the SAE Password Identifiers KDE, the SAE Password Identifier values from that KDE are not necessarily character-based and they are stored without modification as octet strings in the dot11RSNAConfigPasswordValueTable. When a "password identifier" is called for in the description of SAE that follows, the identifier from the dot11RSNAConfigPasswordValueTable is used. A mesh STA shall not use a password identifier.

Insert the following at the end of 12.4.3 (Representation of a passwords and password identifiers):

SAE password identifiers can expose information that a passive attacker could use to identify and track STAs that authenticate to a network. To provide privacy, the password identifier can be changed for each use. A list of new password identifiers can be provided in either message 3 of the 4-way handshake or in the encrypted (Re)Association Response frame after a successful SAE authentication when the non-AP STA or the non-AP MLD and the AP or the AP MLD support the mechanism for updating SAE password identifiers. The non-AP STA or the non-AP MLD indicates its support using the SAE Password Identifier Change

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Support bit in the RSNXE. The AP or the AP MLD indicates its support by providing a list of new password identifiers in an SAE Password Identifiers KDE. The AP or AP MLD shall include the SAE Password Identifiers KDE in message 3 of 4-way handshake or in the encrypted (Re)Association Response frame if the AP or AP MLD supports SAE password identifier change and the non-AP STA or the non-AP MLD has the SAE Password Identifier Change Support bit in the RSNXE equal to 1. When an AP in an ESS provides a list of password identifiers, those new password identifiers can be used with all APs in the ESS. after successful completion of the 4-way handshake or encrypted (Re)Association Request/Response frame exchange. When a new set of password identifiers is delivered in a successfully completed 4-way handshake or encrypted (Re)Association Response frame, (1) the AP or the AP MLD shall retain the actual password identifier value itself if it generates the list of new password identifiers by encrypting the password identifier value in a manner that produces different output for each instance while those encrypted values are deleted in this case, (2) the STA or the MLD shall delete password identifiers that were previously delivered in an SAE Password Identifiers KDE, and (3) the STA or the MLD may retain a character-based password identifier that was not delivered in an SAE Password Identifiers KDE. The AP or the AP MLD indicates whether it deletes the character-based password identifier in the Character-based Password Identifier Deleted field of the SAE Password Identifiers KDE. The AP decision on when to update the password identifiers and how many active password identifiers to provide are outside the scope of this standard. However, an AP or an AP MLD should update the list of password identifiers in a manner that prevents need for the non-AP STA or the non-AP MLD to reuse the same identifier. The AP or the AP MLD shall not share the list of password identifiers with third parties outside the set of APs in the ESS to avoid exposing information that might be used to track the non-AP STA or the non-AP MLD.

NOTE 1—The list of active SAE password identifiers might also be updated using out-of-band mechanisms. Such mechanisms are outside the scope of this standard.

NOTE 2—If the AP or AP MLD deletes the character-based password identifier, possibilities for sharing the same password identifier with multiple STAs are limited and a new character-based password identifier might need to be created for each STA.

NOTE 3—Since non-AP STAs and non-AP MLDs return the exact value provided by an AP or an AP MLD in the SAE Password Identifiers KDE, APs in an ESS might use an approach that generates the new password identifier values that are exchanged over air by encrypting the actual character-based password identifier string using a key that is shared by all APs in an ESS. Such encryption mechanism needs to generate changing output for each instance for the same password identifier. As an example, constructions like AES-SIV(key=AP-shared-key, plaintext=PW ID | random length padding of 0x00 octets | 32-bit counter, aad=<zero components>) might be used for this purpose. When processing a received password identifier, an AP or an AP MLD need to first determine whether they have a matching character-based password identifier string in the dot11RSNAConfigPasswordValueTable and if so, use that entry, and otherwise, try to decrypt the received password identifier interpreted as an octet string and if the decryption succeeds, use the decrypted password identifier to find a match from the dot11RSNAConfigPasswordValueTable. As defined in 12.4.5.4 (Processing of a peer's SAE Commit message), the exact contents of the Password Identifier element, not the decrypted value, is used in construction of the PWE.

When a non-AP STA or a non-AP MLD has more than one password identifier available for a password it uses for SAE authentication, the non-AP STA or the non-AP MLD may use any of the password identifiers to identify the password. In this case, the non-AP STA or the non-AP MLD shall select a password identifier to use in a manner that avoids reuse of the same value.

12.4.7 Framing of SAE

12.4.7.3 Encoding and decoding of SAE Commit messages

Modify the fourth paragraph as follows:

If a password identifier is used in generation of the PWE, a Password Identifier element shall be included and the identifier shall be encoded either as a UTF-8 string or as an octet string from an SAE Password Identifiers KDE in the Identifier field (see 9.4.2.215 (Password Identifier element)); otherwise it shall not be included.

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12.5 RSNA confidentiality and integrity protocols

12.5.2 CTR with CBC-MAC protocol (CCMP)

12.5.2.4 CCMP decapsulation

12.5.2.4.4 PN and replay detection

Change item c) and d) of the third paragraph (not all shown) and create new items as follows:

See 12.5.2.2 (CCMP MPDU format) for a description of how the PN is encoded in the CCMP header. The following processing rules are used to detect replay:

c) If management frame protection is negotiated, the receiver shall maintain a single replay counter for received individually addressed robust PV0 Management frames that are received with the To DS subfield equal to 0, except Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing frame details)), ~~and~~ Protected Sensing frames (see 9.6.36 (Protected Sensing frame details)), and EPP robust beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)), and (S1G STA only) a single replay counter for received individually addressed robust PV1 Management frames.

d) If dot11RSNAProtectedManagementFramesActivated is true and dot11QMFActivated is also true, the receiver shall maintain an additional replay counter for each ACI for received individually addressed robust PV0 Management frames that are received with the To DS subfield equal to 1, except Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing Frame details)), ~~and~~ Protected Sensing frames (see 9.6.39 (Protected Sensing frame details)), and EPP robust beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)).

e) If dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving individually addressed Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing frame details(11az))) and shall use the PN from the received frame to detect replays.

f) If dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving individually addressed Protected Sensing frames (see 9.6.39 (Protected Sensing frame details)) and shall use the PN from the received frame to detect replays.

Insert the following paragraph after item f):

fa) For non-MLO, if dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving EPP robust individually addressed beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)) and shall use the PN from the received frame to detect replays.

fb) For MLO, if dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter in each setup link for receiving EPP robust individually addressed beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)) and shall use the PN from the received frame to detect replays.

12.5.4 GCM protocol (GCMP)

12.5.4.3 GCMP cryptographic encapsulation

12.5.4.3.3 Construct AAD

Change the first paragraph as follows:

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The AAD construction for all frames except for the Privacy Beacon frame is defined in 12.5.2.3.3 (Construct AAD). The AAD construction of the Privacy Beacon frame is defined as shown in Figure 12-29a (AAD construction for Privacy Beacon frame).

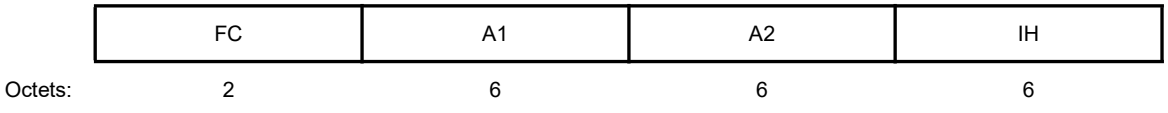


Figure 12-29a—AAD construction for Privacy Beacon frame

a) FC—MPDU Frame Control field

a) A1—MPDU Address 1 field.

b) A2—MPDU Address 2 field.

c) IH—MPDU Identity Hash field.

12.5.4.4 GCMP decapsulation

12.5.4.4.4 PN and replay detection

Change item c) and d) of the first paragraph (not all shown) and create new items as follows:

To effect replay detection, the receiver extracts the PN from the GCMP header. See 12.5.4.2 (GCMP MPDU format) for a description of how the PN is encoded in the GCMP header. The following processing rules are used to detect replay:

c) If management frame protection is negotiated, the receiver shall maintain a single replay counter for received individually addressed robust PV0 Management frames that are received with the To DS subfield equal to 0, except Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing frame details)), ~~and~~ Protected Sensing frames (see 9.6.39 (Protected Sensing frame details)), and EPP robust beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)).

d) If dot11RSNAProtectedManagementFramesActivated is true and dot11QMFActivated is also true, the receiver shall maintain an additional replay counter for each ACI for received individually addressed robust Management frames that are received with the To DS subfield equal to 1, except Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing Frame details)), ~~and~~ Protected Sensing frames (see 9.6.39 (Protected Sensing frame details)), and EPP robust beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)).

e) If dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving individually addressed Protected Fine Timing frames (see 9.6.34 (Protected Fine Timing Frame details)) and shall use the PN from the received frame to detect replays.

f) If dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving individually addressed Protected Sensing frames (see 9.6.39 (Protected Sensing Frame details)) and shall use the PN from the received frame to detect replays.

Insert the following paragraph after item f):

fa) For non-MLO, if dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for receiving EPP robust individually addressed beamforming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)) and shall use the PN from the received frame to detect replays.

fb) For MLO, if dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter in each setup link for receiving EPP robust individually addressed beam-

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forming/CSI/CQI frames (see 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)) and shall use the PN from the received frame to detect replays.

12.6 RSNA security association management

12.6.1 Security associations

12.6.1.1 Security association definitions

12.6.1.1.1 General

Change second paragraph (not all lines shown) by modifying a sub-bullet and adding a sub-bullet as follows:

A security association is a set of policy(ies) and key(s) used to protect information. The information in the security association is stored by each party of the security association, needs to be consistent among all parties, and needs to have an identity. The identity is a compact name of the key and other bits of security association information to fit into a table index or an MPDU. The following types of security associations are supported by an RSNA STA:

- PTKSA: A result of a successful 4-way handshake, FT 4-way handshake, FT authentication sequence, FILS authentication, ~~or~~ PASN authentication, EPPKE authentication, or authentication followed by the encrypted (Re)Association Request/Response frame exchange.
- PGTKSA: A result of a successful group key handshake, the Reassociation Response frame of the fast BSS transition protocol, the encrypted Reassociation Response frame specified in 12.16.6.3 (MLO procedure), or successful FILS authentication.
- Identity Key Security Association: A result of a successful group key handshake with a BPE AP MLD, or the encrypted (Re)association Response frame specified in 12.16.6.3 (MLO procedure).

12.6.1.1.2 PMKSA

Change the third paragraph (not all lines shown) as follows:

A PMKSA association is bidirectional. In other words, both parties use the information in the security association for both sending and receiving. The PMKSA is used to create the PTKSA. PMKSAs have a certain lifetime. The PMKSA consists of the following:

- PMKID, as defined in 12.7.1.3 (Pairwise key hierarchy), ~~or~~ 12.7.1.6.3 (PMK-R0), or 12.16.7.2 (PMKID privacy). The PMKID identifies the security association.

12.6.1.1.3 PMK-R0 security association

Change the first paragraph (not all lines shown) as follows:

The PMK-R0 security association is the result of a successful completion of the IEEE 802.1X authentication, SAE authentication, or use of PSK during the FT initial mobility domain association. This security association is bidirectional. It has a certain lifetime. It consists of the following:

- SSID
- MDID
- PMK-R0
- R0KH-ID
- PMKR0Name

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— If PMKSA caching privacy is used, recomputed PMKR0Name

12.6.1.1.4 PMK-R1 security association

Change the second paragraph (not all lines shown) as follows:

This security association is bidirectional. It has a certain lifetime. It consists of the following:

— SSID
— MDID
— PMK-R1
— PMK-R1 lifetime
— PMKR1Name
— R1KH-ID
— R0KH-ID
— PMKR0Name
— If PMKSA caching privacy is used, recomputed PMKR0Name

12.6.1.1.6 PTKSA

Change first paragraph as follows:

The PTKSA results from a successful 4-way handshake, FT 4-way handshake, FT protocol, FT resource request protocol, FILS authentication, IEEE 802.1X authentication utilizing Authentication frames with PTKSA derivation, ~~or~~ PASN authentication, or EPPKE authentication. This security association is also bidirectional. PTKSAs, except those established using PASN authentication or EPPKE authentication, have the same lifetime as the PMKSA or PMK-R1 security Association, whichever comes first. PTKSAs for PASN authentication or EPPKE authentication have a minimum of the lifetime of the PMKSA used and the timeout negotiated, if any, during PASN authentication. Because the PTKSA is tied to the PMKSA or to a PMK-R1 security association, it only has the additional information from the 4-way handshake, FT Protocol authentication, FILS authentication, IEEE 802.1X authentication utilizing Authentication frames with PTKSA derivation, ~~or~~ PASN authentication, or EPPKE authentication. There shall be only one PTKSA per key ID per band (see 12.6.20 (Multi-band RSNA)) or per MLD setup (see 35.3.5 (ML (re)setup)) with the same Supplicant and Authenticator MAC addresses.

Change fifth paragraph as follows (not all lines shown):

The PTKSA consists of the following:

— PTK, where the PTK includes the KDK only when any of the following is true:~~WUR frame protection is negotiated~~
  - WUR frame protection is negotiated.
  - dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.240 (RSNXE)).
  - Frame anonymization is negotiated (see 10.71 (Frame anonymization)).

Insert a new subclause at the end of 12.6.1.1 (Security association definitions) as follows:

-- Editor Note: 802.11-2024 up to 12.6.1.1.13, 802.11bh-2024, 802.11be-2024, 802.11bk-2025, 802.11bf-2025 do not add new subclause

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12.6.1.1.14 PGTKSA

An Authenticator's SME creates a PGTKSA when dot11EPPEpochActivated is true. A PGTKSA has the lifetime of the AP MLD.

A Supplicant's SME creates a PGTKSA when dot11EPPEpochActivated is true and the SME receives a PGTK from its Authenticator.

A PGTKSA consists of the following:

— PGTK
— Authenticator MAC address

12.6.1.1.15 Identity key security association

An Authenticator's SME creates an Identity Key Security Association for an identity key.

A Supplicant's SME creates an Identity Key Security Association when the SME receives an Identity Key from its Authenticator. The Identity Key Security Association has a lifetime that is signaled by the authenticator together in the Identity Key KDE.

An Identity Key Security Association consists of the following:

— Director vector (whether the identity key is used for transmit or receive).
— Identity key
— Lifetime of the identity key

12.6.1.2 Security association life cycle

12.6.1.2.2 Security association in an ESS

Change item b) and item c) of the second paragraph (not all shown) as follows:

A STA and AP establish an initial security association via the following steps:

b) The STA then performs IEEE 802.11 authentication followed by association to the chosen AP. Confirmation of security parameters takes place during association. A non-DMG STA performing IEEE 802.1X authentication uses Open System authentication or IEEE 802.1X authentication utilizing Authentication frames. A STA performing password-based authentication can use SAE authentication. A STA performing FILS uses FILS authentication. A non-DMG STA executing the OWE exchange uses Open System authentication.

c) SAE authentication, IEEE 802.1X authentication, and FILS authentication provides mutual authentication and derivation of a PMK. If Open System authentication is chosen instead, the Authenticator or the Supplicant initiates IEEE 802.1X authentication after (re)association. The EAP method used by IEEE Std 802.1X-2020 needs to support mutual authentication, as the STA needs assurance that the AP is a legitimate AP.

Change the fourth note as follows:

NOTE 4—Prior to the completion of IEEE 802.1X authentication and the installation of keys, the IEEE 802.1X Controlled Port in the AP blocks all Data frames. The IEEE 802.1X Controlled Port returns to the unauthorized state and blocks all Data frames before invocation of an MLME-DELETEKEYS.request primitive. The IEEE 802.1X Uncontrolled Port allows EAPOL PDUs to pass between the Supplicant and Authenticator. The Supplicant and Authenticator might alternatively use Authentication frames to pass EAPOL PDUs.

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Change item d) of the second paragraph (not all shown) as follows:

d) The last step is key management. The authentication process, whether SAE authentication utilizing Authentication frames, or FILS authentication utilizing Authentication frame, or IEEE 802.1X authentication utilizing Authentication frames, or IEEE 802.1X authentication utilizing Data frames post association, or the OWE exchange utilizing association frames creates cryptographic keys shared between the cryptographic endpoints—the AP and STA, or the IEEE 802.1X AS and the STA, when using SAE/FILS/OWE or IEEE Std 802.1X, respectively. When using IEEE Std 802.1X, the AS transfers these keys to the AP, and if encryption of (Re)Association Request/Response frames is not negotiated, the AP and STA uses one of the key confirmation handshakes, e.g., the 4-way handshake or FT 4-way handshake, to complete security association establishment. When using SAE authentication or OWE there is no AS and therefore no key transfer; if encryption of (Re)Association Request/Response frames is not negotiated, the 4-way handshake is performed directly between the AP and STA. The key confirmation handshake indicates when the link has been secured by the keys and is ready to allow normal data traffic and protected robust Management frames. When FILS authentication is performed or if encryption of (Re)Association Request/Response frames is negotiated, the key confirmation is performed using association frames. Hence, no additional handshake is necessary.

Change second bullet of the third paragraph (not all shown) as follows:

When FT is not enabled, a STA BSS transitioning within an ESS establishes a new PMKSA by one of the five schemes:

— In the case of SAE authentication or IEEE 802.1X authentication followed by (re)association, the STA repeats the same actions as for initial contact association, but the non-AP STA also deletes the PTKSA when it roams from the old AP. Note that a STA can take advantage of the fact that it can perform SAE authentication or IEEE 802.1X authentication to multiple APs while maintaining a single association with one AP, and then use any of the PMKSAs created during authentication to effect a fast BSS transition.

Change the last paragraph as follows:

The MLME-DELETEKEYS.request primitive deletes the temporal key(s) established for the security association so that they cannot be used to protect subsequent IEEE 802.11 traffic. An SME uses this primitive when it deletes a PTKSA, GTKSA, IGTKSA, BIGTKSA, PGTKSA, WIGTKSA or TPKSA.

Insert the following paragraph at the end:

The Identity Key Security Association has a separate lifetime, and the identity key is deleted when its lifetime expires. A STA may use the identity key to identify the APs with that configured identity key from its Privacy beacons after the STA has been disassociated or deauthenticated. An SME uses the MLME DELETEKEYS.request primitive when it deletes an Identity Key.

12.6.2 RSNA Selection

Change the first paragraph as follows:

A STA not affiliated with a BPE MLD and prepared to establish RSNAs shall advertise its capabilities by including the RSNE in Beacon, Information Response, and Probe Response frames and may also include the RSNE in DMG Beacon and Announce frames. The included RSNE shall specify all of the authentication and cipher suites enabled by the STA's policy. A STA shall not advertise any authentication or cipher suite that is not enabled. (11ba)If WUR frame protection is enabled, a WUR AP shall advertise such capability by setting to 1 the Protected WUR Frame Support subfield in the RSNXE in its Beacon and Probe Response frames.

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An AP affiliated with a BPE AP MLD does not transmit Beacon and Probe Response frames. The AP sends Privacy Beacons to be identifiable for the STAs that have the configured identity key and the parameters required to authenticate and associated with the AP MLD as defined in 10.71.8.2 (BPE AP MLD discovery).

12.6.5 RSNA policy selection in an MBSS

Change the fourth paragraph as follows:

When establishing an RSNA in an MBSS, neither PASN authentication nor EPPKE authentication shall ~~not~~ be used.

12.6.7 RSN management of the IEEE 802.1X Controlled Port

Change the first paragraph as follows:

In an RSN, this standard relies on the fact that IEEE 802.1X Supplicants and Authenticators exchange protocol information via the IEEE 802.1X Uncontrolled Port or Authentication frames carrying EAPOL PDUs. The IEEE 802.1X Controlled Port is blocked from passing general data traffic between the STAs until an IEEE 802.1X authentication procedure completes successfully over the authentication frame exchanges carrying EAPOL PDUs (if using IEEE 802.1X authentication utilizing Authentication frames) or alternatively the IEEE 802.1X Uncontrolled Port. The security of an RSNA depends on this being true.

Change the seventh paragraph as follows:

EAPOL PDUs shall be carried in individually addressed MSDUs or individually addressed Authentication frames.

12.6.8 RSNA establishment in an infrastructure BSS

12.6.8.1 General

Change the first paragraph as follows:

When establishing an RSNA in a non-FT environment or during an FT initial mobility domain association, a STA shall use IEEE 802.11 SAE authentication, FILS authentication, IEEE 802.1X authentication, or Open System authentication prior to (re)association. PASN authentication may be used without (re)association.

Change the fifth paragraph as follows:

IEEE 802.1X authentication can be initiated by any one of the following mechanisms:

— If a STA negotiates to use IEEE 802.1X authentication during (re)association, the STA's management entity may respond to the MLME-ASSOCIATE.confirm (or indication) or MLME-REASSOCIATE.confirm (or indication) primitive by requesting the Supplicant (or Authenticator) to initiate IEEE 802.1X authentication if IEEE 802.1X authentication is not performed before (re)association. Thus, in this case, authentication is driven by the STA's decision to associate and the AP's decision to accept the association.

— If a STA's MLME-SCAN.confirm primitive finds another AP within the ESS of which the STA is a member, a STA may signal its Supplicant to use IEEE Std 802.1X-2020 to preauthenticate with that AP (see 12.6.8.2 (Preauthentication and RSNA key management)).

NOTE 2—A BSS transitioning STA's IEEE 802.1X Supplicant can initiate preauthentication by sending an EAPOL-Start PDU (in one or more EAPOL-Start frames) via its old AP, through the DS, to a new AP.

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— If a STA's MLME-SCAN.confirm primitive finds another AP within the ESS that the STA is a member of that advertises support for IEEE 802.1X authentication utilizing Authentication frames in its RSNXE, a STA may signal its Supplicant to use IEEE Std 802.1X-2020 to authenticate with that AP (see 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames)).

— If a STA receives an IEEE 802.1X message, it delivers this to its Supplicant or Authenticator, which may initiate a new IEEE 802.1X authentication.

12.6.8.3 Cached PMKSAs and RSNA key management

Change the first and second paragraph as follows:

A STA might cache PMKSAs it establishes as a result of previous authentication. The PMKSA shall not be changed while cached. The PMK in the PMKSA is used with the 4-way handshake or FILS authentication to establish fresh PTKs or PTKSA derivation with IEEE 802.1X authentication frame exchange as defined in 12.16.8.3 (IEEE 802.1X) or EPPKE authentication as defined in 12.16.9 (Enhanced privacy protection key exchange).

If a STA in an infrastructure BSS has determined it has a valid PMKSA with an AP to which it is about to (re)associate, it performs Open System authentication to the AP, or performs FILS Authentication to the AP or performs PTKSA derivation with IEEE 802.1X Authentication frame exchange or performs EPPKE authentication, or (only in the case of a DMG STA) does not perform IEEE 802.11 authentication. When Open System authentication is used or IEEE 802.11 authentication is not performed, it includes the PMKID for the PMKSA in the RSNE in the (re)association request. When FILS Authentication is performed, it includes the PMKID for the PMKSA in the FILS Authentication frame (see 12.11.2 (FILS authentication protocol)). When PTKSA derivation with IEEE 802.1X Authentication frame exchange and PMKSA caching is performed, it includes the PMKID for the PMKSA in the first 802.1X Authentication frame. When EPPKE authentication with PMKSA caching is performed, it includes the PMKID for the PMKSA in the first EPPKE Authentication frame. When the PMKSA was not created using preauthentication, the AKM indicated in the RSNE by the STA in the (re)association request shall be identical to the AKM used to establish the cached PMKSA in the first place.

Change the fourth paragraph as follows:

If a cached PMKSA is used in FT Initial Mobility Domain Association, the cached MPMK is used to derive the PMK-R0 of a new FT key hierarchy (see 12.7.1.6 (FT key hierarchy)). The PMKID indicated by the STA in the ~~(Re)Association Request frame and message 1 of the FT 4-way handshake (when FILS authentication is not used) or~~ FILS Authentication frame (when FILS authentication is used), IEEE 802.1X Authentication frame (when PTKSA derivation with IEEE 802.1X Authentication frame exchange is used), EPPKE Authentication frame (when EPPKE authentication is used), or (Re)Association Request frame and the first message of the FT 4-way handshake (otherwise) is the PMKID of the cached PMKSA as defined in 12.7.1.6.3 (PMK-R0) (i.e., not the PMKR0Name or PMKR1Name of the FT key hierarchy that was derived when the PMKSA was originally established). The PMKR1Name indicated in the RSNE in messages 2 and 3 of the FT 4 way handshake (when FILS authentication is not used) or in (Re)Association Request and Response frames (when FILS authentication is used) is the PMKR1Name of the newly derived FT key hierarchy (see 13.4 (FT initial mobility domain association)).

Add the following after the sixth paragraph as follows:

Upon receipt of a FILS Authentication frame with one or more PMKIDs, an AP checks whether its Authenticator has cached a PMKSA for the PMKIDs, whether the AKM in the cached PMKSA matches the AKM in the FILS Authentication frame, and whether the PMK is still valid; and if so, it shall assert possession of that PMK by including the PMKID in the FILS Authentication frame sent in response (see 12.11.2.3.4 (AP construction of Authentication frame)). If the Authenticator does not have a PMK for the PMKIDs in the

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FILS Authentication frame or the AKM does not match, the AP may either reply with an EAP-Finish/ Reauth packet to continue FILS Shared Key authentication option if the non-AP STA included sufficient information for that, or the AP rejects the authentication.

The procedure for PTKSA derivation with IEEE 802.1X Authentication frame exchange and PMKSA caching is defined in 12.16.8.3 (IEEE 802.1X).

The procedure for the EPPKE authentication exchange and PMKSA caching is defined in 12.16.9 (Enhanced privacy protection key exchange).

Change the eighth paragraph as follows:

If both sides assert possession of a cached PMKSA, but the 4-way handshake, ~~or~~ FILS authentication, encrypted (re)association exchange with 802.1X Authentication frame exchange, or EPPKE authentication fails, both sides may delete the cached PMKSA for the selected PMKID.

12.6.16 RSNA security association termination

Change the second paragraph as follows:

it deletes some security associations. In the case of an ESS, the non-AP STA's SME shall delete any PTKSA(s), GTKSA(s), IGTKSA(s), BIGTKSA(s), WIGTKSA(s), WTKSA(s), TPKSA(s), the non-AP MLD's SME shall delete any PGTKSA(s) and the AP's SME shall delete the PTKSA. In the case of an IBSS, the SME shall delete the PTKSA(s) and the GTKSA(s) and any IGTKSA(s). Once the security associations have been deleted, the SME then invokes the MLME-DELETEKEYS.request primitive to delete all temporal keys associated with the deleted security associations.

12.6.17 Protection of robust Management frames

Insert the following after "note 6 - The STA is not sent …" as shown below

If management frame protection is negotiated for the link and encryption of (Re)Association Request/ Response frames is negotiated, a STA shall not transmit any of the following, and shall discard all of the following:

— An unprotected Association Request frame, Association Response frame, Reassociation Request frame, or Reassociation Response frame.

12.6.18 Robust management frame selection procedure

Insert the following to the end of Clause 12.6.18 as shown below

The selection rules for EPP robust individually addressed Management frames and EPP robust individually addressed beamforming/CSI/CQI frames are described in 12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames).

12.6.19 RSNA rekeying

Change the fifth paragraph as follows:

For MLO, the AP MLD's Authenticator manages packet number assignment for the PTKSA with a non-AP MLD. For a given link, the affiliated AP's Authenticator manages packet number assignment for the IGTKSA, GTKSA, or BIGTKSA. If an IGTKSA, GTKSA, or BIGTKSA update is triggered, the affiliated AP updates group keys for the given link through a group key handshake between the AP MLD and non-AP

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MLD. If a PGTKSA update is triggered, the AP MLD updates the PGTK through a group key handshake between the AP MLD and non-AP MLD.

Change the seventh paragraph as follows:

An Authenticator may initiate a group key handshake for the purpose of GTK rekeying (with a GTKSA), IGTK keying (with an IGTKSA), BIGTK rekeying (with a BIGTKSA), PGTK rekeying (with a PGTKSA) or WIGTK rekeying (with a WIGTKSA).

12.7 Keys and key distribution

12.7.1 Key hierarchy

12.7.1.1 General

Change the first paragraph as follows (not all lines shown):

RSNA defines the following key hierarchies:

f) WIGTK, a hierarchy consisting of a single key to provide integrity protection for broadcast and group addressed WUR Wake-up frames

g) PGTK, a hierarchy consisting of a single key used for frame anonymization

12.7.1.3 Pairwise key hierarchy

Change the second paragraph (not all lines shown) as follows:

The PTK is partitioned into PTK-KCK, PTK-KEK, a temporal key, and a KDK if derived under the conditions defined in 12.6.1.1.6 (PTKSA). ~~A KDK is derived if any of the following are true:~~

- ~~WUR frame protection is negotiated~~
- ~~dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.240 (RSNXE)))~~

Change the seventh paragraph (not all lines shown) as follows:

The following apply when not using FILS authentication:

- SNonce is a random or pseudorandom value contributed by the Supplicant; its value is taken when a PTK is instantiated and is sent to the PTK Authenticator.
- ANonce is a random or pseudorandom value contributed by the Authenticator.
- The PTK shall be derived from the PMK by

PTK = PRF-Length(PMK, "Pairwise key expansion", Min(AA,SPA) || Max(AA,SPA) || Min(ANonce,SNonce) || Max(ANonce,SNonce) || DHss) if key derivation with authentication frame exchange for IEEE 802.1X is used as defined in 12.16.8.3 (IEEE 802.1X).

Otherwise, PTK = PRF-Length(PMK, "Pairwise key expansion", Min(AA,SPA) || Max(AA,SPA) || Min(ANonce,SNonce) || Max(ANonce,SNonce))

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12.7.1.6 FT key hierarchy

12.7.1.6.5 PTK

Change the third paragraph as follows:

PTK = KDF-*Hash-Length*(PMK-R1, "FT-PTK", SNonce || ANonce || BSSID || STA-ADDR || DHss) if key derivation with Authentication frame exchange for FT is used as defined in 12.16.8.2 (FT protocol).

Otherwise, PTK = KDF-*Hash-Length*(PMK-R1, "FT-PTK", SNonce || ANonce || BSSID || STA-ADDR)

where

KDF-*Hash-Length*	is the key derivation function as defined in 12.7.1.6.2 (Key derivation function (KDE)) using the hash algorithm identified by the AKM suite selector (see Table 9-190 (AKM suite selectors))
PMK-R1	is the key that is shared between the S1KH and the R1KH
SNonce	is a 256-bit random bit string contributed by the S1KH
ANonce	is a 256-bit random bit string contributed by the R1KH
STA-ADDR	is the non-AP STA's MAC address
BSSID	is the BSSID of the target AP's BSS
Length	is the total number of bits to derive, i.e., number of bits of the PTK. The length is dependent on the negotiated cipher suites and AKMP as defined by Table 12-8 (Cipher suite key lengths) in 12.7.2 (EAPOL-Key frames) and Table 12-11 (Integrity and key wrap algorithms) in 12.7.3 (EAPOL-Key PDU construction and processing), and whether a KDK is derived (see 12.6.1.1.6 (PTKSA) for the conditions of derivation of KDK). ~~A KDK shall be derived if any of the following are true:~~ ~~— WUR frame protection is negotiated~~ ~~— dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.240 (RSNXE))~~ ~~Otherwise, it shall not be derived.~~

Insert the following subclause at the end of 12.7.1 (Key hierarchy):

-- Editor Note: 802.11-2024 up to 12.7.1.8, 802.11bh-2024, 802.11be-2024, 802.11bk-2025, 802.11bf-2025 do not add new subclause

12.7.1.9 Privacy group key hierarchy

The Authenticator shall select the PGTK as a random value each time it is generated.

The Authenticator may update the PGTK for any reason, including:

a) The disassociation or deauthentication of a non-AP MLD.

b) An event within the SME that triggers a group key handshake.

The PGTK is configured via the MLME-SETKEYS.request primitive; see 6.5.14 (SetKeys).

12.7.2 EAPOL-Key frames

-- Editor Note: ANA assignment is done.

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Modify Table 12-10 (KDE selectors) (not all lines shown) as follows:

Table 12-10—KDE selectors

OUI	Data type	Meaning
00-0F-AC	23	PGTK KDE
00-0F-AC	25	SAE Password Identifiers KDE

Insert the following at the end of 12.7.2 (EAPOL-Key frames):

-- Editor Note: 802.11-2024 up to Figure 12-50, 802.11bh-2024 up to Figure 12-50c, 802.11be-2024 up to Figure 12-50h, 802.11bk-2025, 802.11bf-2025 no new figure

The format of the PGTK KDE is shown in Figure 12-50i (PGTK KDE format).

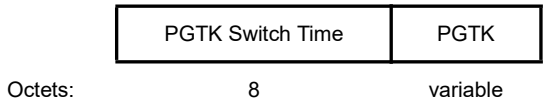


Figure 12-50i—PGTK KDE format

The PGTK Switch Time field indicates the time that the PGTK indicated in the Key field shall be applied to replace the PGTK in use by the EPP AP MLD and EPP non-AP MLDs. The PGTK Switch Time field is set to the time that the PGTK shall be applied by the EPP AP MLD and the EPP non-AP MLDs using, as a time-base, the value of the TSF corresponding to the BSS identified by the BSSID of the frame containing the PGTK KDE.

The PGTK field contains the PGTK. The length of PGTK is equal to the value of PMK_bits (see 12.7.1.3 (Pairwise key hierarchy)).

The format of the SAE Password Identifiers KDE is shown in Figure 12-50j (SAE Password Identifiers KDE format).

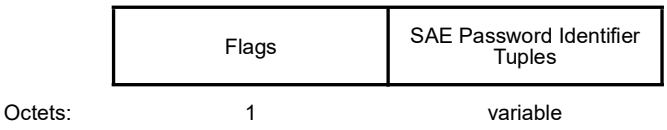


Figure 12-50j—SAE Password Identifiers KDE format

The format of the Flags field is shown in Figure 12-50k (Flags field format).

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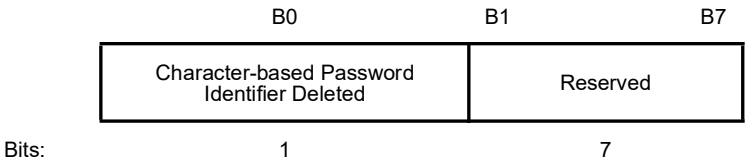


Figure 12-50k—Flags field format

The Flags field is set as follows:

— Bit 0: Character-based Password Identifier Deleted field (0 = not deleted, 1 = deleted).

— Bit 1–7: Reserved.

The SAE Password Identifier Tuples field contains one or more SAE Password Identifier Tuple fields.

The format of the SAE Password Identifiers Tuple field is shown in Figure 12-50l (SAE Password Identifier Tuple field format).

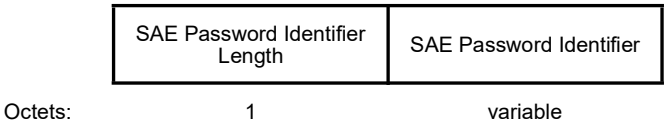


Figure 12-50l—SAE Password Identifier Tuple field format

The SAE Password Identifier Length field indicates the length of the following SAE Password Identifier field.

The SAE Password Identifier field contains an SAE Password Identifier.

12.7.4 EAPOL-Key PDU notation

change the first paragraph as follows (not all lines shown):

The following notation is used throughout the remainder of 12.7 (Keys and key distribution) and 13.4 (FT initial mobility domain association) to represent EAPOL-Key PDUs:

EAPOL-Key(S, M, A, I, K, Reserved, RSC, ANonce/SNonce, MIC, {Key Data})

where

....	.....
WIPN	is the last WIPN, as provided by the WIGTK KDE
PGTK[ST]	is the PGTK KDE, with the PGTK Switch Time field set to ST
Identity Key	is the Identity Key KDE

12.7.6 4-way handshake

12.7.6.2 4-way handshake message 1

change the third paragraph as follows:

The Authenticator sends message 1 to the Supplicant at the end of a successful IEEE 802.1X authentication, after (re)association completes for a STA that has authenticated with SAE or for which PSK authentication is

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negotiated, when a cached PMKSA is used, or after a STA requests a new key. On reception of message 1, the Supplicant determines whether the Key Replay Counter field value has been used before with the current PMKSA. If the Key Replay Counter field value is less than or equal to the current local value, the Supplicant discards the message. Otherwise, the Supplicant:

a) Generates a new nonce SNonce, if no SNonce has yet been generated for this 4-way handshake. The same SNonce is reused within this 4-way handshake until a valid message 3 has been received.

b) Derives PTK~~, the derived PTK including the Key derivation key (KDK) if WUR frame protection is being negotiated~~.

c) Constructs message 2.

12.7.6.4 4-way handshake message 3

Modify the list for Key Data as follows: (not all lines shown)

Message 3 uses the following values for each of the EAPOL-Key PDU fields:

...

Key Data =

— ....

— The SSID element containing the SSID of the BSS when both the Authenticator and the Supplicant have indicated support for SSID protection in the RSNXE.

— When SAE was used with a password identifier and the non-AP STA indicated support for changing the SAE password identifiers, optionally contains the SAE Password Identifiers KDE.

12.7.7 Group key handshake

12.7.7.1 General

Change the first paragraph as follows:

The Authenticator uses the Group key handshake to send a new GTK and, if management frame protection is negotiated, a new IGTK, and if beacon protection is enabled, a new BIGTK, and if WUR frame protection is negotiated, a new WIGTK, to the Supplicant. When the Authenticator is an AP MLD and the Supplicant is a non-AP MLD, the Authenticator may also use the Group key handshake to send new GTK(s) for any of the setup links and, if management frame protection is negotiated, new IGTK(s) for any of the setup links, and if beacon protection is enabled, new BIGTK(s) for any of the setup links to the Supplicant and if EPP epoch operation is supported by both the AP MLD and the non-AP MLD, a new PGTK.

Change the second paragraph as follows:

The Authenticator may initiate the exchange at any time when a Supplicant is disassociated or deauthenticated.

Message 1: Authenticator→Supplicant:

EAPOL-Key(1,1,1,0,G,0,RSC,0, MIC, {[GTK(N)] [, OCI} [, IGTK(M, IPN)] [, BIGTK(Q, BIPN)] [, WIGTK(R, WIPN)] [, MLO GTKn] [, MLO IGTKn] [, MLO BIGTKn] [, PGTK (ST)]})

Message 2: Supplican→Authenticator: EAPOL-Key(1,1,0,0,G,0,0,0,MIC,{ [OCI]})

Change the third paragraph as follows (not all lines shown):

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The following apply:

— WIGTK[R], when present, denotes the WIGTK with its key ID as encapsulated using the KDE as defined in 12.7.2 (EAPOL-Key frames) using the PTK-KEK defined in 12.7.1.3 (Pairwise key hierarchy) and associated IV.

— PGTK, when present, denotes the PGTK with its Switch Time as encapsulated using the KDE as defined in 12.7.2 (EAPOL-Key frames).

12.7.7.2 Group key handshake message 1

Change the first paragraph by inserting a sub-bullet at the end of Key Data as follows (not all lines shown):

Message 1 uses the following values for each of the EAPOL-Key PDU fields:

Key Data =

— For MLO, when present, PGTK and its switch time (see 12.7.2 (EAPOL-Key frames))

— For BPE MLOAP MLD, when present, Identity Key (see 12.7.2 (EAPOL-Key frames))

Change the third paragraph as follows (not all lines shown):

On reception of message 1, the Supplicant:

d) When the Supplicant is not an MLD, uses the MLME-SETKEYS.request primitive to configure the GTK and, the IGTK when present, and the BIGTK if beacon protection is enabled at the non-AP STA, and the WIGTK if WUR frame protection is negotiated, into the MAC. When the Supplicant is a non-AP MLD, uses the MLME-SETKEYS.request primitive to configure the GTK(s) when present and, the IGTK(s) when present, and the BIGTK(s) when present for the indicated link(s) into the MAC of the affiliated non-AP STA(s) operating on the indicated link(s). When the Supplicant is a non-AP MLD, uses the MLME-SETKEYS.request primitive to configure the PGTK when present, into the MAC at the switch time and the identity key when present into the MAC.

12.11 Authentication for FILS

12.11.2 FILS authentication protocol

12.11.2.3 Key establishment with FILS Shared Key authentication

12.11.2.3.2 Non-AP STA construction of Authentication frame

Change the third paragraph (not all shown) as follows:

The STA then constructs an Authentication frame with the Authentication algorithm number set to 4 (FILS Shared Key authentication without PFS) or 5 (FILS Shared Key authentication with PFS) (see 9.4.1.1 (Authentication Algorithm Number field)) depending on whether PFS is used, and the Authentication transaction sequence number set to 1. The random nonce shall be encoded in the ~~FILS~~ Nonce element (see 9.4.2.188 (FILS Nonce element)).

12.11.2.4 Key establishment with FILS Public Key authentication

12.11.2.4.2 Prior to exchange

Change the first paragraph (not all shown) as follows:

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FILS Public Key authentication performs key establishment with a Diffie-Hellman exchange. Prior to beginning the exchange, the non-AP STA performs the following:

a) Selects a finite cyclic group from the dot11RSNAConfigDLCGroupTable to perform the Diffie-Hellman exchange.

b) Generates a random 16-octet nonce, generates an ephemeral private key, and uses the selected group's scalar-op (see 12.4.4.1 (General)) with its private key to generate its ephemeral public key.

c) Determine whether to attempt PMKSA caching and if so, generates a list of PMKSA identifiers.

d) Constructs an Authentication frame (see 9.3.3.11 (Authentication frame format)) as follows:

   1) The Authentication algorithm number shall be set to 6 (FILS Public Key authentication) (see 9.4.1.1 (Authentication Algorithm Number field)) and the Authentication transaction sequence number shall be set to 1.

   2) The random nonce shall be encoded in the ~~FILS~~ Nonce element (see 9.4.2.188 (FILS Nonce element)).

12.11.2.4.4 Post processing

Change the first paragraph (not all shown) as follows:

Next, the AP shall

a) Generate a random 16-octet nonce and a random ephemeral private key, and then uses the agreed-upon group's scalar-op (see 12.4.4.1 (General)) with its private key to generate its ephemeral public key.

b) Construct an Authentication frame (see 9.3.3.11 (Authentication frame format)) as follows:

   1) The Authentication algorithm number is set to 6 (FILS Public Key authentication) (see 9.4.1.1 (Authentication Algorithm Number field)), and the Authentication transaction sequence number is set to 2.

   2) The random nonce is encoded in the ~~FILS~~ Nonce element (see 9.4.2.188 (FILS Nonce element)).

12.11.2.6 Key confirmation with FILS authentication

12.11.2.6.3 (Re)Association Response for FILS key confirmation

Change the second paragraph as follows:

The FILSR constructs a Key Delivery element indicating the current GTK and GTK PN, and the current IGTK and IPN if management frame protection is enabled, and the current BIGTK and BIPN if beacon protection is enabled, and the current WIGTK and WIPN if WUR frame protection is enabled, the current PGTK if EPP epoch operation is supported by both the AP MLD and the non-AP MLD, and the identity key if the AP MLD is a BPE AP MLD. For non-MLO, the GTK is carried in a GTK KDE. The IGTK and IPN are carried in an IGTK KDE, the BIGTK and BIPN are carried in a BIGTK KDE and the WIGTK and WIPN are carried in a WIGTK KDE. For MLO, the PGTK is carried in a PGTK KDE, the identity key is carried in a Identity Key KDE, GTKs for all setup links are carried in MLO GTK KDEs, the IGTKs in MLO IGTK KDEs, and the BIGTKs in MLO BIGTK KDEs.

Change the last paragraph as follows

Upon successful completion of the FILS authentication procedure, the FILSO shall process the Key Delivery element in the (Re)Association Response frame. The FILSO installs the GTK and GTK RSC, and IGTK and IGTK RSC if management frame protection is enabled, and BIGTK and BIGTK RSC if present in the Key Delivery element and dot11BeaconProtectionEnabled is true, and WIGTK and WIGTK RSC if present

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in the Key Delivery element and dot11RSNAWURFrameProtectionActivated is true and PGTK if present in the Key Delivery element and EPP epoch operation is supported by both the AP MLD and the non-AP MLD. For MLO, the FILSO installs the identity key if the identity key has not been installed and the PGTK and installs GTKs, IGTKs and BIGTKs for each setup link.

Insert the following new subclauses at the end of 12 (Security):

-- Editor Note: 802.11-2024 up to 12.15, 802.11bh-2024, 802.11be-2024, 802.11bk-2025, 802.11bf-2025 do not add new subclause

-- Editor Note: 802.11-2024 up to Table 12-13

-- Editor Note: 802.11-2024 up to Equation 12-9, 802.11bh-2024, 802.11be-2024, 802.11bk-2025, 802.11bf-2025 do not add new equation

12.16 Client privacy enhancement

12.16.1 Introduction

When dot11MACPrivacyActivated is true, an EPP STA or an EPP MLD has a MAC that comprises the functions defined in 12.2.11 (Requirements for support of MAC privacy enhancements) except that the functions in 12.16 (Client privacy enhancement) supersede the functions in 12.2.11 (Requirements for support of MAC privacy enhancements).

12.16.2 Contents of Probe Request frame

This subclause defines rules for the contents of a Probe Request frame that is not a multi-link probe request that preserve privacy.

An EPP AP that is not a BPE AP shall follow the rules defined in 11.1.4.3.4 (Criteria for sending a response) when receiving a Probe Request frame addressed to it even if the frame does not contain some of the elements that Table 9-68 (Probe Request frame body) indicates to be present when the soliciting non-AP STA follows the rules described in this subclause.

An EPP non-AP STA follows the rules defined in 35.3.4.5 (Probe Request frame content for a non-AP EHT STA) to determine the contents of a Probe Request frame that is not a multi-link probe request with the following exceptions:

- The EPP non-AP STA may omit the Supported Rates and BSS Membership Selectors element.
- In the 2.4 GHz band, if the Supported Rates and BSS Membership Selectors element is included, the EPP STA should indicate only 1, 2, 5.5, 6, 11, 12, and 24 Mb/s in the Supported Rates and BSS Membership Selectors element and should not include the Extended Supported Rates and BSS Membership Selectors element.
- In the 5 GHz or 6 GHz band, if the Supported Rates and BSS Membership Selectors element is included, the EPP STA should indicate only 6, 12, and 24 Mb/s in the Supported Rates and BSS Membership Selectors element and should not include the Extended Supported Rates and BSS Membership Selectors element.

NOTE 1—The inclusion of the Request element, the SSID List element, the Extended Request element, the FILS Request Parameters element, the Short SSID List element, the Known BSSID element, and the Vendor Specific elements, is optional as described in Table 9-68 (Probe Request frame body) and an EPP non-AP STA can omit these elements to preserve privacy.

NOTE 2—An EPP STA can follow the rules defined in 12.2.10 (Requirements for support of MAC privacy enhancements) to avoid leakage of possibly sensitive network identifying information in the SSID element.

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12.16.3 EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames

This subclause defines rules for the individually addressed Management frames described in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust) and the individually addressed beamforming/CSI/CQI frames described in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust).

Table 12-13a—EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust

Robust	Not Robust
Protected Notify Channel Width frame (see 9.6.40.2 (Protected Notify Channel Width frame format))	Notify Channel Width frame (see 9.6.11.2 (Notify Channel Width frame format))
Protected SM Power Save frame (see 9.6.40.3 (Protected SM Power Save frame format))	SM Power Save frame (see 9.6.11.3 (SM Power Save frame format))
Protected Group ID Management frame (see 9.6.41.3 (Protected Group ID Management frame format))	Group ID Management frame (see 9.6.22.3 (Group ID Management frame format))
Protected Operating Mode Notification frame (see 9.6.41.4 (Protected Operating Mode Notification frame format))	Operating Mode Notification frame (see 9.6.22.4 (Operating Mode Notification frame format))
Protected Quiet Time Period frame (see 9.6.32.5 (Protected Quiet Time Period frame format))	Quiet Time Period frame (see 9.6.31.3 (Quiet Time Period action frame format))

When performing operations that need to use any individually addressed Management frame that is in the Not Robust column in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust), if management frame protection is negotiated and both STAs set the EPP Robust Individually Addressed Management Frame Support field in the RSNXE that they transmit to 1, the STAs shall

— use the corresponding robust individually addressed Management frame described in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust) instead of the individually addressed Management frame that is in the Not Robust column in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust) and

— discard any individually addressed Management frame that is in the Not Robust column in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust) from the peer STA.

If management frame protection is not negotiated or either STA does not indicate support for the EPP robust individually addressed Management frame, the STAs shall not use any robust individually addressed Management frame described in Table 12-13a (EPP robust individually addressed Management frame and its corresponding individually addressed Management frame that is not robust).

When performing operations that need to use any individually addressed beamforming/CSI/CQI frame that is in the Not Robust column in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust), if management frame protection is negotiated, the receiving STA sets the EPP Robust Individually Addressed Beamforming/CSI/CQI Frame Rx Support field in the RSNXE that it transmits to 1, and the transmitting STA sets either EPP Robust Individually Addressed Beamforming/CSI/CQI TB Frame Tx Support field in

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Table 12-13b—EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust

Robust	Not Robust
Protected CSI frame (see 9.6.40.4 (Protected CSI frame format))	CSI frame (see 9.6.11.4 (CSI frame format))
Protected Noncompressed Beamforming frame (see 9.6.40.5 (Protected Noncompressed Beamforming frame format))	Noncompressed Beamforming frame (see 9.6.11.5 (Noncompressed Beamforming frame format))
Protected Compressed Beamforming frame (see 9.6.40.6 (Protected Compressed Beamforming frame format))	Compressed Beamforming frame (see 9.6.11.6 (Compressed Beamforming frame format))
Protected VHT Compressed Beamforming frame (see 9.6.41.2 (Protected VHT Compressed Beamforming frame format))	VHT Compressed Beamforming frame (see 9.6.22.2 (VHT Compressed Beamforming frame format))
Protected HE Compressed Beamforming/CQI frame (see 9.6.32.4 (Protected HE Compressed Beamforming/CQI frame format))	HE Compressed Beamforming/CQI frame (see 9.6.31.2 (HE Compressed Beamforming/CQI frame format))
Protected EHT Compressed Beamforming/CQI frame (see 9.6.38.15 (Protected EHT Compressed Beamforming/CQI frame format))	EHT Compressed Beamforming/CQI frame (see 9.6.37.2 (EHT Compressed Beamforming/CQI frame format))

the RSNXE that it transmits to 1 or EPP Robust IndividuallyAddressed Beamforming/CSI/CQI Non-TB Frame Tx Support field in the RSNXE that it transmits to 1 or both of them to 1 then

— the transmitting STA shall use the corresponding robust individually addressed beamforming/CSI/CQI frame described in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust) instead of the individually addressed beamforming/CSI/CQI frame in the Not Robust column in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust) and

— the receiving STA shall discard any individually addressed beamforming/CSI/CQI frame that is in the Not Robust column in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust) from the peer STA with negotiated management frame protection.

The RSNXE capabilities signal the beamforming allowed to a STA as shown in Table 12-13c (RSNXE beamforming TX capability fields settings and allowed beamforming).

Table 12-13c—RSNXE beamforming TX capability fields settings and allowed beamforming

EPP Robust IndividuallyAddressed Beamforming/CSI/CQI Frame TB Tx Support field	EPP Robust IndividuallyAddressed Beamforming/CSI/CQI Frame Non-TB Tx Support field	Beamforming allowed to the STA
0	0	Non protected beamforming only
0	1	Protected non-TB beamforming only

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Table 12-13c—RSNXE beamforming TX capability fields settings and allowed beamforming

EPP Robust IndividuallyAddressed Beamforming/CSI/CQI Frame TB Tx Support field	EPP Robust IndividuallyAddressed Beamforming/CSI/CQI Frame Non-TB Tx Support field	Beamforming allowed to the STA
1	0	Protected TB beamforming only
1	1	Protected non-TB and TB beamforming only

NOTE —The AP might trigger only one STA, if only one STA uses the protected TB beamforming.

If management frame protection is not negotiated or the transmitting STA does not indicate support for the transmission of the EPP robust individually addressed beamforming/CSI/CQI frame, or the receiving STA does not indicate support for the reception of the EPP robust individually addressed beamforming/CSI/CQI frame, the transmitting STA shall not transmit any robust individually addressed beamforming/CSI/CQI frame described in Table 12-13b (EPP robust individually addressed beamforming/CSI/CQI frame and its corresponding individually addressed beamforming/CSI/CQI frame that is not robust) to the receiving STA.

12.16.4 Capabilities and operation parameters request and response procedure

12.16.4.1 Introduction

This subclause defines rules to request and respond with capabilities and operation parameters using an EPP Capabilities And Operation Parameters Request frame and an EPP Capabilities And Operation Parameters Response frame.

12.16.4.2 Non-MLO procedure

For non-MLO, a non-AP STA that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1 may send an EPP Capabilities And Operation Parameters Request frame without a Basic Multi-Link element to request capabilities and operation parameters from an associated AP that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1.

An AP that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1 and receives an individually addressed EPP Capabilities And Operation Parameters Request frame without a Basic Multi-Link element shall respond with an individually addressed EPP Capabilities And Operation Parameters Response frame without a Basic Multi-Link element.

An AP that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1 may transmit an unsolicited individually addressed EPP Capabilities And Operation Parameters Response frame without a Basic Multi-Link element to an associated non-AP STA that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1. The EPP Capabilities And Operation Parameters Response frame shall include all elements that would be included in a Probe Response frame except the Multi-Link element and the Multiple BSSID element, and the elements shall be in the order defined for a Probe Response frame. The EPP Capabilities And Operation Parameters Response frame shall also include the Group Parameter update element.

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12.16.4.3 MLO procedure

For MLO, all STAs affiliated with an MLD shall set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to the same value.

A non-AP STA affiliated with a non-AP MLD that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1 may send an EPP Capabilities And Operation Parameters Request frame with a Basic Multi-Link element to request capabilities and operation parameters of APs affiliated with an associated AP MLD if APs affiliated with the associated AP MLD set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1.

If APs affiliated with an AP MLD set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1 and the AP MLD receives through a setup link from an associated non-AP MLD an individually addressed EPP Capabilities And Operation Parameters Request frame with a Basic Multi-Link element, then the AP MLD shall respond with an individually addressedEPP Capabilities And Operation Parameters Response frame through an affiliated AP over an enabled link to the non-AP MLD.

If APs affiliated with an AP MLD set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1, the AP MLD may send an unsolicited individually addressed EPP Capabilities And Operation Parameters Response frame to an associated non-AP MLD through a setup link, where non-AP STAs affiliated with the non-AP MLD set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1. If BPE APs affiliated with a BPE AP MLD set the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1, a BPE AP affiliated with the BPE AP MLD may send an unsolicited broadcast addressed EPP Capabilities And Operation Parameters Response frame if there exists at least one BPE non-AP STA affiliated with an associated BPE non-AP MLD corresponding to the setup link that sets the EPP Capabilities And Operation Parameters Request/Response Support field in the RSNXE to 1. The EPP Capabilities And Operation Parameters Response frame shall include the Group Parameter update element.

An unsolicited EPP Capabilities And Operation Parameters Response frame shall include a Basic Multi-Link element, and the Basic Multi-Link element shall include a Per-STA Profile subelement with the Complete Profile subfield set to 1 for each AP affiliated with the AP MLD (see 9.4.2.322.2.4 (Link Info field of the Basic Multi-Link element)). The individually addressed EPP Capabilities And Operation Parameters Response frame sent in response to an EPP Capabilities And Operation Parameters Request frame shall include a Basic Multi-Link element, and the Basic Multi-Link element shall include a Per-STA Profile subelement with the Complete Profile subfield set to 1 for each requested AP affiliated with the associated AP MLD (see 9.4.2.322.2.4 (Link Info field of the Basic Multi-Link element)). If the Basic Multi-Link element in the EPP Capabilities And Operation Parameters Request frame does not include any per-STA profiles, then all APs affiliated with the associated AP MLD shall be the requested APs. If the Basic Multi-Link element in the EPP Capabilities And Operation Parameters Request frame includes one or more per-STA profiles, then only the APs affiliated with the associated AP MLD with a link ID equal to the value in the Link ID field in a per-STA profile in the Basic Multi-Link element in the EPP Capabilities And Operation Parameters Request frame shall be the requested APs.

The STA Profile field in the Per-STA Profile subelement for each AP affiliated with the AP MLD shall include the following in order and shall not follow 35.3.3.3 (Advertisement of complete or partial per-link information):

— The Capability Information field as defined in 9.4.1.4 (Capability Information field).

— All elements that are not inherited and would be included in a Probe Response frame except the Multi-Link element and the Multiple BSSID element, and the elements shall be in the order defined for a Probe Response frame.

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Except for the Vendor Specific element, if an element, identified by an Element ID and Element ID Extension (if applicable), is carried in the first per STA profile subelement in the Basic Multi-Link element of the EPP Capabilities And Operation Parameters Response frame and there is no element having the same Element ID and Element ID Extension (if applicable) in a complete profile of a following per STA profile subelement in the same Basic Multi-Link element, then the element is inherited and is considered to be part of the following per STA profile subelement and the value of the element to use in the following per STA profile subelement is the same as that of the corresponding element carried in the first per STA profile subelement unless the following per STA profile subelement carries the Non-Inheritance element (see 9.4.2.239 (Non-Inheritance element)) and the element is listed in that Non-Inheritance element.

The Vendor Specific elements are not inherited in any per STA profile subelement in the Basic Multi-Link element of the EPP Capabilities And Operation Parameters Response frame. If there is a Vendor Specific element that is required to be part of a per STA profile subelement in the Basic Multi-Link element of the EPP Capabilities And Operation Parameters Response frame, the Vendor Specific element is included in that per STA profile subelement.

12.16.5 IEEE 802.1X authentication utilizing Authentication frames

If an AP sets the IEEE 802.1X Authentication Utilizing Authentication Frame Support field in the RSNXE that it transmits to 1, then a non-AP STA (originator) with dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated equal to true may signal its Supplicant to authenticate with the AP (responder) using IEEE Std 802.1X-2020 utilizing Authentication frames by transmitting the Authentication frames to the AP.

If any AP affiliated with an AP MLD sets the IEEE 802.1X Authentication Utilizing Authentication Frame Support field in the RSNXE that it transmits to 1, then a non-AP MLD (originator) with dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated equal to true may signal its Supplicant to authenticate with the AP MLD (responder) using IEEE Std 802.1X-2020 utilizing Authentication frames by transmitting the Authentication frames to the AP through a non-AP STA affiliated with the non-AP MLD.

When the originator is a non-AP MLD and the responder is an AP MLD, the RA field of an Authentication frame in response to an Authentication frame from the peer shall be set to the TA field of the Authentication frame from the peer.

If an originator chooses to initiate IEEE 802.1X authentication utilizing Authentication frames, it first selects an IEEE 802.1X AKM that is supported by the responder.

The originator then shall construct the first Authentication frame of the exchange as follows:

- Authentication Algorithm Number field is set to 8 (IEEE 802.1X authentication).
- Authentication Transaction Sequence Number field is set to 1.
- The Encapsulation field carries an EAPOL PDU from the PAE.
- Include the AKM Suite Selector element indicating the selected IEEE 802.1X AKM.

The originator sends the first Authentication frame to the responder.

Upon receiving the first Authentication frame, the responder shall:

- Validate that the AKM indicated in AKM Suite Selector element is an IEEE 802.1X AKM. Otherwise, processing status is set to STATUS_INVALID_AKMP.
- Validate that the selected IEEE 802.1X AKM indicated in AKM Suite Selector element is supported. Otherwise processing status is set to STATUS_INVALID_AKMP.
- If the validation is successful, extract an EAPOL PDU from the Encapsulation field, and forward the EAPOL PDU to the PAE.

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The responder then shall construct the second Authentication frame of the exchange as follows:

- Authentication Algorithm Number field is set to 8 (IEEE 802.1X authentication).
- Authentication Transaction Sequence Number field is set to 2.
- Status Code field indicates the processing status.
- The Encapsulation Length field indicates 0 if the status is equal to STATUS_INVALID_AKMP.
- The Encapsulation field (if present) carries an EAPOL PDU from the PAE.
- Include the AKM Suite Selector element indicating the same IEEE 802.1X AKM indicated in the first Authentication frame.

Once the processing is complete, the responder sends the second Authentication frame to the originator. If the processing status returned in the frame was not SUCCESS, the responder shall terminate the authentication.

Upon receiving the second Authentication frame, the originator shall:

- Validate that the Status Code field is SUCCESS. Otherwise, the originator shall terminate the authentication.
- Validate that the AKM indicated in AKM Suite Selector element is the same as the one indicated in the first Authentication frame. Otherwise, processing status is set to STATUS_INVALID_AKMP.
- If the validation is successful, extract an EAPOL PDU from the Encapsulation field (if present), and forward the EAPOL PDU to the PAE.

The originator then shall construct the third Authentication of the exchange as follows:

- Authentication Algorithm Number field is set to 8 (IEEE 802.1X authentication).
- Authentication Transaction Sequence Number field is set to 3.
- Status Code field indicates the processing status.
- The Encapsulation Length field indicates 0 if the status is equal to STATUS_INVALID_AKMP.
- The Encapsulation field (if present) carries an EAPOL PDU from the PAE.

Once the processing is complete, the originator sends the third Authentication frame to the responder. If the processing status returned in the frame was not SUCCESS, the originator shall terminate the authentication.

Upon receiving the Authentication frame with Authentication Transaction Sequence Number field larger than or equal to 3, the originator or the responder:

- Extract an EAPOL PDU (if present) from the Encapsulation field, and forward the EAPOL PDU to the PAE.
- Validate that the Status Code field is SUCCESS or 802_1_X_AUTH_SUCCESS. Otherwise, the originator shall terminate the authentication after forwarding the EAPOL PDU (if present) to the PAE.

If needed by the EAP method, the originator or the responder then shall construct the Authentication frame of the exchange in response to the Authentication frame with Authentication Transaction Sequence Number field larger than or equal to 3, as follows:

- Authentication Algorithm Number field is set to 8 (IEEE 802.1X authentication).
- Authentication Transaction Sequence Number field is set to the value of the Authentication Transaction Sequence Number field of the Authentication frame being responded to +1.
- The Encapsulation field (if present) carries an EAPOL PDU from the PAE.
- Status Code field indicates the processing status.

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- For the responder, if the 802.1X authentication is successful, the Status Code field is set to 802_1_X_AUTH_SUCCESS.
- For the responder, if the 802.1X authentication fails, the Status Code field is set to 802_1_X_-AUTH_FAILED.
- Otherwise, the Status Code field is set to the appropriate status code.

NOTE 1—The number of Authentication frame exchanges depends on the EAP method in use.

Once the processing is complete, the originator or the responder sends the Authentication frame in response to the Authentication frame with Authentication Transaction Sequence Number field set to a value that is larger than or equal to 3, to its peer (if needed by the EAP method). If the processing status returned in the frame was not SUCCESS or 802_1_X_AUTH_SUCCESS, the originator or the responder shall terminate the authentication.

NOTE 2—For MLO, Basic Multi-link element is included in every Authentication frame as defined in Table 9-70 (Authentication frame body).

12.16.6 (Re)Association Request/Response frame encryption

12.16.6.1 General

This subclause defines rules to encrypt the Frame Body field of the (Re)Association Request/Response frame and to include a DS MAC Address element in the encrypted (Re)Association Request frame.

An EPP STA that sets the (Re)Association Frame Encryption Support field in the RSNXE to 1 shall set the MFPC subfield in the RSN Capabilities field in the RSNE to 1.

12.16.6.2 Non-MLO procedure

Encryption of (Re)Association Request/Response frames is negotiated if both an EPP non-AP STA and an EPP AP set the (Re)Association Frame Encryption Support field in the RSNXE to 1.

An EPP non-AP STA that negotiates encryption of (Re)Association Request/Response frames shall indicate a pairwise cipher, establish a PTKSA, and derive a temporal key (TK) through the authentication frame exchange with an EPP AP (see 12.16.8 (Key derivation with authentication frame exchange) and 12.16.9 (Enhanced privacy protection key exchange)).

An EPP non-AP STA shall randomize its STA MAC address during a BSS transition if the BSS transition procedure uses an encrypted (Re)Association Request frame to carry the DS MAC Address element.

After a pairwise cipher is indicated by the EPP non-AP STA and a temporal key (TK) is derived during the authentication frame exchange between the EPP non-AP STA and an EPP AP, the EPP non-AP STA shall encrypt the (Re)Association Request frame transmitted to the EPP AP using the TK and the pairwise cipher indicated in the authentication frame exchange.

If the FT initial mobility domain association is used, then the EPP non-AP STA shall include the MDE and the FTE in the (Re)Association Request frame. MDE and the FTE shall be the same as the ones in the second Authentication frame as described in 12.16.8.4 (FT initial mobility domain association).

If the FT initial mobility domain association is used, then the PMKR1Name shall be included in the RSNE in the (Re)Association Request frame. The PMKR1Name shall be as calculated by the S1KH according to the procedures of 12.7.1.6.4 (PMK-R1).

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For the RSNE included in the (Re)Association Request frame, other than the Length field, the PMKID Count field and the PMKID List field, shall be identical to those in the RSNE present in the first Authentication frame.

If the FT protocol is used, then the EPP non-AP STA shall not calculate the MIC for the MIC field of the FTE in the Reassociation Request frame. The length of the MIC field of the FTE in the Reassociation Request frame shall be 0 (i.e., the MIC Length subfield of the MIC Control field of the FTE is set to 3). The Element Count subfield of the MIC Control field of the FTE shall be set to 0.

If the FILS authentication protocol is used, then the EPP non-AP STA shall not encrypt the (Re)Association Request frame using the AEAD algorithm as described in 12.11.2.6.2 (Re)Association Request for FILS key confirmation.

If dot11DSMACAddressActivated is true and the EPP AP sets the DS MAC Address Support field in the RSNXE to 1, the EPP non-AP STA shall include the DS MAC Address element in the (Re)Association Request frame to indicate the DS MAC address to be used by the EPP AP for the mapping to the DS.

The EPP non-AP STA may randomize the DS MAC address. To construct a random DS MAC address, the EPP non-AP STA shall select the randomized DS MAC address according to IEEE Std 802-2014 and IEEE Std 802c-2017. If dot11DSMACAddressActivated is true, the EPP non-AP STA shall use the same DS MAC address for the duration of its connection across an ESS.

NOTE 1—Detection and remediation of possible DS MAC address collisions are outside the scope of this standard.

The EPP AP shall decrypt the (Re)Association Request frame received from the EPP non-AP STA using the TK and the pairwise cipher indicated in the authentication frame exchange. If there is no output from the decryption algorithm because of unsuccessful MIC check (see 12.5.2.4.2 (CCM recipient processing) and 12.5.4.4.2 (GCM recipient processing)), then the EPP AP shall discard the frame.

If the FT initial mobility domain association is used, then the FTE and the MDE in the (Re)Association Request frame are checked to be the same as those provided in the second Authentication frame as defined in 12.16.8.4 (FT initial mobility domain association). If the MDE check fails, the (Re)Association Request frame is rejected with status code STATUS_INVALID_MDE. If the FTE check fails, the (Re)Association Request frame is rejected with status code STATUS_INVALID_FTE.

If the FT initial mobility domain association is used, then the PMKR1Name in the RSNE in the (Re)Association Request frame is checked to be included and calculated according to the procedures of 12.7.1.6.4 (PMK-R1). If the check fails, the (Re)Association Request frame is rejected with status code STATUS_INVALID_PMKID.

The EPP AP shall verify that the RSNE fields other than the Length field, the PMKID Count field and the PMKID List field in the (Re)Association Request frame is identical to the RSNE fields included in the first Authentication frame. The EPP AP shall also verify that the RSNXE in the (Re)Association Request frame is identical to the RSNXE included in the first Authentication frame. If the validation fails, the EPP AP shall reject the association.

If the FT protocol is not used and the FT initial mobility domain association is not used, the RSNE included in the (Re)Association Response frame shall be identical to the corresponding RSNE fields in the Beacon or Probe Response frames from the EPP AP.

If the FT initial mobility domain association is used, then EPP AP shall include the MDE, the FTE, TIE[ReassociationDeadline], and TIE[KeyLifetime] in the (Re)Association Response frame. The MDE and the FTE shall be the same as the ones in the second Authentication frame as described in 12.16.8.4 (FT ini-

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tial mobility domain association). The reassociation deadline timeout is set to the minimum of dot11FTReassociationDeadline and the key lifetime in the TIE[ReassociationDeadline], and the PTK lifetime in the TIE[KeyLifetime].

If the FT initial mobility domain association is used, then the PMKR1Name shall be included in the RSNE in the (Re)Association Response frame. The PMKR1Name shall be as calculated by the R1KH according to the procedures of 12.7.1.6.4 (PMK-R1). The RSNE fields other than the Length field, the PMKID Count field and the PMKID List field shall be identical to the corresponding RSNE fields in the Beacon or Probe Response frames from the EPP AP.

If the FT protocol is used, then the EPP AP shall not wrap the Key field of the subelements in the FTE in the Reassociation Response frame and shall not calculate the MIC for the MIC field of the FTE in the Reassociation Response frame. The length of the MIC field of the FTE in the Reassociation Response frame shall be 0 (i.e., the MIC Length subfield of the MIC Control field of the FTE is set to 3). The Element Count subfield of the MIC Control field of the FTE shall be set to 0.

If the FILS authentication protocol is used, then the EPP AP shall not encrypt the (Re)Association Response frame using the AEAD algorithm as described in 12.11.2.6.3 ((Re)Association Response for FILS key confirmation).

The EPP AP shall encrypt the (Re)Association Response frame transmitted to the EPP non-AP STA in response to the (Re)Association Request frame using the TK and the pairwise cipher indicated in the authentication frame exchange.

If the FILS authentication protocol and the FT protocol are not used, the EPP AP shall include a Key Delivery element in the (Re)Association Response frame.

If a Key Delivery element is included in the (Re)Association Response frame, the EPP AP shall construct the Key Delivery element indicating the current GTK PN in the RSC subfield, with the GTK KDE, with the IGTK KDE if management frame protection is enabled, with the BIGTK KDE if beacon protection is enabled, with the WIGTK KDE if WUR frame protection is enabled, and with the SAE Password Identifiers KDE if the EPP AP supports SAE password identifier change and the non-AP STA sets the SAE Password Identifier Change support bit in the RSNXE to 1.

The EPP non-AP STA shall decrypt the (Re)Association Response frame received from the EPP AP using the TK and the pairwise cipher indicated in the authentication frame exchange. If there is no output from the decryption algorithm because of unsuccessful MIC check (see 12.5.2.4.2 (CCM recipient processing) and 12.5.4.4.2 (GCM recipient processing)), then the EPP non-AP STA shall discard the frame.

If the FT protocol is not used, the FT initial mobility domain association is not used, and in the (Re)Association Response frame the RSNE fields are not identical to the corresponding RSNE fields in the Beacon or Probe Response frames from the EPP AP, the EPP non-AP STA shall discard the response.

If the FT protocol is not used, the EPP non-AP STA shall verify that the RSNXE included in the (Re)Association Response frame is identical to the RSNXE included in the Beacon or Probe Response frames from the EPP AP. If those frames did not include the RSNXE or if the RSNXEs are not identical, the EPP non-AP STA shall discard the response.

If the FT initial mobility domain association is used, then the FTE and the MDE in the (Re)Association Response frame are checked to be the same as those provided in the second Authentication frame as defined in 12.16.8.4 (FT initial mobility domain association). If the check fails, the EPP non-AP STA shall discard the response.

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If the FT initial mobility domain association is used, then the PMKR1Name in the RSNE in the (Re)Association Response frame is checked to be included and identical to the value that is sent in the Association Request frame. If the check fails, the EPP non-AP STA shall discard the response.

If the FT initial mobility domain association is used and in the (Re)Association Response frame the RSNE fields other than the Length field, the PMKID Count field and the PMKID List field are not identical to the corresponding RSNE fields in the Beacon or Probe Response frames from the EPP AP, the EPP non-AP STA shall discard the response.

If IEEE 802.1X Authentication utilizing Authentication frame is used, and the RSN capabilities fields of the RSNE received in the (Re)Association Response frame is not identical to the RSN capabilities fields of the RSNE received in the second Authentication frame, the EPP non-AP STA shall discard the response.

On a successful (re)association,

— The EPP non-AP STA shall process the Key Delivery element in the (Re)Association Response frame if present.

— The EPP non-AP STA shall install the GTK and GTK RSC, and IGTK and IGTK RSC if management frame protection is enabled, and BIGTK and BIGTK RSC if present in the Key Delivery element and dot11BeaconProtectionEnabled is true, and WIGTK and WIGTK RSC if present in the Key Delivery element and dot11RSNAWURFrameProtectionActivated is true.

— The EPP AP and the EPP non-AP STA shall transition to State 4 (as defined in 11.3 (STA authentication and association)).

— If the DS MAC Address element is included in the (Re)Association Request frame, the EPP non-AP STA shall use the indicated DS MAC address rather than the MAC address of the EPP non-AP STA for the EPP non-AP STA to the EPP AP mapping to the DS.

— If the DS MAC Address element is included in the (Re)Association Request frame, the EPP AP shall process the DS MAC Address element and use the indicated DS MAC address rather than the MAC address of the EPP non-AP STA to establish the EPP non-AP STA to the EPP AP mapping to the DS.

NOTE 2—If the DS MAC Address element is included in the (Re)Association Request frame, the source address or destination address parameters of the MAC service tuples (see 5.2.4.2 (Semantics of the service primitive)) for the EPP non-AP STA are set to the DS MAC address, which is the identity of the non-AP STA known by the DS.

On a failed (re)association, the established PTKSA shall be irretrievably deleted.

12.16.6.3 MLO procedure

Encryption of (Re)Association Request/Response frames is negotiated if both an EPP non-AP MLD and an EPP AP MLD set the (Re)Association Frame Encryption Support field in the RSNXE to 1.

A non-AP MLD that negotiates encryption of (Re)Association Request/Response frames shall indicate a pairwise cipher, establish a PTKSA, and derive a temporal key (TK) through the authentication frame exchange with an EPP AP MLD (see 12.16.8 (Key derivation with authentication frame exchange) and 12.16.9 (Enhanced privacy protection key exchange)).

NOTE 1—For MLO, all STAs affiliated with an MLD set the RSNXE to the same value.

An EPP non-AP MLD shall randomize the STA MAC addresses of its affiliated STAs and its MLD MAC address during a BSS transition if the BSS transition procedure uses an encrypted (Re)Association Request frame to carry the DS MAC Address element.

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After a pairwise cipher is indicated by the EPP non-AP MLD and a TK is derived during the authentication frame exchange between the EPP non-AP MLD and an EPP AP MLD, the EPP non-AP MLD shall encrypt the (Re)Association Request frame transmitted to the EPP AP MLD using the TK and the pairwise cipher indicated in the authentication frame exchange.

If the FT initial mobility domain association is used, then the EPP non-AP MLD shall include the MDE and the FTE in the (Re)Association Request frame. The MDE and the FTE shall be the same as the ones in the second Authentication frame as defined in 12.16.8.4 (FT initial mobility domain association).

If the FT initial mobility domain association is used, then the PMKR1Name shall be included in the RSNE in the (Re)Association Request frame. The PMKR1Name shall be as calculated by the S1KH according to the procedures of 12.7.1.6.4 (PMK-R1).

For the RSNE included in the (Re)Association Request frame, other than the Length field, the PMKID Count field and the PMKID List field, shall be identical to the RSNE fields in the first Authentication frame.

The (Re)Association Request frame shall:

— Have the Address 1 field set to the Address 1 field of the Authentication frame used by the non-AP MLD to establish the PTKSA.

— Have the Address 2 field set to the Address 2 field of the Authentication frame used by the non-AP MLD to establish the PTKSA.

— Include the DS MAC Address element in the (Re)Association Request frame to indicate the DS MAC address to be used by the EPP AP MLD for the mapping to the DS if dot11DSMACAddressActivated is true and the APs affiliated with the EPP AP MLD set the DS MAC Address Support field in the RSNXE to 1.

If the FT protocol is used, then the EPP non-AP MLD shall not calculate the MIC for the MIC field of the FTE in the Reassociation Request frame. The length of the MIC field of the FTE in the Reassociation Request frame shall be 0 (i.e., the MIC Length subfield of the MIC Control field of the FTE is set to 3). The Element Count subfield of the MIC Control field of the FTE shall be set to 0.

If the FILS authentication protocol is used, then the EPP non-AP MLD shall not encrypt the (Re)Association Request frame using the AEAD algorithm as described in 12.11.2.6.2 (Re)Association Request for FILS key confirmation.

An EPP non-AP MLD should not use the MLD MAC address as the DS MAC address. The EPP non-AP MLD may randomize its DS MAC address. To construct a random DS MAC address, the EPP non-AP MLD shall select the randomized DS MAC address according to IEEE Std 802-2014 and IEEE Std 802c-2017. If dot11DSMACAddressActivated is true, the EPP non-AP MLD shall use the same DS MAC address for the duration of its connection across an ESS.

NOTE 2—Detection and remediation of possible DS MAC address collisions are outside the scope of this standard.

The EPP AP MLD shall decrypt the (Re)Association Request frame received from the EPP non-AP MLD using the TK and the pairwise cipher indicated in the authentication frame exchange. If there is no output from the decryption algorithm because of unsuccessful MIC check (see 12.5.2.4.2 (CCM recipient processing) and 12.5.4.4.2 (GCM recipient processing)), then the EPP AP MLD shall discard the frame.

If the FT initial mobility domain association is used, then the FTE and the MDE in the (Re)Association Request frame are checked to be the same as those provided in the second Authentication frame as defined in 12.16.8.4 (FT initial mobility domain association). If the MDE check fails, the (Re)Association Request

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frame is rejected with status code STATUS_INVALID_MDE. If the FTE check fails, the (Re)Association Request frame is rejected with status code STATUS_INVALID_FTE.

If the FT initial mobility domain association is used, then the PMKR1Name in the RSNE in the (Re)Association Request frame is checked to be included and calculated according to the procedures of 12.7.1.6.4 (PMK-R1). If the check fails, the (Re)Association Request frame is rejected with status code STATUS_INVALID_PMKID.

The EPP AP MLD shall verify that the RSNE fields other than the Length field, the PMKID Count field and the PMKID List field in the (Re)Association Request frame is identical to the RSNE fields included in the first Authentication frame. The EPP AP MLD shall also verify that the RSNXE in the (Re)Association Request frame is identical to the RSNXE included in the first Authentication frame. If the validation fails, the EPP AP MLD shall reject the association.

If the FT protocol is not used and the FT initial mobility domain association is not used, the RSNE fields corresponding to each link included in the (Re)Association Response frame shall be identical to the corresponding RSNE fields of the link in the Beacon or Probe Response frames from the corresponding AP affiliated with the EPP AP MLD or in the multi-link probe response received from the EPP AP MLD.

If the FT initial mobility domain association is used, then EPP AP MLD shall include the MDE, the FTE, TIE[ReassociationDeadline], and TIE[KeyLifetime] in the (Re)Association Response frame. The MDE and the FTE shall be the same as the ones in the second Authentication frame as described in 12.16.8.1a (FT initial mobility domain association). The reassociation deadline timeout is set to the minimum of dot11FTReassociationDeadline and the key lifetime in the TIE[ReassociationDeadline], and the PTK lifetime in the TIE[KeyLifetime].

If the FT initial mobility domain association is used, then the PMKR1Name shall be included in the RSNE in the (Re)Association Response frame. The PMKR1Name shall be as calculated by the R1KH according to the procedures of 12.7.1.6.4 (PMK-R1). The RSNE fields other than the Length field, the PMKID Count field and the PMKID List field corresponding to each link shall be identical to the corresponding RSNE fields of the link in the Beacon or Probe Response frames from the corresponding AP affiliated with the EPP AP MLD or in the multi-link probe response received from the EPP AP MLD.

If the FT protocol is used, then the EPP AP MLD shall not wrap the Key field of the subelements in the FTE in the Reassociation Response frame and shall not calculate the MIC for the MIC field of the FTE. The length of the MIC field shall be 0 (i.e., the MIC Length subfield of the MIC Control field of the FTE is set to 3). The Element Count subfield of the MIC Control field of the FTE shall be set to 0.

If the FILS authentication protocol is used, then the EPP AP MLD shall not encrypt the (Re)Association Response frame using the AEAD algorithm as described in 12.11.2.6.3 ((Re)Association Response for FILS key confirmation).

The EPP AP MLD shall encrypt the (Re)Association Response frame transmitted to the EPP non-AP MLD in response to the (Re)Association Request frame using the TK and the pairwise cipher indicated in the authentication frame exchange.

If the FILS authentication protocol and the FT protocol are not used, the EPP AP MLD shall include a Key Delivery element in the (Re)Association Response frame.

If a Key Delivery element is included in the (Re)Association Response frame, the EPP AP MLD shall construct the Key Delivery element with the RSC field set to 0, with the MLO GTK KDE for each setup link, with the MLO IGTK KDE for each setup link if management frame protection is negotiated, with the MLO BIGTK KDE for each setup link if beacon protection is enabled, with the PGTK KDE if the Group EPP Epoch Supported field in the RSNXE of both the APs affiliated with the AP MLD and the non-AP MLD is

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equal to 1, with the Identity Key KDE if the AP MLD is a BPE AP MLD, and with the SAE Password Identifiers KDE if the EPP AP MLD supports SAE password identifier change ands the non-AP MLD sets the SAE Password Identifier Change Support bit in the RSNXE to 1.

The EPP non-AP MLD shall decrypt the (Re)Association Response frame received from the EPP AP MLD using the TK and the pairwise cipher indicated in the authentication frame exchange. If there is no output from the decryption algorithm because of unsuccessful MIC check (see 12.5.2.4.2 (CCM recipient processing) and 12.5.4.4.2 (GCM recipient processing)), the EPP non-AP MLD shall discard the frame.

If the FT protocol is not used, the FT initial mobility domain association is not used, and in the (Re)Association Response frame the RSNE fields corresponding to each link are not identical to the corresponding RSNE fields of the link in the Beacon or Probe Response frames from the corresponding AP affiliated with the EPP AP MLD or in the multi-link probe response received from the EPP AP MLD, the EPP non-AP MLD shall discard the response.

If the FT protocol is not used, the EPP non-AP MLD shall verify that the RSNXE corresponding to each link in the (Re)Association Response frame is identical to the corresponding RSNXE of the link in the Beacon or Probe Response frames from the corresponding AP affiliated with the EPP AP MLD or in the multi-link probe response received from the EPP AP MLD. If those frames did not include the RSNXE or if the RSNXEs are not identical, the EPP non-AP MLD shall discard the response.

If the FT initial mobility domain association is used, then the FTE and the MDE in the (Re)Association Response frame are checked to be the same as those provided in the second Authentication frame as defined in 12.16.8.4 (FT initial mobility domain association). If the check fails, the EPP non-AP MLD shall discard the response.

If the FT initial mobility domain association is used, then the PMKR1Name in the RSNE in the (Re)Association Response frame is checked to be included and identical to the value that is sent in the (Re)Association Request frame. If the check fails, the EPP non-AP MLD shall discard the response.

If the FT initial mobility domain association is used and in the (Re)Association Response frame the RSNE fields other than the Length field, the PMKID Count field and the PMKID List field corresponding to each link are not identical to the corresponding RSNE fields of the link in the Beacon or Probe Response frames from the corresponding AP affiliated with the EPP AP MLD or in the multi-link probe response received from the EPP AP MLD, the EPP non-AP MLD shall discard the response.

If IEEE 802.1X Authentication utilizing Authentication frame is used, and the RSN capabilities fields of the RSNE received in the (Re)Association Response frame is not identical to the RSN capabilities fields of the RSNE received in the second Authentication frame, the EPP non-AP MLD shall discard the response.

On successful (re)association,

- The EPP non-AP MLD shall process the Key Delivery element in the (Re)Association Response frame if present.
- The EPP non-AP MLD shall install the GTK and GTK RSC, and IGTK and IGTK RSC if management frame protection is enabled, and BIGTK and BIGTK RSC if present in the Key Delivery element and dot11BeaconProtectionEnabled is true, and PGTK if the Group EPP Epoch Supported field in the RSNXE of both the APs affiliated with the AP MLD and the non-AP MLD is equal to 1, and identity key if the AP MLD is a BPE AP MLD.
- The EPP AP MLD and the EPP non-AP MLD shall transition to State 4 (as defined in 11.3 (STA authentication and association)).
- If the DS MAC Address element is included in the (Re)Association Request frame, the EPP non-AP MLD shall use the indicated DS MAC address rather than the MLD MAC address of the non-AP MLD for the EPP non-AP MLD to the EPP AP MLD mapping to the DS.

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— If the DS MAC Address element is included in the (Re)Association Request frame, the EPP AP MLD shall process the DS MAC Address element and use the indicated DS MAC address rather than the MLD MAC address of the EPP non-AP MLD to establish the EPP non-AP MLD to the EPP AP MLD mapping to the DS.

NOTE 3—If the DS MAC Address element is included in the (Re)Association Request frame, the source address or destination address parameters of the MAC service tuples (see 5.2.4.2 (Semantics of the service primitive)) for the EPP non-AP MLD are set to the DS MAC address, which is the identity of the non-AP MLD known by the DS.

On failed (re)association, the established PTKSA shall be irretrievably deleted.

12.16.7 PMKSA caching privacy

12.16.7.1 General

This subclause defines rules for PMKSA caching privacy such that the identifier related to PMKSA caching can be changed, thus, cannot be used for tracking.

A STA that sets the PMKSA Caching Privacy Support field in the RSNXE to 1 shall set the (Re)Association Frame Encryption Support field in the RSNXE to 1.

12.16.7.2 PMKID privacy

For non-MLO, if the EPP non-AP STA and the EPP AP set the PMKSA Caching Privacy Support field in the RSNXE to 1,

— The EPP non-AP STA shall include the Nonce element in the (Re)Association Request frame.

— The EPP AP shall include the Nonce element in the (Re)Association Response frame.

For MLO, if the EPP non-AP STA(s) affiliated with an EPP non-AP MLD and the EPP AP(s) affiliated with an EPP AP MLD set the PMKSA Caching Privacy Support field in the RSNXE to 1,

— The EPP non-AP MLD shall include the Nonce element in the (Re)Association Request frame.

— The EPP AP MLD shall include the Nonce element in the (Re)Association Response frame.

After a PTKSA is established,

- For non-MLO, both the EPP non-AP STA and the EPP AP shall change the PMKID for the identified PMKSA to be used next time.
- For MLO, both the EPP non-AP MLD and the EPP AP MLD shall change the PMKID for the identified PMKSA to be used next time.

NOTE 1—For MLO, all STAs affiliated with an MLD set the RSNXE to the same value.

NOTE 2—When PMKSA Caching Privacy Support field in the RSNXE is equal to 1, then the AP stores only the anonymized PMKID created by the PMKID Nonces in the (Re)Association Request and Response frames.

The PMKID shall be changed as:

PMKID = Truncate-128(Hash("PMK Name" || PMKIDANonce || PMKIDSNonce))

where:

Hash	is the hash algorithm from the key derivation type (see Table 9-190 (AKM suite selectors)) for each AKM
PMKIDANonce	is the Authenticator nonce indicated in the Nonce element in the (Re)Association Response frame used to compute the changed PMKID.

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PMKIDSNonce	is the Supplicant nonce indicated in the Nonce element in the (Re)Association Request frame used to compute the changed PMKID.

NOTE 3—For a different PMKID indicated in a frame to ensure privacy, the MAC address in the TA field of the frame and the MLD MAC address (if present) in the Frame Body field of the frame need to be randomized to avoid tracking based on the MAC address. During the BSS transition, the DS MAC address can remain the same to preserve the mapping to the DS.

12.16.7.3 PMKR0Name privacy

APs in the same mobility domain shall set the PMKSA Caching Privacy Support field in the RSNXE to the same value.

If both an FTO and a target FTR set the PMKSA Caching Privacy Support field in the RSNXE to 1, after the target FTR uses the indicated PMKR0Name to identify the PMK-R1 (see 13.8.1 (Overview)), and a PTKSA is established using the identified PMK-R1,

— the R1KH of the target FTR shall send the latest ANonce and SNonce to the R0KH and

— both the S0KH of the FTO and the R0KH shall recompute the PMKR0Name.

NOTE 1 —For MLO, all STAs affiliated with an MLD set the RSNXE to the same value.

The PMKR0Name shall be recomputed as follows:

PMKR0Name = Truncate-128(HMAC-Hash(XXKey, “FT-R0N” || ANonce || SNonce))

where:

Hash	is the hash algorithm from the key derivation type (see Table 9-190 (AKM suite selectors)) for each AKM
“FT-R0N”	is treated as an ASCII string
XXKey	is defined in 12.7.1.6.3 (PMK-R0)
ANonce	is the Authenticator nonce used when the current PTKSA was established
SNonce	is the Supplicant nonce used when the current PTKSA was established

NOTE 2—For a different PMKR0Name indicated in a frame to ensure privacy, the MAC address in the TA field of the frame and the MLD address (if present) in the Frame Body field of the frame need to be randomized to avoid tracking based on the MAC address. During the BSS transition, the DS MAC address can remain the same to preserve the mapping to the DS.

NOTE 3—PMKR1Name is still derived based on the indicated PMKR0Name with the same formula defined in 12.7.1.6.4 (PMK-R1) for the first time and PMKR1Name once derived is not recomputed because the Reassociation Request and Response frame that carry PMKR1Name are encrypted.

The R0KH may then deliver the latest PMKR0Name to other R1KHs with corresponding PMK-R1 SA in the same mobility domain. The R1KH of the target FTR may also retrieve the latest PMKR0Name from the R0KH.

12.16.8 Key derivation with authentication frame exchange

12.16.8.1 Introduction

This subclause defines rules to derive a PTK through the authentication frame exchange, and the corresponding TK is used to encrypt the Frame Body field of the (Re)Association Request/Response frame as defined in 12.2.7 (Requirements for management frame protection).

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12.16.8.2 FT protocol

If an FTO or FTR (see 13 (Fast BSS transition)) sets the (Re)Association Frame Encryption Support field in the RSNXE to 1, then the FTO or FTR supports the additional rules defined in this subclause.

An FTO that sets the (Re)Association Frame Encryption Support field in the RSNXE to 1 and receives an RSNXE from the FTR with the (Re)Association Frame Encryption Support field equal to 1 shall:

- Include a Diffie-Hellman Parameter element in the first message of the FT protocol (see 13.8 (FT authentication sequence)).
- Select a finite cyclic group in the Diffie-Hellman Parameter element from the dot11RSNAConfig-DLCGroupTable that is at least of the security strength provided by the AKM and cipher suites.
- Generate an ephemeral (random) private key with the chosen finite cyclic group, use the selected group's scalar operation (see 12.4.4.1 (General)) with the private key to generate its ephemeral public key, and indicate the ephemeral public key in the Diffie-Hellman Parameter element.

Otherwise, an FTO that sets dot11EPPReAssociationFrameEncryptionSupportActivated to false or does not receive the RSNXE from the FTR with the (Re)Association Frame Encryption Support field equal to 1 shall not include a Diffie-Hellman Parameter element in the first message of the FT protocol.

For the purpose of interoperability, an FTO and an FTR shall support group 19, an ECC group defined over a 256-bit prime order field.

An FTR that sets the (Re)Association Frame Encryption Support field in the RSNXE to 1 and receives the first message of the FT protocol with the (Re)Association Frame Encryption Support field in the RSNXE set to 1 shall:

- Validate that finite cyclic group indicated in the Diffie-Hellman Parameter element in the first message is supported (present in dot11RSNAConfigDLCGroupTable). Otherwise, the FTR shall reject the first message with status code set to UNSUPPORTED_FINITE_CYCLIC_GROUP.
- Verify the public key indicated in the Diffie-Hellman Parameter element in the first message as specified in 5.6.2.3 of NIST SP 800-56A R2. If verification fails, the FTR shall reject the first message with status code set to INVALID_PUBLIC_KEY.
- Generate an ephemeral (random) private key with the chosen finite cyclic group and use the selected group's scalar operation with the private key to generate its ephemeral public key if the first message is not rejected. Perform the group's scalar-op (see 12.4.4.1 (General)) with the FTO's ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss.
- Derive PTK with DHss as defined in 12.7.1.6.5 (PTK).
- Irretrievably delete the shared secret, DHss, upon completion of PTK generation.
- Include a Diffie-Hellman Parameter element in the second message of the FT protocol (see 13.8 (FT authentication sequence)).
- Indicate the chosen finite cyclic group in the Diffie-Hellman Parameter element of the second message, which is the same as the finite cyclic group in the Diffie-Hellman Parameter element of the first message.
- Indicate its ephemeral public key in the Diffie-Hellman Parameter element of the second message.
- Calculate the MIC in the FTE by using the key, the algorithm, and the MIC size as defined in 13.8.5 (FT authentication sequence: contents of fourth message) on the concatenation of the following data, in the order given here as the input:
    i) FTO's MAC address.
    ii) FTR's MAC address.

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iii) RSNE sent in the Beacons transmitted by the AP with MAC address equal to Address 1 field of the first message.

iv) RSNXE sent in the Beacons transmitted by the AP with MAC address equal to Address 1 field of the first message.

v) the Frame Body field of the second message with MIC field of the FTE set to 0.

— Set the MIC field in the FTE as the calculated MIC.

Otherwise, an FTR that sets dot11EPPReAssociationFrameEncryptionSupportActivated to false or does not receive the RSNXE in the first message of the FT protocol with the (Re)Association Frame Encryption Support field set to 1 shall not include a Diffie-Hellman Parameter element in the second message of the FT protocol.

After receiving the second message of the FT protocol with the status code equal to SUCCESS, an FTO shall:

— Validate that there is a Diffie-Hellman Parameter element included in the second message of the FT protocol if the FTO included a Diffie-Hellman Parameter element in the first message of the FT protocol. If the validation fails, the FTO shall discard the frame and terminate further protocol processing.

— Validate that there is no Diffie-Hellman Parameter element included in the second message of the FT protocol if the FTO did not include a Diffie-Hellman Parameter element in the first message of the FT protocol. If the validation fails, the FTO shall discard the frame and terminate further protocol processing.

— Validate that the finite cyclic group indicated in the Diffie-Hellman Parameter element in the second message is the same as the finite cyclic group indicated in the Diffie-Hellman Parameter element in the first message if the FTO included a Diffie-Hellman Parameter element in the first message of the FT protocol. If the validation fails, the FTO shall discard the frame and terminate further protocol processing.

— Verify the public key indicated in the Diffie-Hellman Parameter element in the second message as specified in 5.6.2.3 of NIST SP 800-56A R2. If the verification fails, the FTO shall discard the frame and terminate further protocol processing.

— Perform the group's scalar-op (see 12.4.4.1 (General)) with the FTR's ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss, if the second message is not discarded.

— Derive PTK with DHss as defined in 12.7.1.6.5 (PTK).

— Irretrievably delete the shared secret, DHss, upon completion of PTK generation.

— Verify the MIC in the FTE using the S1KH of the FTO. If the verification fails, the FTO shall discard the frame and terminate further protocol processing.

12.16.8.3 IEEE 802.1X

If an originator or a responder defined in 12.16.5 (IEEE 802.1X authentication utilizing Authentication frames) has the (Re)Association Frame Encryption Support field in the RSNXE equal to 1, then the originator or the responder supports the additional rules defined in this subclause when performing IEEE 802.1X authentication frame exchange.

An originator that has the (Re)Association Frame Encryption Support field in the RSNXE equal to 1 receives an RSNXE from the responder with the (Re)Association Frame Encryption Support field equal to 1, and intends to continue association after authentication shall do the following in the first Authentication frame:

— Include a Nonce element to indicate SNonce.

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— Include an RSNE to indicate the AKM and the pairwise cipher suite. The Version field shall be set to 1. The Pairwise Cipher Suite Count field shall be set to 1. The AKM Suite Count field shall be set to 1. The PMKID count field and the PMKID List field is set corresponding to PMKSA identifiers if exists. All other fields shall be as specified in 9.4.2.5 (TIM element) and 12.6.3 (RSNA policy selection in an infrastructure BSS).

— Not include an AKM Suite Selector element.

— Include an RSNXE.

— Include a Diffie-Hellman Parameter element.

  • Select a finite cyclic group in the Diffie-Hellman Parameter element from the dot11RSNAConfigDLCGroupTable that is at least of the security strength provided by the AKM and cipher suites.
  • With the chosen finite cyclic group, generate an ephemeral (random) private key, use the selected group's scalar operation (see 12.4.4.1 (General)) with the private key to generate its ephemeral public key, and indicate the ephemeral public key in the Diffie-Hellman Parameter element.

Otherwise, an originator that sets dot11EPPReAssociationFrameEncryptionSupportActivated to false or does not receive the RSNXE from the responder with the (Re)Association Frame Encryption Support field equal to 1 shall not include a Diffie-Hellman Parameter element nor an RSNE nor an RSNXE nor a Nonce element in the first Authentication frame for IEEE 802.1X authentication.

For the purpose of interoperability, an authenticator and a supplicant shall support group 19, an ECC group defined over a 256-bit prime order field.

A responder that sets the (Re)Association Frame Encryption Support field in the RSNXE to 1 receives the first Authentication frame with a Nonce element, RSNE, RSNXE, and a Diffie-Hellman Parameter element shall:

— Verify that the AKM indicated in the RSNE is supported. Otherwise, the responder shall reject the first message with status code set to STATUS_INVALID_AKMP.

— Verify that the pairwise cipher indicated in the RSNE is supported. Otherwise, the responder shall reject the first message with status code set to STATUS_INVALID_PAIRWISE_CIPHER.

— Validate that the finite cyclic group indicated in the Diffie-Hellman Parameter element in the first Authentication frame is supported (present in dot11RSNAConfigDLCGroupTable). Otherwise, the responder shall reject the first message with status code set to UNSUPPORTED_FINITE_CYCLIC_GROUP.

— Verify the public key indicated in the Diffie-Hellman Parameter element in the first message as specified in 5.6.2.3 of NIST SP 800-56A R2. If verification fails, the responder shall reject the first Authentication frame with status code set to INVALID_PUBLIC_KEY.

— Verify that a PMKSA named via a PMKID in the RSNE exists for the specified AKM in the RSNE if one or more PMKIDs are included.

  • If a PMKSA is identified, the responder shall use PMKSA caching, shall not process the EAPOL PDU in the first Authentication frame, and shall not include the EAPOL PDU in the second Authentication frame.
  • If no PMKSA is identified, continue the IEEE 802.1X authentication.

If the first Authentication frame is not rejected, the responder shall:

— Store the indicated SNonce and generate an ephemeral (random) private key with the chosen finite cyclic group and use the selected group's scalar operation with the private key to generate its ephemeral public key.

— Perform the group's scalar-op (see 12.4.4.1 (General)) with the originator's ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss.

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— Use PMKSA caching if a PMKSA is identified via a PMKID in the RSNE in the first Authentication frame and before sending the second Authentication frame:
  • Derive PTK with the identified PMKSA and DHss as defined in 12.7.1.3 (Pairwise key hierarchy).
  • Irretrievably delete the shared secret, DHss, upon completion of PTK generation.

The responder shall do the following in the second Authentication frame:

— Include an RSNE to indicate the AKM and pairwise cipher indicated in the first Authentication frame.
  • If a PMKSA is identified via a PMKID in the RSNE in the first Authentication frame, the responder shall include the PMKID corresponding to the PMKSA in the RSNE.
  • Otherwise, the responder shall not include any PMKID in the RSNE.
— Not include an AKM Suite Selector element.
— Include a Diffie-Hellman Parameter element.
  • Indicate the chosen finite cyclic group in the Diffie-Hellman Parameter element, which is the same as the finite cyclic group in the Diffie-Hellman Parameter element of the first Authentication frame.
  • Indicate its ephemeral public key in the Diffie-Hellman Parameter element.
— Include a Nonce element to indicate ANonce.
— If a PMKSA is identified via a PMKID in the RSNE in the first Authentication frame, include a MIC element where the MIC field of the MIC element is set to the first MMM octets of:

HMAC-HASH (KCK, AA || SPA || AP RSNE || AP RSNXE || Frame Data) (12-10)

where

HASH	is the hash algorithm from the key derivation type (see Table 9-190 (AKM suite selectors)) for each AKM.
KCK	is the key confirmation key for the PTKSA.
AA	is the authenticator address.
SPA	is the supplicant address.
AP RSNE	is the RSN element sent by the AP (e.g. RSNE in the Beacon frames or Probe Response frames) that sends the second Authentication frame.
AP RSNXE	is the RSNXE sent by the AP (e.g. RSNXE in the Beacon frames or Probe Response frames) that sends the second Authentication frame.
Frame Data	is the body of the second Authentication frame including the MIC element with the octets in the MIC field of the MIC element set to 0.
MMM	is half of the output length in octets for the hash function used, that is, 16 or 24 octets for SHA-256 and SHA-384, respectively.

Otherwise, a responder that sets dot11EPPReAssociationFrameEncryptionSupportActivated to false or does not receive the RSNXE in the first Authentication frame with the (Re)Association Frame Encryption Support field equal to 1 shall not include a Diffie-Hellman Parameter element nor a Nonce element nor an RSNE in the second Authentication frame for IEEE 802.1X authentication.

After receiving the second Authentication frame with the status code equal to SUCCESS, an originator shall:

— Validate that there is a Diffie-Hellman Parameter element and an RSNE included in the second Authentication frame and there is no AKM Suite Selector element in the second Authentication frame if the originator included a Diffie-Hellman Parameter element in the first Authentication frame. If the validation fails, the originator shall discard the frame and terminate further protocol processing.

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— Validate that there is no Diffie-Hellman Parameter element and no RSNE included in the second Authentication frame if the originator did not include a Diffie-Hellman Parameter element in the first Authentication frame. If the validation fails, the originator shall discard the frame and terminate further protocol processing.

— Validate that the finite cyclic group indicated in the Diffie-Hellman Parameter element in the second Authentication frame is the same as the finite cyclic group indicated in the Diffie-Hellman Parameter element in the first Authentication frame if the originator included a Diffie-Hellman Parameter element in the first Authentication frame. Validate that the pairwise cipher suite and the AKM indicated in the second Authentication frame are the same as the pairwise cipher suite and the AKM indicated in the first Authentication frame if the originator includes a Diffie-Hellman Parameter element in the first Authentication frame. If the validation fails, the originator shall discard the frame and terminate further protocol processing.

— Verify the public key indicated in the Diffie-Hellman Parameter element in the second Authentication frame as specified in 5.6.2.3 of NIST SP 800-56A R2. If verification fails, the originator shall discard the frame and terminate further protocol processing.

— Validate that the Encapsulation Length field is equal to 0 and validate that the PMKID included in the second Authentication frame matches one of the PMKID(s) indicated in the first Authentication frame if the originator includes one or more PMKIDs in the first Authentication frame, and the second Authentication frame includes a PMKID. If verification succeeds, the originator shall use PMKSA caching with the PMKSA identified by the PMKID indicated in the second Authentication frame and shall not continue the IEEE 802.1X authentication frame exchange. If verification fails, the originator shall discard the frame and terminate further protocol processing.

— Validate that there is no PMKID included in the second Authentication frame if the originator does not include any PMKID in the first Authentication frame. If verification fails, the originator shall discard the frame and terminate further protocol processing.

— Store the indicated ANonce, perform the group's scalar-op (see 12.4.4.1 (General)) with the originator’s ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss, if the second Authentication frame is not discarded.

— Derive the PTK with the identified PMKSA and DHss as defined in 12.7.1.3 (Pairwise key hierarchy) if a PMKSA is identified. Irretrievably delete the shared secret, DHss, upon completion of PTK generation. Compute the MIC as specified in Equation (12-10) and verify it to be the same as the MIC provided in the MIC element. If validation fails, the originator shall discard the frame.

If a PMKSA is not identified through PMKSA caching, before sending the Authentication frame carrying EAP Success, a responder shall:

— Derive the PTK with DHss as defined in 12.7.1.3 (Pairwise key hierarchy).

— Irretrievably delete the shared secret, DHss, upon completion of PTK generation.

If a PMKSA is not identified throughPMKSA caching, after receiving the Authentication frame carrying EAP Success, an originator shall:

— Derive the PTK with DHss as defined in 12.7.1.3 (Pairwise key hierarchy).

— Irretrievably delete the shared secret, DHss, upon completion of PTK generation.

12.16.8.4 FT initial mobility domain association

If an FTO and an FTR (see 13 (Fast BSS transition)) has the (Re)Association Frame Encryption Support field in the RSNXE equal to 1, then the FTO and the FTR support the additional rules defined in this subclause to initiate FT initial mobility domain association.

For the FTO, the first Authentication frame carries an MDE. The contents of the MDE shall be the values advertised by the AP that receives the first Authentication frame in its Beacon or Probe Response frames.

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For the FTR, if the contents of the MDE received in the first Authentication frame do not match the contents advertised in the Beacon or Probe Response frames, the processing status is set to STATUS_INVALID_MDE. If an MDE is present in the first Authentication frame and the base AKMP does not indicate the single FT AKMP, the processing status is set to STATUS_INVALID_AKMP.

For the FTR, the second Authentication frame carries an MDE and an FTE. The MDE has contents as presented in Beacon and Probe Response frames. The FTE shall include the key holder identities, the R0KH-ID and R1KH-ID, set to the values of dot11FTR0KeyHolderID and dot11FTR1KeyHolderID, respectively. The Element Count subfield of the MIC Control field of the FTE shall be set to 0. The length of the MIC field of the FTE shall be 0 (i.e., the MIC Length subfield of the MIC Control field of the FTE is set to 3). The FTE shall have ANonce and SNonce set to 0.

Upon successful completion of Authentication frame processing to establish PMKSA, the R0KH establishes key hierarchy. If a key hierarchy already exists for the FTO belonging to the same mobility domain (i.e., having the same MDID), the R0KH shall delete the existing PMK-R0 security association and PMK-R1 security associations. It then calculates the PMK-R0, PMKR0Name, and PMK-R1 and makes the PMK-R1 available to the R1KH.

12.16.8.5 FILS Authentication

If a FILSO and a FILSR (see 12.11 (Authentication for FILS)) set the (Re)Association Frame Encryption Support field in the RSNXE to 1, then the FILSO and FILSR shall perform FILS authentication with PFS.

12.16.9 Enhanced privacy protection key exchange

12.16.9.1 General

If dot11EPPKEActivated is true, then dot11EPPReAssociationFrameEncryptionSupportActivated and dot11KEKPASNActivated shall be set to true.

Enhanced privacy protection key exchange (EPPKE) is an RSNA authentication protocol that uses the PASN procedures (see 12.12 (Preassociation security negotiation)) with the following differences:

— The base AKM shall be one of:

  • SAE AKMP 00-0F-AC:8, 00-0F-AC:9, 00-0F-AC:24, or 00-0F-AC:25.

— The three Authentication frames have the Authentication Algorithm Number field set to 9 (EPPKE Authentication).

— The generated PTK is used once associated.

— For MLO, Basic Multi-link element is included in every Authentication frame as defined in Table 9-70 (Authentication frame body).

NOTE—The PTK-KEK derived in EPPKE will be used for group key handshake after association even if PTK-KEK is not used in EPPKE frame exchange.

12.16.9.2 Discovery of an EPPKE capable AP

An AP indicates it is capable of performing EPPKE authentication by including the EPPKE AKMP as part of the RSNE. When the EPPKE AKMP is advertised, the AP shall also include at least one additional AKMP in the RSNE.

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12.16.9.3 Key establishment with EPPKE authentication

12.16.9.3.1 Overview

This subclause defines the procedures for establishing a PTKSA and the corresponding shared keys between an EPPKE capable STA and an EPPKE AP (for non-MLO) as well as between an EPPKE capable non-AP MLD and an EPPKE capable AP MLD (for MLO). The same procedures as specified in 12.13.3.1 (Overview) are used with the following differences:

— The three Authentication frames have the Authentication Algorithm Number field set to 9 (EPPKE Authentication).

— For MLO, the PMKSA association is between the AP MLD and the non-AP MLD.

12.16.9.3.2 EPPKE frame construction and processing

The same procedures as specified in 12.12.3.2 (PASN Frame Construction and Processing) are used with the following differences:

— The three Authentication frames have the Authentication Algorithm Number field set to 9 (EPPKE Authentication).

— The PTK is generated as specified in 12.16.9.3.4 (PTKSA derivation and MIC computation with EPPKE authentication).

— The PASN requirement of group addressed traffic not allowed, i.e., setting the Group Data Cipher Suite and the Group Management Cipher Suite to 00-0F-AC:7, does not apply for EPPKE.

— The PASN requirement of setting MFPR to 1 in the RSN Capabilities field does not apply for EPPKE.

— The PASN requirement of setting No Pairwise bit to 0 in the RSNE does not apply for EPPKE.

NOTE—In RSN Capabilities field, there is no No Pairwise bit. (See 9.4.2.23.4 (RSN capabilities)).

For MLO, the first Authentication frame can be sent by any of the non-AP STA affiliated with the non-AP MLD. The RA field of the second Authentication frame in response to the first Authentication frame shall be set to the TA field of the first Authentication frame. The RA field of the third Authentication frame in response to the second Authentication frame shall be set to the TA field of the second Authentication frame.

12.16.9.3.3 EPPKE authentication with SAE

The same procedures as specified in 12.13.5 (PASN authentication with SAE) are used. If the Base AKMP is 00-0F-AC:9 or 00-0F-AC:25, the additional rules defined in 12.16.8.4 (FT initial mobility domain association) is followed.

12.16.9.3.4 PTKSA derivation and MIC computation with EPPKE authentication

The same procedures as specified in 12.13.8 (PTKSA derivation with PASN authentication) shall be used except that “EPPKE PTK Derivation” string is used rather than “PASN PTK Derivation” and for MLO:

— The AP MLD MAC address shall be used instead of the BSSID.

— The non-AP MLD MAC address shall be used instead of the SPA.

The same procedures as specified in 12.13.9.2 (MIC computation for third PASN frame) are used. For MLO, the following modifications shall be used for HMAC-Hash computation:

— The AP MLD MAC address is used instead of the BSSID.

— The non-AP MLD MAC address is used instead of the SPA.

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The Key ID in the PTKSA (see 12.6.1.1.6 (PTKSA)) resulting from EPPKE authentication shall be 0.

NOTE—The MLD MAC address corresponding to the affiliated STA that transmits the Authentication frame is included in the Basic Multi-link element in the three Authentication frames. See 35.3.5.4 (Basic Multi-Link element usage in the context of ML (Re)Setup, Authentication, and FT Action frame exchanges between two MLDs).

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13. Fast BSS transition

13.2 Key holders

13.2.1 Introduction

Insert the following paragraph at the end:

The derived PTK includes a Key derivation key (KDK) under the conditions described in 12.6.1.1.6 (PTKSA).

13.2.2 Authenticator key holders

Change the fourth paragraph as follows

The R1KH and S1KH each derive the PTK~~, the derived PTK including the Key derivation key (KDK) if WUR frame protection is negotiated~~. If WUR frame protection is negotiated, the R1KH and S1KH each derive a WTK from the KDK, which is part of the derived PTK.

Change the seventh paragraph as follows

The R1KH shall meet the following requirements:

- The R1KH-ID shall be set to a MAC address of the physical entity that stores the PMK-R1 and uses it to generate the PTK. That same MAC address shall be used to advertise the PMK-R1 identity to the STA or non-AP MLD and the R0KH.
- For non-MLO, the R1KH shall derive and distribute the GTK and IGTK to all connected STAs. For MLO, the R1KH shall distribute the GTKs and IGTKs for setup links to all connected non-AP MLDs.
- If WUR frame protection is enabled, the R1KH shall derive and distribute the IWGTK and WIPN to all WUR non-AP STAs with which the R1KH has negotiated WUR frame protection.
- For non-MLO, if beacon protection is enabled, the R1KH shall derive and distribute the BIGTK and BIPN to all connected STAs. For MLO, the R1KH shall derive and distribute the BIGTKs and BIPNs for setup links to all connected non-AP MLDs.
- For MLO, if the Group EPP Epoch Supported field in the RSNXE of the APs affiliated with the AP MLD is 1, the R1KH shall derive and distribute the PGTK to the non-AP MLDs that have the Group EPP Epoch Supported field in the RSNXE equal to 1.
- For BPE AP MLD, the R1KH shall derive and distribute the identity key to all connected BPE non-AP MLDs.
- When the PMK-R1 lifetime expires, the R1KH shall delete the PMK-R1 PMKSA and shall revoke all PTKSAs derived from the PMK-R1 using the MLME-DELETEKEYS primitive.
- The R1KH shall not expose the PMK-R1 to other parties.

13.2.3 Supplicant key holders

Change the fourth paragraph as follows

The S1KH shall derive the PTK mutually with the R1KH~~, the derived PTK including the Key derivation key (KDK) if WUR frame protection is negotiated~~. If WUR frame protection is negotiated, the S1KH shall derive a WTK from the KDK, which is part of the derived PTK.

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13.5 FT protocol

13.5.2 Over-the-air FT protocol authentication in an RSN

Change eighth paragraph as follows:

If PMKSA caching privacy is not used, t~~T~~he R1KH of the target FTR uses the value of PMKR0Name and other information in the frame to calculate PMKR1Name and check if a PMK-R1 can be identified with the PMKR1Name. If PMKSA caching privacy is used, then the R1KH of the target FTR uses the value of PMKR0Name to check if a PMK-R1 and corresponding PMKR1Name can be identified (see 12.6.1.1.4 (PMK-R1 security association)). If the target FTR does not ~~have the key identified by PMKR1Name~~identify a PMK-R1, it may retrieve that key from the R0KH identified by the FTO. See 13.2 (Key holders). Upon receiving a new PMK-R1 for a STA, the target AP shall delete the prior PMK-R1 security association and PTKSAs derived from the prior PMK-R1.

13.6 FT resource request protocol

13.6.2 Over-the-air fast BSS transition with resource request

Change the twelfth paragraph as follows:

In an RSN, on successful completion of the FT authentication exchange of the FT resource request protocol, the PTKSA has been established and proven live. The key replay counter shall be initialized to 0, and the subsequent EAPOL-Key PDUs (e.g., GTK, IGTK, BIGTK, PGTK and WIGTK updates) shall use the key replay counter to detect and discard replays. The PTKSA shall be deleted by the target FTR if it does not receive a Reassociation Request frame from the FTO within the reassociation deadline timeout value.

13.6.3 Over-the-DS fast BSS transition with resource request

Change the twelfth paragraph as follows:

In an RSN, on successful completion of the FT Confirm/Acknowledgment frame exchange, the PTKSA has been established and proven live. The key replay counter shall be initialized to 0, and the subsequent EAPOL-Key frames (e.g., GTK, IGTK, BIGTK, ~~and~~ WIGTK and PGTK updates) shall use the key replay counter to detect and discard replays. The PTKSA shall be deleted by the target FTR if it does not receive a Reassociation Request frame from the FTO within the reassociation deadline timeout value. Resource request procedures are specified in 13.11 (Resource request procedures).

13.7 FT reassociation

13.7.1 FT reassociation in an RSN

Change the second paragraph as follows:

The FTO shall perform a reassociation directly with the target FTR via the following exchange:

FTO→Target FTR: Reassociation Request(RSNE[PMKR1Name], MDE, FTE[MIC, ANonce, SNonce, R1KH-ID, R0KH-ID], RIC-Request, RSNXE, Basic Multi-Link element)

Target FTR→FTO: Reassociation Response(RSNE[PMKR1Name], MDE, FTE[MIC, ANonce, SNonce, R1KH-ID, R0KH-ID, GTK[N], IGTK[M], BIGTK[Q], WIGTK[R], PGTK, MLO GTKn, MLO IGTKn, MLO BIGTKn], RIC-Response, RSNXE, Basic Multi-Link element)

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where

— MLO GTK is the MLO GTK subelement for the AP affiliated with the AP MLD for the link specified by the value in the Link ID field,

— MLO IGTK is the MLO IGTK subelement for the AP affiliated with the AP MLD for the link specified by the value in the Link ID field,

— MLO BIGTK is the MLO BIGTK subelement for the AP affiliated with the AP MLD for the link specified by the value in the Link ID field.

— The GTK[N], IGTK[M], and BIGTK[Q] are present when the FTR is an AP.

— The MLO GTKn, MLO IGTKn, MLO BIGTKn, PGTK and the Basic Multi-Link element are present when the FTR is an AP MLD.

13.8 FT authentication sequence

13.8.1 Overview

Change sixth paragraph as follows:

The first message is used by the FTO to initiate a fast BSS transition. When RSNA is enabled, the FTO shall include the R0KH-ID and the SNonce in the FTE and the PMKR0Name in the RSNE. If PMKSA caching privacy is not used, t~~T~~he target FTR can use the PMKR0Name to derive the PMKR1Name and check if a PMK-R1 can be identified. If PMKSA caching privacy is used, then the R1KH of the target FTR uses the value of PMKR0Name to check if a PMK-R1 and corresponding PMKR1Name can be identified (see 12.6.1.1.4 (PMK-R1 security association)). ~~, and~~ ~~I~~if the target FTR does not identify~~have the~~ a PMK-R1 ~~identified by PMKR1Name~~, it may attempt to retrieve that key from the R0KH identified by R0KH-ID. See 13.2 (Key holders). The FTO includes a fresh SNonce as its contribution to the association instance identifier and to provide key separation of the derived PTK; it is selected randomly to serve as a challenge that demonstrates the liveness of the peer in the fourth message.

13.8.5 FT authentication sequence: contents of fourth message

Change the fifth paragraph as follows:

If present, the FTE shall be set as follows:

— When this message of the authentication sequence appears in a Reassociation Response frame, the Optional Parameter(s) field in the FTE may include the GTK, IGTK, BIGTK, WIGTK subelements or identity key, PGTK, MLO GTK, MLO IGTK and MLO BIGTK subelements. If a GTK, an IGTK, a BIGTK, a WIGTK, a PGTK, an identity key, an MLO GTK, an MLO IGTK or an MLO BIGTK are included, the Key field of the subelement shall be wrapped using PTK-KEK or KEK2 and the appropriate key wrap algorithm, as specified in Table 12-11 (Integrity and key wrap algorithms) and 12.7.2 (EAPOL-Key frames). The padding consists of appending a single octet 0xdd followed by zero or more 0x00 octets. When processing a received message, the receiver shall ignore this trailing padding. Addition of padding does not change the value of the Key Length field. Note that the length of the encrypted Key field can be determined from the length of the GTK, IGTK, BIGTK, PGTK, identity key, WIGTK, MLO GTK, MLO IGTK, or MLO BIGTK subelement.

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Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.2 Abbreviations and special symbols

B.2.2 General abbreviations for Item and Support columns

Insert the following abbreviations (maintaining alphabetical order):

EPPM Enhanced privacy protection MAC

B.4 PICS proforma—IEEE Std 802.11-2020

B.4.3 IUT configuration

Insert the following rows at the end of the table:

Item	IUT configuration	References	Status	Support
…				
*CFEPPCPE	EPP features for client privacy enhancements	9.4.2.240 (RSNXE), 10.71 (Frame anonymization), 12.16 (Client privacy enhancement)	O	Yes ☐ No ☐ N/A ☐
*CFEPPBPE	EPP features for basic service set (BSS) privacy enhancements	9.4.2.240 (RSNXE), 10.71 (Frame anonymization), 12.16 (Client privacy enhancement)	O	Yes ☐ No ☐ N/A ☐

B.4.4 MAC protocol

B.4.4.2 MAC frames

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Modify the table as follows (maintaining item order):

Item	MAC frame	References	Status	Support
	Is transmission of the following MAC frames supported?	Clause 9		
…	…	…	…	…
FR77.13	Link Reconfiguration Response frame	9.6.38.14	CFEHTMLD-nonAP AND EHTM10.14.3: M	Yes ☐ No ☐ N/A ☐
FR77.14	Protected EHT Compressed Beamforming/CQI frame	9.6.38.15	EPPM2.2: M	Yes ☐ No ☐ N/A ☐
…	…	…	…	…
FT97	EPP Action	9.6.42 (EPP Action frame details)		Yes ☐ No ☐ N/A ☐
FT97.1	EPP Capabilities And Operation Parameters Request frame	9.6.42.2 (EPP Capabilities And Operation Parameters Request frame format)	EPPM3: M	Yes ☐ No ☐ N/A ☐
FT97.2	EPP Capabilities And Operation Parameters Response frame	9.6.42.3 (EPP Capabilities And Operation Parameters Response frame format)	EPPM3: M	Yes ☐ No ☐ N/A ☐
FT97.3	EPP Groups Parameter frame	9.6.42.4 (EPP Groups Parameter frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.4	EPP Group Request frame	9.6.42.5 (EPP Group Request frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.5	EPP Group Response frame	9.6.42.6 (EPP Group Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.6	EPP Group Assignment frame	9.6.42.7 (EPP Group Assignment frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐

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Item	MAC frame	References	Status	Support
FT97.7	OTA MAC Collision Notification frame	9.6.42.8 (OTA MAC Collision Notification frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.8	OTA MAC Collision Response frame	9.6.42.9 (OTA MAC Collision Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.9	AID Assignment Request frame	9.6.42.10 (AID Assignment Request frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.10	AID Assignment Response frame	9.6.42.11 (AID Assignment Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT97.11	EPP Group Parameter Update frame format	9.6.42.12 (EPP Group Parameter Update frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FT98	Privacy Beacon frame	9.3.4.4 (Privacy Beacon frame format)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐
	Is reception of the following MAC frames supported?	Clause 9		
…	…	…	…	…
FR98	EPP Action	9.6.42 (EPP Action frame details)		Yes ☐ No ☐ N/A ☐
FR98.1	EPP Capabilities And Operation Parameters Request frame	9.6.42.2 (EPP Capabilities And Operation Parameters Request frame format)	EPPM3: M	Yes ☐ No ☐ N/A ☐

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Item	MAC frame	References	Status	Support
FR98.2	EPP Capabilities And Operation Parameters Response frame	9.6.42.3 (EPP Capabilities And Operation Parameters Response frame format)	EPPM3: M	Yes ☐ No ☐ N/A ☐
FR98.3	EPP Groups Parameter frame	9.6.42.4 (EPP Groups Parameter frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.4	EPP Group Request frame	9.6.42.5 (EPP Group Request frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.5	EPP Group Response frame	9.6.42.6 (EPP Group Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.6	EPP Group Assignment frame	9.6.42.7 (EPP Group Assignment frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.7	OTA MAC Collision Notification frame	9.6.42.8 (OTA MAC Collision Notification frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.8	OTA MAC Collision Response frame	9.6.42.9 (OTA MAC Collision Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.9	AID Assignment Request frame	9.6.42.10 (AID Assignment Request frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR98.10	AID Assignment Response frame	9.6.42.11 (AID Assignment Response frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐

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Item	MAC frame	References	Status	Support
FR98.11	EPP Group Parameter Update frame format	9.6.42.12 (EPP Group Parameter Update frame format)	EPPM9.1: M	Yes ☐ No ☐ N/A ☐
FR99	Privacy Beacon frame	9.3.4.4 (Privacy Beacon frame format)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐

Insert the following new subclause at the end of Annex B.4:

-- Editor Note: 802.11bf-2025 up to B.4.41

B.4.42 Enhanced privacy protection (EPP) features

B.4.42.1 EPPMAC features

Item	Protocol capability	References	Status	Support
	Are the following MAC protocol features supported?			
EPPM1	Contents of Probe Request frame	12.16.2 (Contents of Probe Request frame)	CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM2	EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames			
EPPM2.1	EPP robust individually addressed Management frames	12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐

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Item	Protocol capability	References	Status	Support
*EPPM2.2	EPP robust individually addressed beamforming/CSI/CQI frames	12.16.3 (EPP robust individually addressed Management frames and robust individually addressed beamforming/CSI/CQI frames)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
*EPPM3	Capabilities and operation parameters request and response procedure	12.16.4 (Capabilities and operation parameters request and response procedure)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM4	IEEE 802.1X authentication utilizing Authentication frames	12.16.5 (IEEE 802.1X authentication utilizing Authentication frames)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM5	(Re)Association Request/Response frame encryption	12.16.6 ((Re)Association Request/Response frame encryption)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM6	PMKSA caching privacy	12.16.7 (PMKSA caching privacy)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM7	Key derivation with authentication frame exchange	12.16.8 (Key derivation with authentication frame exchange)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐
EPPM8	Enhanced privacy protection key exchange	12.16.9 (Enhanced privacy protection key exchange)	PC34 AND CFEPPCPE: O	Yes ☐ No ☐ N/A ☐

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Item	Protocol capability	References	Status	Support
EPPM9	Frame anonymization	10.71 (Frame anonymization)		Yes ☐ No ☐ N/A ☐
*EPPM9.1	EPP epoch operation	10.71.2 (EPP epoch operation)	PC34 AND CFEHTMLD AND CFEPPCPE: O PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.2	Establishing CPE MAC header anonymization parameter sets	10.71.3 (Establishing CPE MAC header anonymization parameter sets)	PC34 AND CFEHTMLD AND CFEPPCPE: O PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.3	Establishing BPE MAC header anonymization parameter sets	10.71.4 (Establishing BPE MAC header anonymization parameter sets)	PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.4	MAC header anonymization and transmitting functions	10.71.5 (MAC header anonymization and transmitting functions)	PC34 AND CFEHTMLD AND CFEPPCPE: O PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.5	MAC header anonymization and receiving functions	10.71.6 (MAC header anonymization and receiving functions)	PC34 AND CFEHTMLD AND CFEPPCPE: O PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.6	Frame anonymization and AID	10.71.7 (Frame anonymization and AID)	PC34 AND CFEHTMLD AND CFEPPCPE: O PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐

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Item	Protocol capability	References	Status	Support
*EPPM9.7	BSS privacy enhancement operations	10.71.8 (BSS privacy enhancements operations)	PC34 AND CFEHTMLD AND CFEPPBPE: M	Yes ☐ No ☐ N/A ☐
EPPM9.7.1	BPE AP MLD discovery	10.71.8.2 (BPE AP MLD discovery)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐
EPPM9.7.2	BPE AP MLD beaconing	10.71.8.3 (BPE AP MLD beaconing)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐
EPPM9.7.3	Group addressed frames anonymization	10.71.8.4 (Group addressed frames anonymization)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐
EPPM9.7.4	Steering to a BPE AP MLD	10.71.8.5 (Steering to a BPE AP MLD)	EPPM9.7: M	Yes ☐ No ☐ N/A ☐

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Annex C

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ASN.1 encoding of the MAC and PHY MIB

C.3 MIB detail

Change the comment list following the dot11smt definition (not all lines shown) as follows:

-- Editor Note: ANA assignment is done

```
-- ****************************************************************
-- *  Major sections
-- ****************************************************************

--  Station ManagementT (SMT) Attributes
    --  DEFINED AS "The SMT object class provides the necessary support
    --  at the station to manage the processes in the station such that
    --  the station may work cooperatively as a part of an IEEE 802.11
    --  network."

   dot11smt OBJECT IDENTIFIER ::= { ieee802dot11 1 }
   --  dot11smt GROUPS
   -- ...
   --  dot11EPPStationConfigTable                    ::= { dot11smt 50 }
```

Insert a new "dot11EPPStationConfig" Table and associated MIBs as shown below

```
-- ****************************************************************
-- * dot11EPPStationConfig TABLE
-- ****************************************************************

dot11EPPStationConfigTable OBJECT-TYPE
   SYNTAX SEQUENCE OF Dot11EPPStationConfigEntry
   MAX-ACCESS not-accessible
   STATUS current
   DESCRIPTION
      "EPP station configuration attributes. In tabular form to allow for multi-
      ple instances on an agent"
   ::= { dot11smt 50 }

dot11EPPStationConfigEntry OBJECT-TYPE
   SYNTAX Dot11EPPStationConfigEntry
   MAX-ACCESS not-accessible
   STATUS current
   DESCRIPTION
      "An entry in the dot11EPPStationConfigTable. The entries address configu-
      ration options for EPP operation."
   INDEX { ifIndex }
   ::= { dot11EPPStationConfigTable 1 }

Dot11EPPStationConfigEntry ::= SEQUENCE
   {
      dot11EPPKEActivated                                TruthValue,
      dot11EPPEpochActivated                             TruthValue,
      dot11EPPEpochStartTimeMargin                       Unsigned32,
      dot11EPPEpochTransitionTime                        Unsigned32,
```

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```
        dot11EPPRobustIndividuallyAddressedManagementFrameActivated
                                                                TruthValue,
        dot11EPPCapabilitiesAndOperationParametersExchangeActivated
                                                                TruthValue,
        dot11EPPReAssociationFrameEncryptionSupportActivated  TruthValue,
        dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated
                                                                TruthValue,
        dot11EPPPMKSACachingPrivacySupportActivated           TruthValue,
        dot11DSMACAddressActivated                            TruthValue,
        dot11PrivacyBeaconResponseTime                        Unsigned32,
        dot11EPPRobustIndividuallyAddressedBFCSICQITBTxActivated
                                                                TruthValue,
        dot11EPPRobustIndividuallyAddressedBFCSICQINonTBTxActivated
                                                                TruthValue
    }


dot11EPPKEActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute indicates whether or not EPPKE authentication is enabled."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 1 }


dot11EPPEpochActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect as soon as practical in the implementation.

        This attribute, when true, indicates that the station capability of group
        epochs is enabled. false indicates that the capability is present but is
        disabled. This attribute is not present if the capability is not present."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 2 }


dot11EPPEpochStartTimeMargin OBJECT-TYPE
    SYNTAX Unsigned32 (1..100)
    UNITS "0.1 milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect as soon as practical in the implementation.

        This attribute indicates the duration, before an epoch boundary, during

        which a STA receiving individually addressed frames accepts frames that

        use the current or the next epoch settings."
```


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```
    DEFVAL { 100 }
    ::= { dot11EPPStationConfigEntry 3 }


dot11EPPEpochTransitionTime OBJECT-TYPE
    SYNTAX Unsigned32 (1..20470)
    UNITS "TUs"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect as soon as practical in the implementation.

        This attribute indicates the duration, after an epoch boundary, during
        which a STA receiving individually addressed frames accepts frames that
        use the previous or the current epoch settings."
    DEFVAL { 300 }
    ::= { dot11EPPStationConfigEntry 4 }


dot11EPPRobustIndividuallyAddressedManagementFrameActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support EPP robust
        individually addressed Management frames is enabled. The capability is
        disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 5 }


dot11EPPCapabilitiesAndOperationParametersExchangeActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support capabili-
        ties and operation parameters request and response is enabled. The capa-
        bility is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 6 }


dot11EPPReAssociationFrameEncryptionSupportActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.
```

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```
        This attribute, when true, indicates the capability to support encryption
        of the (Re)Association frame is enabled. The capability is disabled other-
        wise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 7 }


dot11EPPIEEE8021XUtilizingAuthenticationFrameActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support IEEE 802.1X
        authentication utilizing Authentication frame is enabled. The capability
        is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 8 }


dot11EPPPMKSACachingPrivacySupportActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support PMKSA cach-
        ing privacy is enabled. The capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 9 }

dot11DSMACAddressActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support use of a DS
        MAC address is enabled. The capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 10 }

dot11PrivacyBeaconResponseTime OBJECT-TYPE
SYNTAX Unsigned32 (1..100)
    UNITS "0.05 milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect as soon as practical in the implementation.
```


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```
        This attribute defines the maximum time after receiving a Privacy Beacon
        Solicit Request frame that a Privacy Beacon frame is scheduled for trans-
        mission."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 11 }

dot11EPPRobustIndividuallyAddressedBFCSICQITBTxActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support EPP robust
        individually addressed trigger-based beamforming Management frame trans-
        mission is enabled. The capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 12 }

dot11EPPRobustIndividuallyAddressedBFCSICQINonTBTxActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.
        Changes take effect for the next MLME-START.request primitive or MLME-
        JOIN.request primitive.

        This attribute, when true, indicates the capability to support EPP robust
        individually addressed non-trigger-based beamforming Management frame
        transmission is enabled. The capability is disabled otherwise."
    DEFVAL { false }
    ::= { dot11EPPStationConfigEntry 13 }

-- ************************************************************************
-- * End of dot11EPPStationConfig TABLE
-- ************************************************************************
```

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